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Classical and Rough Volatility Models

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Aus dem Paradies, das Cantor uns geschaffen hat, soll uns niemand vertreiben können.

We shall never be expelled from the paradise Cantor created for us.

David Hilbert

Preface

This document constitutes the Master's Thesis of Héctor Zalama Alonso, prepared as the final requirement for the Master's Degree in Advanced Mathematics at UNED, under the supervision of Carlos Escudero Liébana, Full Professor at UNED.

The aim of this work is to present a natural progression from the classical foundations of stochastic calculus to the study and application of modern tools such as Rough Path Theory, with the ultimate goal of employing these tools in the field of financial modelling — particularly in the calibration of advanced volatility models. The intention is to illustrate how techniques originally developed in purely analytical contexts can find natural and effective applications in real-world problems in quantitative finance.

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Notation

A recurring difficulty in a text of this kind is that probabilists and practitioners often write the same object in different ways, and occasionally write different objects in the same way. The following conventions are fixed once and used without exception throughout. Where a financial convention differs from the probabilistic one, the discrepancy is pointed out in a footnote at the place where it first matters.

Probability space and information	
$(\Omega, \mathcal{F}, \mathbb{P})$	Complete probability space; $\mathbb{P}$ is always the *physical* measure.
$\mathbb{Q}$	Risk-neutral measure associated with the numeraire S^0.
$\mathbb{Q}^T$	T-forward measure, associated with the numeraire $P(\cdot, T)$.
$(\mathcal{F}_t)_{t\in[0,T]}$	Filtration, always satisfying the usual conditions (Definition 1.1.6).
$(\mathcal{F}_t^W)_{t\in[0,T]}$	Augmented filtration generated by the Brownian motion W.
$\mathbb{E}[\,\cdot\,]$, $\mathbb{E}^{\mathbb{Q}}[\,\cdot\,]$	Expectation under $\mathbb{P}$, and under a named measure.
$\mathbb{V}[\,\cdot\,]$	Variance. (Not to be confused with V, an option value.)
$\mathcal{B}(E)$	Borel σ-algebra of the topological space E.
$\mathbb{1}_A$	Indicator function of the event A.
Time	
$[0, T]$	Time interval. Always anchored at 0; T is simultaneously the time horizon and the maturity of the contract under consideration, which is exactly why the same letter is used.[1]
τ	A stopping time; $\mathcal{T}_{[t,T]}$ is the set of stopping times valued in $[t, T]$.
Processes	
W	Brownian motion. The letter B is never used for it.
$\mathbf{W}$	Multidimensional Brownian motion with independent components.
W^H	Fractional Brownian motion with Hurst parameter H.
μ	A *drift*, always. Abstract measures are denoted ν, λ.
σ	A *diffusion coefficient*; $\Sigma = \sigma\sigma^\top$ is the diffusion matrix.
ρ	A *correlation*, always. Densities are denoted p or f.
$p_{s,x}(t, y)$	Transition density of a Markov process.
$[X]$, $\langle X\rangle$	Realised (pathwise) and predictable quadratic variation of X; the two coincide for the continuous processes considered here.
$\mathcal{A}$, $\mathcal{A}^*$	Infinitesimal generator of a Markov semigroup and its adjoint.
$\mathcal{E}_\alpha$	Stochastic (Doléans-Dade) exponential of $\int \alpha\, dW$.
Finance	
S_t	Spot price of the underlying.
F_t or $F(t, T)$	Forward price for delivery at T.
$P(t, T)$	Price at t of a zero-coupon bond maturing at T.
S^0	Numeraire asset.
H_T, Φ	Terminal payoff (a random variable) and payoff function.
K	Strike price; $k = \log(K/F)$ is log-moneyness.
r, q	Short rate and continuous dividend (or repo) yield.
σ_{imp}, σ_{loc}	Implied and local volatility; $w = \sigma_{\text{imp}}^2 T$ is total implied variance.
V	The value of a derivative. (Not to be confused with $\mathbb{V}$, variance.)

[1] Probability texts customarily work on a generic interval $[a, b]$. We deliberately fix $a = 0$: in a financial context the present instant is the natural origin, every quoted quantity is a function of time-to-maturity, and carrying an arbitrary left endpoint through the pricing formulas obscures rather than generalises. Nothing is lost, since any statement on $[a, b]$ transfers by translation.

Part I

Foundations and Classical Models

Introduction

0.1 A worthy note on context

The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a gift which we neither understand nor deserve.

Eugene Wigner expressed his admiration in his 1960 article in *Communications in Pure and Applied Mathematics.* It is a sentiment many would agree with: the effectiveness of mathematics in describing Nature is both remarkable and fascinating. Moreover, over the last century its branches have reached into areas where, not so long ago, mathematicians were considered too abstract to be of much use in business terms.

In 1827, Robert Brown, while studying the behaviour of pollen grains suspended in water, noticed something unexpected: the grains wandered along irregular, hard-to-describe paths. Physics was not prepared to tackle such a problem yet. What was needed belonged to what we now call *statistical mechanics*, the study of collective behaviour of particles and molecules, intimately tied to *thermodynamics.* The decisive step came later, in 1905, when Albert Einstein provided a theory of this motion — the same year he also explained the photoelectric effect, introduced the Special Relativity theory and provided the first appearance of the mass-energy equivalence formula, $E = mc^2$. What a year.

Regarding the polen grains, a parallel story took place in finance. Louis Bachelier, a doctoral student of Henri Poincaré, was interested in modelling stock prices. His model, which we now recognise as essentially Gaussian, worked with some shaky concepts on probability theory (Kolmogorov and his peers were not born yet, even Lebesgue had not published his work on measure theory yet). Worse, it allowed negative prices — a point critics did not miss.

The common thread in both stories is *Brownian motion.* This stochastic process is a central object in probability: geometers find its fractal dimension fascinating, while probabilists value the rich family of processes built upon it. The idea of treating "random noise" as a genuine mathematical object took shape gradually, but it was Kiyoshi Itô's work on *stochastic integration* that truly changed the picture.

Finance joined somewhat later, but with undeniable impact. Today mathematics is everywhere in banks, consultancy and audit firms, underpinning the valuation of derivatives, that is, assigning a *fair price* (according to market data) to contracts involving random future cash flows. This pricing orientation will be the focus of the present thesis, among many other applications in finance, such as risk management and portfolio optimization.

0.2 Objective and Scope

The primary objective of this Thesis is to provide a solid review of *Volatility*, aiming to fill the gap between those who are strictly Mathematics-oriented—who often lack expertise in Finance—and those who are Finance-oriented and do not fully appreciate the depth of the ideas they are working with. For this reconciliatory purpose, the Thesis embraces a clear and rigorous mathematical mindset, while maintaining continuous references to practical and financial interpretations.

Part I aims to provide a thorough introduction to the subject. The core mathematical ideas, rooted in Probability Theory and forming the basis of modern Financial Theory and basic Pricing Theory, are presented in Chapter 1. Particular attention is paid there to the regularity of Brownian trajectories, since it is precisely the delicate balance between roughness and tractability — non-vanishing quadratic variation on the one hand, unbounded total variation on the other — that both makes Itô's theory necessary and marks the limits of its reach.

Part II is devoted to more recent developments in volatility modelling, including Rough Paths and Rough Volatility theories (Chapter 2). The latter is introduced from a general and mathematical perspective: although it finds direct applications in Finance, the theory of rough paths has far-reaching implications in other areas of mathematics as well. These chapters aim to highlight the necessity of more sophisticated tools to describe and operate in the market.

Finally, these theoretical constructs are brought to practice in Chapter 3, where a calibration of XXX is performed. Here, the models are numerically implemented and compared, providing an applied perspective and illustrating the strengths and challenges associated with each approach. Some final conclusions are left for Chapter 4, where a general overview of the work done will be given, along with reflections on possible directions for further research.

Chapter 1

Stochastic and Financial Foundations

We assume the reader is familiar with the foundations of Probability Theory. Standard references include, for example, Billingsley's classic text [4], as well as the treatments in [20] and [38]. Nevertheless, we briefly recall the basic probabilistic framework in order to fix notation and ensure clarity.

Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with Ω the sample space, $\mathcal{F}$ a σ-algebra of events, and $\mathbb{P}$ a probability measure. A **random variable** is a measurable map $X : (\Omega, \mathcal{F}) \to (E, \mathcal{E})$, the target being $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ in most of our applications.

Convention 1.0.1. Every probability space in this dissertation is assumed *complete*: if $A \in \mathcal{F}$ has $\mathbb{P}(A) = 0$, then every $B \subseteq A$ lies in $\mathcal{F}$ and has $\mathbb{P}(B) = 0$.

Completeness is not a formality, and we shall see in Definition 1.1.6 exactly what it buys. For the moment it is enough to record that this is the first of several points at which the measure-theoretic language earns its place: it is not a technical convenience but the only setting in which path regularity, filtrations and martingale properties can be discussed at all.

Convention 1.0.2 (The symbol $\blacksquare$)**.** A result whose statement ends with $\blacksquare$ is quoted **without proof**, a reference being given in every case. A result without the symbol is one we prove, either immediately below it or — when the argument is routine but too long for the body — in Appendix B. The convention is followed throughout Part I and Part II.

1.1 Probabilistic Preliminaries

The notion of *stochastic process* generalizes that of a single random variable by allowing evolution in an external parameter, usually interpreted as time. To accommodate both discrete-time and continuous-time settings in a unified framework, we let $\mathbb{T}$ denote the **index set**, representing time. Typically, $\mathbb{T}$ is a subset of $\mathbb{R}$, such as:

$$\mathbb{T} \in \{\mathbb{N}, \mathbb{N}_0, \mathbb{Z}, [0, T], [0, \infty), \mathbb{R}\},$$

depending on the application.

Definition 1.1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space, and let $\mathbb{T}$ be a time index set. A **stochastic process** is a collection of random variables $\{X_t : t \in \mathbb{T}\}$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$ with values in a measurable space $(E, \mathcal{E})$, such that the map

$$X : \mathbb{T} \times \Omega \to E, \quad (t, \omega) \mapsto X_t(\omega)$$

is jointly measurable with respect to $\mathcal{B}(\mathbb{T}) \otimes \mathcal{F}$. In particular:

- For each fixed $t \in \mathbb{T}$, the function $\omega \mapsto X_t(\omega)$ is a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$.
- For each fixed $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ is called the *sample path* or *trajectory* of the process.

We may denote the process by $X = \{X_t\}_{t\in\mathbb{T}}$, and abbreviate $X_t(\omega)$ as $X(t)$ or simply X_t when context allows. We say X is *discrete* whenever $\mathbb{T}$ is $\mathbb{N}$ or $\mathbb{Z}$ and *continuous* if $\mathbb{T}$ is $[0, T]$ for some $T > 0$ or $[0, \infty)$.

This formulation includes many important examples such as Markov chains, Poisson processes, and Brownian motion. Note that the nature of the index set $\mathbb{T}$ often dictates the regularity assumptions or path properties we may require (e.g., right-continuity, càdlàg, continuity in probability, etc.). Analyzing the dynamic evolution of these processes requires us to quantify their expected future behavior given currently available information. This idea is contained in the **conditional expectation**. Its existence relies on the following profound result:

Theorem 1.1.2 (Radon–Nikodym)**.** *Let ν and λ be two σ-finite measures on the same measurable space $(\Omega, \mathcal{F})$. Suppose that ν is* absolutely continuous *with respect to λ, denoted $\nu \ll \lambda$; that is,*

$$\lambda(A) = 0 \quad \Rightarrow \quad \nu(A) = 0, \quad \textit{for all } A \in \mathcal{F}. \tag{1.1}$$

Then there exists a measurable function $f : \Omega \to [0, \infty)$, unique up to λ-null sets, such that for all $A \in \mathcal{F}$,

$$\nu(A) = \int_A f \, d\lambda. \tag{1.2}$$

This function f is called the Radon–Nikodym derivative *of ν with respect to λ, and is denoted by $f = d\nu/d\lambda$.* ■

The letter μ is deliberately avoided here. From Section 1.2 onwards it will denote a drift coefficient and nothing else; abstract measures are ν and λ throughout.

We shall need the statement in slightly greater generality than above, since the measure we are about to construct is not positive.

Corollary 1.1.3 (Signed version)**.** *Let ν be a finite* signed *measure on $(\Omega, \mathcal{F})$ and λ a σ-finite positive measure with $\nu \ll \lambda$. Then there exists $f \in L^1(\Omega, \mathcal{F}, \lambda)$, real-valued and unique up to λ-null sets, satisfying* (1.2)*.*

Proof: Apply the Jordan decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. Both $\nu^\pm \ll \lambda$, so Theorem 1.1.2 yields non-negative densities $f^\pm$, and $f := f^+ - f^-$ works; integrability follows from $\int (f^+ + f^-) \, d\lambda = |\nu|(\Omega) < \infty$. ■

As a consequence, if $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ (that is, $\int |X| \, d\mathbb{P}$ is finite) and letting $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra, the *conditional expectation* of X given $\mathcal{G}$, denoted $\mathbb{E}[X \mid \mathcal{G}]$, is the unique (up to $\mathbb{P}$-a.e. equality) $\mathcal{G}$-measurable function satisfying:

$$\int_G \mathbb{E}[X \mid \mathcal{G}] \, d\mathbb{P} = \int_G X \, d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}. \tag{1.3}$$

This definition is structurally equivalent to the Radon–Nikodym theorem. Define the set function $\nu : \mathcal{G} \to \mathbb{R}$ by

$$\nu(G) := \int_G X \, d\mathbb{P}, \quad \text{for } G \in \mathcal{G}. \tag{1.4}$$

Since X is integrable, ν is a finite *signed* measure on $(\Omega, \mathcal{G})$, and it is absolutely continuous with respect to $\mathbb{P}|_{\mathcal{G}}$, the restriction of $\mathbb{P}$ to $\mathcal{G}$. It is Corollary 1.1.3, rather than Theorem 1.1.2 itself, that applies here: X need not be non-negative, so neither is ν, and the density produced must be allowed to change sign. With that proviso (see [4] or [55]), there exists a $\mathcal{G}$-measurable function, namely $\mathbb{E}[X \mid \mathcal{G}]$, such that:

$$\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P}, \quad \text{for all } G \in \mathcal{G}. \tag{1.5}$$

Conditional expectation inherits the properties one would expect of an averaging operator: it preserves expectations and order, it is linear, it acts as the identity on $\mathcal{G}$-measurable variables and as the plain expectation on independent ones, known factors may be taken out of it, and it satisfies the tower property $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}]$ for $\mathcal{H} \subseteq \mathcal{G}$. These, together with the conditional forms of the Fatou, monotone and dominated convergence theorems, are recorded as Proposition A.1.1 in Appendix A; we shall use them throughout without comment. Two further properties, however, are used more than all the others combined and are stated here.

Proposition 1.1.4 (Conditional Jensen and the L^2 projection)**.** *Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-field. Then, almost surely:*

8. ***(Conditional Jensen's inequality)*** *If $\phi : \mathbb{R} \to \mathbb{R}$ is convex and both X and $\phi(X)$ are integrable, then*
$$\phi\Big(\mathbb{E}[X \mid \mathcal{G}]\Big) \leq \mathbb{E}[\phi(X) \mid \mathcal{G}].$$

9. *(L^2 **projection**) If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace $L^2(\Omega, \mathcal{G}, \mathbb{P})$. Equivalently, it is the unique $\mathcal{G}$-measurable Y minimising $\mathbb{E}[(X - Y)^2]$, and*
$$\mathbb{E}\Big[(X - \mathbb{E}[X \mid \mathcal{G}])\, Z\Big] = 0 \quad \textit{for every } Z \in L^2(\Omega, \mathcal{G}, \mathbb{P}).$$

■

The numbering is retained from Proposition A.1.1 for ease of reference. Property 8 is what guarantees that X^2 is a submartingale whenever X is a martingale — a fact invoked twice in what follows, in Subsection 1.1.3 and again in the Doob-Meyer decomposition — and it is the reason convexity is such a persistent theme in option pricing.

Property 9 deserves more than emphasis. It says that conditioning is nothing more exotic than orthogonal projection in a Hilbert space, and once this is seen, a good deal of what follows stops looking like a series of coincidences. The Itô isometry of Theorem 1.2.5 will be an isometry between two Hilbert spaces; the density argument that extends the stochastic integral from step processes to all of L^2_{ad} is the standard extension of a bounded operator from a dense subspace; and the Martingale Representation Theorem of Subsection 1.2.5 will say that a certain family of stochastic integrals is not merely orthogonal but *complete*. The geometry is the same throughout.

There is one final structural component that needs to be described. In the study of stochastic processes, it is essential to formalize how information accumulates over time. This is done through the concept of a **filtration**, which models the *evolution of knowledge* or *information available* up to each time t.

Definition 1.1.5 (Filtration)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be a time index set. A **filtration** is a family $\{\mathcal{F}_t\}_{t\in\mathbb{T}}$ of sub-σ-algebras of $\mathcal{F}$ satisfying:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \text{for all } s \leq t. \tag{1.6}$$

That is, the filtration is *increasing* in t. We interpret $\mathcal{F}_t$ as the collection of events whose outcomes are known or observable up to t. A stochastic process $X = \{X_t\}_{t\in\mathbb{T}}$ is said to be **adapted** to the filtration $\{\mathcal{F}_t\}$ if, for each $t \in \mathbb{T}$, the random variable X_t is measurable with respect to $\mathcal{F}_t$. We say that $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ is a **filtered probability space**.

Bare adaptedness turns out to be slightly too weak for comfort. Two further requirements are so universally imposed that they have acquired a name of their own, and we impose them here once and for all rather than invoking them piecemeal later.

Definition 1.1.6 (The usual conditions)**.** A filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ is said to satisfy the **usual conditions** if

i) $\mathcal{F}_0$ contains every $\mathbb{P}$-null set of $\mathcal{F}$ (*completeness*); and

ii) the filtration is *right-continuous*, that is,

$$\mathcal{F}_t = \mathcal{F}_{t^+} := \bigcap_{n\geq 1} \mathcal{F}_{t+1/n}, \qquad t \in [0, T). \tag{1.7}$$

Convention 1.1.7. From this point onwards, every filtered probability space is assumed to satisfy the usual conditions, and this will not be restated.

Neither requirement is cosmetic. Completeness is what allows us to modify a process on a null set — as we shall do whenever we choose a continuous version of an object defined only up to indistinguishability — without leaving the filtration. Right-continuity is what makes first hitting times of open sets stopping times, and it is used explicitly in Proposition 1.1.11 below and again in the Doob-Meyer decomposition of Subsection 1.2.5. Neither is automatic for a natural filtration $\mathcal{F}_t^X$, but both can always be arranged: one replaces $\mathcal{F}_t^X$ by the $\mathbb{P}$-completion of $\mathcal{F}_{t^+}^X$, the so-called *augmented* filtration, and for the filtrations generated by the processes we shall study this augmentation changes nothing of substance.

Remark 1.1.8 (Modification versus indistinguishability). A related piece of hygiene, which we shall need whenever a theorem produces "a continuous version" of a process. Two processes X and Y on the same space are said to be *modifications* of each other if $\mathbb{P}(X_t = Y_t) = 1$ for every fixed t, and *indistinguishable* if $\mathbb{P}(X_t = Y_t \text{ for all } t) = 1$. The second is strictly stronger: the exceptional null set is allowed to depend on t in the first case, and a union of uncountably many null sets need not be null. The two notions do coincide when both processes have right-continuous paths, which is the situation we shall almost always be in — but the coincidence is a small theorem, not a definition, and we shall be careful to say which of the two we mean.

Recall that if χ is a real-valued random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then the σ-algebra generated by χ is denoted $\sigma(\chi)$ and defined as:

$$\sigma(\chi) := \{A \in \mathcal{F} : A = \chi^{-1}(B) \text{ for some } B \in \mathcal{B}(\mathbb{R})\}. \tag{1.8}$$

Example 1.1.9 (Natural Filtration)**.** Given a stochastic process $X = \{X_t\}_{t\in\mathbb{T}}$, the **natural filtration** generated by X is defined by:

$$\mathcal{F}_t^X := \sigma(X_s : s \leq t), \tag{1.9}$$

the smallest σ-algebra with respect to which all variables X_s for $s \leq t$ are measurable. This filtration encodes exactly the information revealed by observing the process up to time t.

Definition 1.1.10 (Stopping time)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying a filtration $\{\mathcal{F}_t\}_{t\in\mathbb{T}}$. A mapping $\tau : \Omega \to \mathbb{T}$ is a *stopping time* if

$$\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t \qquad \text{for each } t \in \mathbb{T}.$$

Stating the definition on a general index set costs nothing and saves a parenthesis. Throughout this dissertation $\mathbb{T} = [0, T]$, so that every stopping time is automatically bounded by T; the single place where the infinite horizon $\mathbb{T} = [0, \infty)$ is needed is Theorem 1.1.28, which says so explicitly, and the reason it is needed there is exactly that boundedness is what would otherwise make its hypotheses vacuous.

Proposition 1.1.11. *Let $(\mathcal{F}_t)_{t\in[0,T]}$ be a right-continuous filtration. A random variable $\tau : \Omega \to [0, T]$ is a stopping time with respect to $(\mathcal{F}_t)_{t\in[0,T]}$ if and only if*

$$\{\, \omega \in \Omega : \tau(\omega) < t \,\} \in \mathcal{F}_t, \quad \textit{for all } t \in [0, T].$$

Proof: Suppose first that τ is a stopping time. Since $\{\tau < t\} = \bigcup_{n\geq 1}\{\tau \leq t - 1/n\}$ and each $\{\tau \leq t - 1/n\}$ lies in $\mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t$, the union lies in $\mathcal{F}_t$. This implication uses nothing beyond monotonicity of the filtration.

Conversely, suppose $\{\tau < t\} \in \mathcal{F}_t$ for every t. Then for each $n \geq 1$ we have $\{\tau < t + 1/n\} \in \mathcal{F}_{t+1/n}$, and since $\{\tau \leq t\} = \bigcap_{n\geq 1}\{\tau < t + 1/n\}$,

$$\{\tau \leq t\} \;\in\; \bigcap_{n\geq 1} \mathcal{F}_{t+1/n} \;=\; \mathcal{F}_{t^+} \;=\; \mathcal{F}_t,$$

the last equality being exactly the right-continuity (1.7). ■

Note where the work was done: the converse consumes right-continuity in full, and fails without it. This is the first of several places where the usual conditions of Definition 1.1.6 do real work rather than merely tidy up.

Example 1.1.12. Let X be a measurable adapted process on $[0, T]$ with $\int_0^T |X_s|^2\, ds < \infty$ almost surely. For $n \in \mathbb{N}$ set

$$A_n(\omega) = \left\{ t \in [0, T] : \int_0^t |X_s(\omega)|^2\, ds > n \right\},$$

and let $\tau_n(\omega) = \inf A_n(\omega)$ if $A_n(\omega) \neq \emptyset$, and $\tau_n(\omega) = T$ otherwise. Since $t \mapsto \int_0^t |X_s|^2 ds$ is continuous and non-decreasing, the threshold n is exceeded strictly before t if and only if it is exceeded at some rational time strictly before t, so that for $t \in (0, T]$

$$\{\tau_n < t\} = \bigcup_{\substack{0<r<t \\ r\in\mathbb{Q}}} \left\{ \int_0^r |X_s|^2\, ds > n \right\} \in \mathcal{F}_t, \tag{1.10}$$

whereas $\{\tau_n < 0\} = \emptyset \in \mathcal{F}_0$. By Proposition 1.1.11, τ_n is a stopping time.

This sequence is not an idle illustration. It is exactly the localising sequence that will allow us, in Subsection 1.2.1, to define the stochastic integral for integrands that are square-integrable pathwise but not in expectation, and it is the prototype of every localisation argument in this chapter.

Example 1.1.13 (First hitting time)**.** Let X be a right-continuous process adapted to $\{\mathcal{F}_t\}$. If $U \subseteq \mathbb{R}$ is an *open* set, then the *first hitting time*

$$\tau_U(\omega) = \inf\{t > 0 : X_t(\omega) \in U\} \tag{1.11}$$

is a stopping time. Indeed, since U is open and X is right-continuous, the path enters U strictly before t if and only if it lies in U at some *rational* time strictly before t: if $X_r \in U$ for some $r < t$, right-continuity places X in the open set U throughout a nonempty interval $[r, r+\epsilon)$, which contains a rational. Hence

$$\{\tau_U < t\} = \bigcup_{\substack{0<r<t \\ r\in\mathbb{Q}}} \{X_r \in U\} \ \in \ \mathcal{F}_t, \tag{1.12}$$

and Proposition 1.1.11 applies.

The hypotheses are doing real work, and it is worth seeing where. Openness of U is what allows the union to be taken over a *countable* set of times: for a *closed* target the same argument fails, since a path may touch a closed set at a single irrational instant and at no rational one, and one needs continuity of the paths rather than mere right-continuity to recover the conclusion. For a general Borel U neither hypothesis suffices, and that τ_U is a stopping time at all is the *début theorem* — a genuinely deep result resting on the measurable projection theorem, and one of the few places in this subject where the measure-theoretic machinery is not merely bookkeeping. We shall need only the open case, which is the elementary one.

1.1.1 Fundamental Distributions

Before presenting some cornerstone distributions, recall that the **characteristic function** of a (real) random variable X is defined by

$$\varphi_X(u) := \mathbb{E}[e^{iuX}], \quad u \in \mathbb{R}. \tag{1.13}$$

It is the Fourier transform of the probability distribution of X, and it uniquely determines the law of X.

Its elementary properties — normalisation, uniform continuity, Hermitian symmetry, behaviour under affine maps and under sums of independent variables, and the correspondence between differentiability at the origin and existence of moments — are collected in Proposition A.2.1 of Appendix A, and are used below without further comment.

Theorem 1.1.14 (Fourier inversion)**.** *Let X be a random variable with distribution function F_X. If φ_X is integrable, then F_X admits a continuous density f_X given by*

$$f_X(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-iux}\varphi_X(u)\,du. \tag{1.14}$$

■

In the general case, without assuming a density, the distribution function F_X can still be recovered via the *Lévy inversion formula*: for any $\alpha < \beta$ that are *continuity points* of F_X,

$$F_X(\beta) - F_X(\alpha) = \lim_{R\to\infty} \frac{1}{2\pi}\int_{-R}^{R} \frac{e^{-i\alpha u} - e^{-i\beta u}}{iu}\,\varphi_X(u)\,du. \tag{1.15}$$

The restriction to continuity points is necessary and not a technicality: at a jump of F_X the right-hand side converges to the average of the left and right limits rather than to F_X itself. Since F_X is monotone it has at most countably many discontinuities, so continuity points are dense and the formula determines F_X everywhere by right-continuity. Note also that the truncation parameter R is unrelated to the time horizon T fixed in the notation table; the symbol has been changed to avoid the collision.

We deduce that if two random variables X and Y have the same characteristic function, i.e. $\varphi_X(u) = \varphi_Y(u)$ for all $u \in \mathbb{R}$, then X and Y have the same distribution.

Theorem 1.1.15 (Lévy's Continuity Theorem)**.** *Let $\{X_n\}_{n\in\mathbb{N}}$ be a sequence of random variables with characteristic functions φ_{X_n}.*

- *If $X_n \xrightarrow{d} X$ for some random variable X, then $\varphi_{X_n}(u) \to \varphi_X(u)$ for every $u \in \mathbb{R}$.*
- *Conversely, suppose $\varphi_{X_n}(u) \to \varphi(u)$ for every $u \in \mathbb{R}$, where $\varphi : \mathbb{R} \to \mathbb{C}$ is* some *function continuous at $u = 0$. Then φ is the characteristic function of a random variable X, and $X_n \xrightarrow{d} X$.* ■

The asymmetry between the two halves is deliberate and is where the theorem earns its keep. In the converse direction one is *not* allowed to assume that the limit φ is a characteristic function — that is the conclusion, not a hypothesis. Continuity at the origin is what rules out mass escaping to infinity, and it is a genuine restriction on φ; note that if one already knew φ to be a characteristic function the requirement would be vacuous, since characteristic functions are automatically uniformly continuous. It is precisely this form of the statement that makes the Central Limit Theorem below a two-line computation: one expands φ_{X_n}, observes that the limit is $e^{-u^2/2}$, and only then concludes that the limit is Gaussian.

We now describe certain distributions of widespread importance and usefulness. The first place must be given to the normal distribution.

Example 1.1.16 (Normal Distribution)**.** Let $\mu \in \mathbb{R}$ and $\sigma > 0$, and define the real-valued random variable $X \colon \Omega \to \mathbb{R}$ with distribution $X \sim \mathcal{N}(\mu, \sigma^2)$, that is, X has the probability density function

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \tag{1.16}$$

Then X is said to be normally distributed with mean μ and variance σ^2; thus, a normal distribution is characterized by its first two moments.

The normal distribution is fundamental in probability theory and statistics. It appears as the limiting distribution in the Central Limit Theorem and models many naturally occurring phenomena. Its characteristic function adopts the form

$$\varphi_X(u) = \exp\left(iu\mu - \tfrac{1}{2}\sigma^2 u^2\right). \tag{1.17}$$

Its resemblance to the *heat kernel* (see [22]) is not a coincidence and will be explored in the next section.

Let $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ be two *independent* normal random variables. The following are easily verified through Theorem 1.1.15:

- For $a, b \in \mathbb{R}$, $aX + bY \sim \mathcal{N}(a\mu_X + b\mu_Y,\, a^2\sigma_X^2 + b^2\sigma_Y^2)$; if instead X and Y are jointly normal with correlation ρ, the variance becomes $a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\,\sigma_X\sigma_Y$. We write $Z \sim \mathcal{N}(\mu, 0)$ for the constant variable μ, so that the identity persists when constants are combined with normal variables.

- Its characteristic function is entire analytic, and $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$ converges in distribution to $\mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$ if and only if $\mu_n \to \mu$ and $\sigma_n^2 \to \sigma^2$.

- The normal distribution maximises **entropy** among distributions with fixed first two moments, making it the *least informative* among them.

- Independence implies zero correlation, as always; for normal variables the converse holds too, *provided the pair is jointly normal.*

The independence and joint-normality hypotheses above deserve a word, because the two statements they guard are among the most frequently misquoted in the subject, and a single counterexample disposes of both at once.

Remark 1.1.17. Let $X \sim \mathcal{N}(0, 1)$ and let ε be independent of X with $\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}$. Put $Y := \varepsilon X$. Conditioning on ε shows at once that $Y \sim \mathcal{N}(0, 1)$, and

$$\mathbb{E}[XY] = \mathbb{E}[\varepsilon]\,\mathbb{E}[X^2] = 0,$$

so X and Y are uncorrelated standard normal variables. They are very far from independent: $|X| = |Y|$ almost surely. Moreover $X + Y = (1 + \varepsilon)X$ vanishes with probability $\frac{1}{2}$, hence is not normal at all.

The moral is that "normal" is a property of a *law on* $\mathbb{R}^d$, not a property of a list of marginals. The pair (X, Y) above has normal marginals but its joint law is not multivariate normal in the sense of Example 1.1.19, and every statement in the list above fails for it. Whenever we invoke Gaussianity for several variables at once — as we shall constantly do for the increments of Brownian motion, and for the driving noises of the multi-factor models of Section 1.5 — it is joint normality that is meant.

However, the most compelling reason why the normal distribution is so important rests on the following fundamental result, in a relatively weak form:

Theorem 1.1.18 (Central Limit Theorem)**.** *Let* $\{X_n\}_{n=1}^{\infty}$ *be a sequence of independent, identically distributed (hereafter,* i.i.d.*) random variables with finite mean* μ *and variance* $\sigma^2 > 0$*. Then*

$$\frac{\sum_{k=1}^{n} X_k - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad \blacksquare \tag{1.18}$$

The multivariate case is the one we shall actually need, since Brownian increments and multi-factor noises are always Gaussian several variables at a time. It is obtained from the one-dimensional statement by the following device, which also explains why Example 1.1.19 below defines multivariate normality through one-dimensional projections rather than through a density.

Example 1.1.19 (Multivariate Normal Distribution)**.** An $\mathbb{R}^d$-valued random vector X follows a **multivariate normal distribution** with mean $\mu \in \mathbb{R}^d$ and covariance $\Sigma \in \mathbb{R}^{d\times d}$, written $X \sim \mathcal{N}_d(\mu, \Sigma)$, if $a^\top X \sim \mathcal{N}(a^\top\mu,\, a^\top\Sigma a)$ for every $a \in \mathbb{R}^d$.

Here Σ is necessarily symmetric positive semi-definite. If it is invertible the law is absolutely continuous on $\mathbb{R}^d$ with density $f_X(x) = (2\pi)^{-d/2}\det(\Sigma)^{-1/2}\exp\Big(-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\Big)$; if it is singular, X is supported on a lower-dimensional affine subspace. In either case the characteristic function is

$$\varphi_X(u) = \mathbb{E}[e^{iu^\top X}] = \exp\left(iu^\top\mu - \tfrac{1}{2}u^\top\Sigma u\right), \quad u \in \mathbb{R}^d. \tag{1.19}$$

Theorem 1.1.20 (Cramér–Wold device)**.** *Let X_n, X be $\mathbb{R}^d$-valued random vectors. Then $X_n \xrightarrow{d} X$ if and only if $a^\top X_n \xrightarrow{d} a^\top X$ for every $a \in \mathbb{R}^d$.*

The proof is immediate from Theorem 1.1.15: the characteristic function of $a^\top X_n$ evaluated at 1 is $\varphi_{X_n}(a)$, so convergence of all one-dimensional projections is precisely pointwise convergence of φ_{X_n}, and the limit is continuous at the origin because it is a characteristic function. Combining this with the one-dimensional Central Limit Theorem gives the form we shall use.

Corollary 1.1.21 (Multivariate Central Limit Theorem)**.** *Let $\{X_n\}_{n\geq 1}$ be i.i.d. $\mathbb{R}^d$-valued random vectors with mean μ and finite covariance Σ. Then*

$$\frac{1}{\sqrt{n}}\sum_{k=1}^{n}\Big(X_k-\mu\Big) \xrightarrow{d} \mathcal{N}_d(0,\Sigma). \tag{1.20}$$

Proof: Fix $a \in \mathbb{R}^d$. The variables $a^\top X_k$ are i.i.d. real, with mean $a^\top\mu$ and variance $a^\top\Sigma a$, so the one-dimensional Central Limit Theorem gives $n^{-1/2}\sum_k a^\top(X_k-\mu) \xrightarrow{d} \mathcal{N}(0, a^\top\Sigma a)$. By Example 1.1.19 that limit is the law of $a^\top Z$ for $Z \sim \mathcal{N}_d(0,\Sigma)$. Since a was arbitrary, Theorem 1.1.20 yields (1.20). The degenerate case $a^\top\Sigma a = 0$ is included, the limit then being the constant 0. ■

For a multivariate normal vector — and, as Remark 1.1.17 showed, only then — uncorrelatedness does imply independence. Indeed, if Σ is block-diagonal then (1.19) factorises as a product of the characteristic functions of the corresponding sub-vectors, which is exactly independence.

Example 1.1.22 (Log-Normal Distribution)**.** A random variable X is said to follow a **log-normal distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted

$$X \sim \mathrm{Log}\mathcal{N}(\mu,\sigma^2), \tag{1.21}$$

if the logarithm of X is normally distributed, i.e.,

$$\log X \sim \mathcal{N}(\mu,\sigma^2). \tag{1.22}$$

The probability density function of $X \sim \mathrm{Log}\mathcal{N}(\mu,\sigma^2)$, after a straightforward change-of-variable calculation, is

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}}\exp\left(-\frac{(\log x-\mu)^2}{2\sigma^2}\right), \quad x > 0. \tag{1.23}$$

The formula $\mathbb{E}[X^k] = \exp(k\mu + \frac{1}{2}k^2\sigma^2)$ holds for every $k \in \mathbb{R}$, not merely for $k \in \mathbb{N}$: it is obtained by writing $X^k = e^{kZ}$ with $Z \sim \mathcal{N}(\mu,\sigma^2)$ and evaluating the Gaussian moment generating function, which is finite for every real argument. Moments of *all* real orders, negative ones included, therefore exist and are finite — a point worth making explicitly, since it is what makes the failure below so surprising. In particular $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ and $\mathbb{V}[X] = (e^{\sigma^2}-1)e^{2\mu+\sigma^2}$. And yet the log-normal distribution has two unpleasant analytical features that will matter later.

The first is that its characteristic function admits no closed-form expression. This is a genuine inconvenience — it is precisely why the Black-Scholes model, despite its explicit pricing formula, is not amenable to the Fourier techniques that make the Heston model of Subsection 1.5.2 tractable — but it is not a pathology. Note that φ_X is defined and bounded by 1 for every *real* u, as it is for any random variable; what fails is the *moment generating function* $\mathbb{E}[e^{uX}]$, which is infinite for every $u > 0$. The two objects are easily confused and we shall keep them apart.

The second is more surprising. Despite having moments of all orders, **the log-normal distribution is not determined by them**: there is an explicit one-parameter family of mutually distinct densities on $(0, \infty)$ sharing every moment with it, exhibited by Heyde [36] and reproduced in Appendix A.3. It is worth being precise about what does *not* fail here, since the point is easy to garble. Lévy's inversion formula (1.15) holds without exception, and the characteristic function does determine the log-normal law uniquely; what fails is the passage from the *moment sequence* to the characteristic function.

The logical status of that failure deserves care, because it is routinely overstated. Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$ is *sufficient* for a law to be determined by its moments, not necessary. The log-normal moments grow as $e^{k^2\sigma^2/2}$, which is fast enough to violate the condition — but a violated sufficient condition establishes only that the criterion is silent, never that the conclusion is false. What establishes indeterminacy here is nothing to do with growth rates: it is Heyde's explicit exhibition of a one-parameter family of distinct densities sharing the moment sequence. The growth rate tells us why no elementary criterion could have saved us; the construction tells us that nothing could.

The lesson is one we shall meet again in Chapter 2: knowing every moment of an object is strictly weaker than knowing the object, and the signature transform will be interesting precisely because, for paths, the analogous sufficient condition is *satisfied* rather than violated, so that the corresponding statement can be made to go the right way.

Two elementary processes will be needed later, and we record them together.

Example 1.1.23 (Random walk and Poisson process)**.** i) Let $\{\xi_n\}$ be i.i.d. with $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = \frac{1}{2}$. The **simple symmetric random walk** is $X_0 := 0$, $X_n := \sum_{k=1}^{n} \xi_k$: a discrete-time, integer-valued process with stationary and independent increments. Since $\varphi_\xi(u) = \cos(u)$, independence gives $\varphi_{X_n}(u) = \cos^n(u)$.

ii) Fix $\lambda > 0$. A process $N = \{N_t\}_{t \geq 0}$ with values in $\mathbb{N}_0$ is a **Poisson process with rate** λ if $N_0 = 0$ almost surely, N has independent increments, and

$$\mathbb{P}(N_{t+s} - N_s = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \qquad k \in \mathbb{N}_0,\ s, t \geq 0. \tag{1.24}$$

Its paths are piecewise constant and non-decreasing, with unit jumps at random times; $\mathbb{E}[N_t] = \mathbb{V}[N_t] = \lambda t$ and $\varphi_{N_t}(u) = e^{\lambda t(e^{iu}-1)}$.

We shall return to i) in Theorem 1.1.39, where the jump size and the waiting time between jumps are adjusted simultaneously and a new process emerges in the limit. Part ii) will serve throughout as the standard example of a process which is *continuous in time*, since $\mathbb{T} = [0, \infty)$, while having discontinuous sample paths — a distinction worth keeping straight. It is the elementary building block for markets in which prices genuinely jump, such as electricity, and it will reappear as the canonical process that is *not* a diffusion.

Remark 1.1.24. *Despite the importance of discrete processes,* ***for most of this dissertation we will assume that the process is continuous****, unless otherwise stated.*

1.1.2 Markov Processes

Markov processes are characterized by a specific form of *memorylessness*. They are particularly suited to describe the evolution of systems where the future depends only on the present state and not on the past history. Some of these processes arise as *diffusions* and in that case, a lot can be said about the distributions.

Definition 1.1.25 (Markov Process)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X = \{X_t\}_{t\in\mathbb{T}}$ be a stochastic process taking values in a measurable space $(E, \mathcal{E})$. We say that X is a **Markov process** if for every $s < t$ and every measurable set $A \in \mathcal{E}$,

$$\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s) \quad \text{almost surely,} \tag{1.25}$$

where $\mathcal{F}_s = \sigma(X_r : r \leq s)$ is the σ-algebra generated by the history of the process up to time s.

The **transition probability** of a Markov process is $\mathbb{P}(X_t \in A \mid X_s = x)$; in order to integrate against the measure it defines we write $P_{s,x}(t, dy) = \mathbb{P}(X_t \in dy \mid X_s = x)$.

There is a rather sharp distinction between finite-state Markov processes and those with infinitely many states. In the finite case, the theory is built around transition matrices: the probability of moving from one state to another in a single step is recorded in a matrix P_t, so that the evolution is the product $\prod_{i=1}^n P_i$, and long-run behaviour can be analysed by classical linear-algebraic tools, especially in the stationary case, that is, when the transition probability is independent of t.

In the infinite-state setting the dynamics are instead described by operators acting on spaces of functions or measures, so that the study of these processes relies on functional analysis and differential equations as much as on probability. That apparatus — the transition semigroup $(P_t)_{t\geq 0}$, its infinitesimal generator, and the Kolmogorov equations they satisfy — is developed in Subsection 1.2.6, once diffusions are available to make it concrete; we do not anticipate it here.

1.1.3 Martingales, semimartingales and local martingales

Martingales are processes for which the expected value of a future instant is precisely the state of the process up to present time. Martingales are absolutely key in the study of stochastic processes. Some generalizations are needed as well. We keep the notation used in the previous section.

Definition 1.1.26 (Martingale)**.** Let X_t be a stochastic process, adapted to a given filtration $\mathcal{F}_t$. We say it is a **martingale** provided:

- $\mathbb{E}[|X_t|] < \infty$ for all t.
- $X_s = \mathbb{E}[X_t|\mathcal{F}_s]$ for $s < t$.

We say that X_t is a **supermartingale** if the second condition is replaced by $X_s \geq \mathbb{E}[X_t|\mathcal{F}_s]$ and a **submartingale** if $X_s \leq \mathbb{E}[X_t|\mathcal{F}_s]$.

The martingale property is stated at deterministic times, but a great deal of what we shall do — localisation in Section 1.2, and the early exercise of American options in Subsection 1.5.3 — involves evaluating processes at *random* times. For this to make sense we must first say what information is available at a stopping time.

Definition 1.1.27 (The σ-algebra $\mathcal{F}_\tau$)**.** Let τ be a stopping time with respect to $(\mathcal{F}_t)_{t\in[0,T]}$. The σ-algebra of events observable up to τ is

$$\mathcal{F}_\tau := \Big\{A \in \mathcal{F} \ : \ A \cap \{\tau \le t\} \in \mathcal{F}_t \ \text{ for every } t \in [0,T]\Big\}. \tag{1.26}$$

The definition reads oddly at first but says something simple: an event belongs to $\mathcal{F}_\tau$ if, whenever τ has occurred by time t, an observer at time t can already tell whether the event happened. One checks readily that $\mathcal{F}_\tau$ is indeed a σ-algebra, that τ is $\mathcal{F}_\tau$-measurable, that $\mathcal{F}_\tau = \mathcal{F}_t$ when $\tau \equiv t$ is constant, and that $\sigma \le \tau$ implies $\mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$. If X is right-continuous and adapted, then X_τ is $\mathcal{F}_\tau$-measurable on $\{\tau < \infty\}$.

Theorem 1.1.28 (Doob's optional stopping theorem)**.** For this theorem only, we work on the infinite horizon: *let $X = \{X_t\}_{t\ge 0}$ be a right-continuous martingale with respect to a filtration $(\mathcal{F}_t)_{t\ge 0}$, and let $\sigma \le \tau$ be stopping times with values in $[0,\infty]$. If any one of the following holds —*

i) τ is bounded, i.e. there is $c < \infty$ with $\tau \le c$ almost surely;

ii) X is uniformly integrable (Definition A.6.1); or

iii) there exists $Y \in L^1(\Omega)$ with $|X_t| \le Y$ for all t,

— then X_σ and X_τ are integrable and

$$\mathbb{E}\Big[X_\tau \mid \mathcal{F}_\sigma\Big] = X_\sigma \qquad \textit{almost surely.} \tag{1.27}$$

In particular $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$. The corresponding statements with $=$ replaced by $\le$ or $\ge$ hold for sub- and supermartingales respectively. ■

Two remarks on the hypotheses before we use the theorem. They are not independent: iii) implies ii), since a family dominated by a single integrable variable is uniformly integrable, so iii) is recorded only because it is the form in which the hypothesis is usually easiest to check. More importantly, under the finite-horizon convention of Definition 1.1.10 — which is in force everywhere else in this dissertation, and under which every stopping time satisfies $\tau \le T$ — hypothesis i) holds automatically, and the theorem applies unconditionally. The hypotheses do work only on an infinite horizon, and that is precisely why the theorem has been stated there: the phenomenon they exclude is invisible on $[0,T]$.

Equation (1.27) is the assertion that a martingale remains a martingale when observed along a random clock, and it is one of those results whose hypotheses are more instructive than its conclusion. That some hypothesis is indispensable is shown by the following example, which requires the infinite horizon and is the reason for the convention adopted in the theorem. Let W be a Brownian motion and $\tau = \inf\{t : W_t = 1\}$, which is finite almost surely. Then $W_\tau = 1$ identically, so $\mathbb{E}[W_\tau] = 1 \ne 0 = \mathbb{E}[W_0]$, and (1.27) fails. Nothing is wrong with the theorem; the stopping time is simply unbounded and $W_{t\wedge\tau}$ is not uniformly integrable. The financial reading of this example is exact and worth stating: a strategy that waits however long is necessary to be up by one unit does indeed win with certainty, and it is not an arbitrage only because executing it may require unbounded losses along the way. That is precisely what the admissibility constraint of Definition 1.3.2 will forbid.

Because $\{X_{t\wedge\tau}\}$ is a martingale whenever X is — which is case i) applied at each t — optional stopping is also the mechanism behind localisation. A local martingale is by Definition 1.1.31

a process which becomes a martingale after stopping; the content of that definition is that all the martingale machinery, including (1.27) and Proposition 1.1.29, remains available on each piece.

The following result will be useful when constructing stochastic processes from Itô integrals. Notice that if X_t is a martingale, then by Jensen's inequality (Proposition 1.1.4, item 8) X_t^2 is a submartingale.

Proposition 1.1.29 (Doob's submartingale inequality)**.** *Let $Z = \{Z_t\}_{t\in[0,T]}$ be a right-continuous submartingale. Then, for any $\varepsilon > 0$,*

$$\mathbb{P}\Big(\sup_{0\le t\le T} Z_t \ge \varepsilon\Big) \le \frac{\mathbb{E}[\max\{Z_T, 0\}]}{\varepsilon}.$$

In particular, for a right-continuous martingale X, $\mathbb{P}(\sup_{0\le t\le T} |X_t| \ge \varepsilon) \le \mathbb{E}[|X_T|]/\varepsilon$.

Proof: The argument is given in Appendix B.1; see also [40, Thm. 4.5.1]. ■

The bound is in terms of the *terminal* value X_T, not of a running X_t; this is the whole content of the inequality, since it converts information at a single time into control of the entire path. For the L^2 theory of the Itô integral we shall need the following stronger, L^p form, which is proved by integrating the tail estimate above.

Proposition 1.1.30 (Doob's L^p maximal inequality)**.** *Let X be a right-continuous martingale and let $p > 1$. Then*

$$\mathbb{E}\Big[\sup_{0\le t\le T} |X_t|^p\Big] \le \left(\frac{p}{p-1}\right)^p \mathbb{E}\Big[|X_T|^p\Big].$$

In particular, for $p = 2$, $\mathbb{E}\Big[\sup_{t\le T} |X_t|^2\Big] \le 4\,\mathbb{E}[|X_T|^2]$. ■

The case $p = 2$ is the one we shall use repeatedly. It says that for a square-integrable martingale, controlling the terminal second moment controls the second moment of the whole trajectory — which is exactly what is needed to upgrade convergence of the Itô integral in $L^2(\Omega)$, at each fixed time, to uniform convergence along a subsequence, and hence to continuity of sample paths (Theorem 1.2.7).

The notion of a martingale is often too restrictive, particularly when considering processes such as stochastic exponentials or certain diffusion models. A relaxation is therefore introduced.

Definition 1.1.31 (Local Martingale)**.** A process $X = \{X_t\}_{t\in[0,T]}$ is a **local martingale** if there exists an increasing sequence of stopping times $(\tau_n)_{n\ge1}$ with $\tau_n \uparrow T$ almost surely, such that each stopped process $X_t^{\tau_n} := X_{t\wedge\tau_n}$ is a martingale. On an infinite horizon the condition reads $\tau_n \uparrow \infty$; we shall use the finite-horizon form, which is the one that appears in the proof of Theorem 1.2.18.

The idea is that X behaves like a martingale, but only up to random horizons. An analogy could be drawn from ODEs: there are solutions that explode to infinity in finite time, and those that can be extended to all times. Here, the integrability condition can be somewhat replaced if we work with sequences of stopping times. This mathematical object deserves its own study but lies beyond the scope of this dissertation.

Finally, there is another related concept, broad enough to serve as a class of *good integrators*. In particular, it extends the role played by Brownian motion as a canonical integrator, while retaining enough structure to allow a general theory of stochastic calculus.

Definition 1.1.32 (Semimartingale)**.** A process X is called a **semimartingale** if it can be decomposed as

$$X_t = M_t + A_t,$$

where M_t is a local martingale and A_t is adapted, *continue à droite, avec limite à gauche* (càdlàg), of locally bounded variation and $A_0 = 0$.

This last definition is the correct level of generality for stochastic integration, and it is worth understanding why the decomposition takes the shape it does. The two summands are exactly the two mechanisms by which an integrator can be usable: A has bounded variation, so $\int g\, dA$ exists pathwise as a Riemann–Stieltjes integral in the classical sense; and M is a local martingale, so $\int g\, dM$ exists in the Itô sense. The last clause deserves a caveat, since the construction of Section 1.2 is carried out for $M = W$ specifically and uses $\mathbb{E}[(W_{t_i} - W_{t_{i-1}})^2] = t_i - t_{i-1}$ at its one essential step, the isometry of Theorem 1.2.5. For a general continuous local martingale the same construction goes through with $\int_0^T \mathbb{E}[f^2]\, dt$ replaced throughout by $\mathbb{E} \int_0^T f^2\, d\langle M \rangle_t$, the compensator supplied by Theorem 1.2.28 and identified with the pathwise quadratic variation by Proposition 1.1.47; localisation then removes the integrability hypotheses. We do not carry this out, and shall integrate only against Brownian motion.

Granting it, a semimartingale is precisely a process that can be integrated against by splitting the work between deterministic analysis and probability, and the Bichteler–Dellacherie theorem makes the converse precise: semimartingales are, in a suitable topological sense, the *only* processes admitting a reasonable theory of stochastic integration; see [52].

We shall not develop that general theory — every integrator in Part I is continuous, and for continuous processes the decomposition is unique — but the definition is recorded because of what happens when it fails. The uniqueness just invoked is worth a line, since Chapter 2 will lean on it: if $M + A = \widetilde{M} + \widetilde{A}$ are two such decompositions, then $M - \widetilde{M} = \widetilde{A} - A$ is simultaneously a continuous local martingale and a continuous process of finite variation, hence has vanishing quadratic variation; a continuous local martingale with vanishing quadratic variation and null initial value is identically zero, by Proposition 1.1.47 and the fact that $M^2 - \langle M \rangle$ is a local martingale. (That last fact is Theorem 1.2.28 localised: the decomposition is stated there for square-integrable martingales, and one recovers the local statement by stopping along a localising sequence and using uniqueness of the decomposition on each piece.) Not every process of interest is a semimartingale: fractional Brownian motion with Hurst parameter $H \neq \frac{1}{2}$ is not one, a result of Rogers [54], and consequently neither the Itô integral nor any of the machinery built on it is available for it. Since such processes are exactly what the empirical evidence on volatility will point towards, the reader should register Definition 1.1.32 less as a technical generalisation than as a boundary: it marks the outer limit of the theory developed in this chapter, and Part II begins where it ends.

1.1.4 Brownian Motion

Brownian motion is a fundamental stochastic process whose importance has grown throughout many areas of mathematics. It is complex enough to generate a new class of *stochastic differential equations* and to address numerous applied problems involving randomness, yet simple enough to allow explicit calculations. Its properties also enable efficient simulation, making it a cornerstone of Monte Carlo methods across disciplines.

Definition 1.1.33 (Brownian Motion)**.** A stochastic process $W : \mathbb{R}^+ \times \Omega \to \mathbb{R}$ is called a **Brownian motion** (or **Wiener process**) if the following conditions hold:

i)
$$\mathbb{P}(W_0 = 0) = 1. \tag{1.28}$$

ii)
$$\mathbb{P}\Big(\{\omega : t \mapsto W_t(\omega) \text{ is continuous}\}\Big) = 1. \tag{1.29}$$

iii) For $0 \le s < t$,
$$W_t - W_s \sim \mathcal{N}(0, t-s). \tag{1.30}$$

iv) The process has independent increments: for $0 \le t_1 < t_2 < \cdots < t_n$, the random variables
$$\{W_{t_1},\ W_{t_2} - W_{t_1},\ W_{t_3} - W_{t_2},\ \ldots,\ W_{t_n} - W_{t_{n-1}}\} \tag{1.31}$$
are independent.

Some properties of Brownian motion follow immediately from the definition and are presented below. The last one illustrates the property known as *stochastic self-similarity.*

In order to work with the broadest possible class of stochastic processes, we introduce the following general result.

Proposition 1.1.34. *Let $S(t)$, $t \ge 0$, be a process adapted to the filtration $(\mathcal{F}_t)_{t\ge 0}$ such that for any $0 \le s < t$, the increment $S(t) - S(s)$ is independent of $\mathcal{F}_s$. Then $S(t)$ has independent increments.*

Proof. Let $0 \le t_1 < t_2 < \cdots < t_n$ and write $\Delta_k = S(t_k) - S(t_{k-1})$. We show by induction on n that the vector $(\Delta_1, \ldots, \Delta_n)$ has independent components. For $n = 1$ there is nothing to prove. Assume the claim for $n-1$. Each of $\Delta_1, \ldots, \Delta_{n-1}$ is $\mathcal{F}_{t_{n-1}}$-measurable, since S is adapted and $t_k \le t_{n-1}$ for $k \le n-1$; hence so is any Borel function of them. By hypothesis Δ_n is independent of $\mathcal{F}_{t_{n-1}}$, and therefore independent of $(\Delta_1, \ldots, \Delta_{n-1})$. For bounded Borel $g_1, \ldots, g_n$ this gives
$$\mathbb{E}\Big[\prod_{k=1}^{n} g_k(\Delta_k)\Big] = \mathbb{E}\Big[g_n(\Delta_n)\Big]\ \mathbb{E}\Big[\prod_{k=1}^{n-1} g_k(\Delta_k)\Big] = \prod_{k=1}^{n} \mathbb{E}\Big[g_k(\Delta_k)\Big],$$
the last equality by the induction hypothesis. Since this holds for all such g_k, the increments are independent. ■

Proposition 1.1.34 suggests the notion we shall actually work with. Definition 1.1.33 describes a process in isolation; every statement in the remainder of this chapter, however, involves a filtration as well — the martingale property, the construction of the Itô integral, Lévy's characterisation and Girsanov's theorem are all assertions about a process *relative to a flow of information.* We therefore fix the compound notion once.

Definition 1.1.35 (Brownian motion with respect to a filtration)**.** Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ be a filtered probability space. A process W is a $(\mathcal{F}_t)_{t\in[0,T]}$-**Brownian motion** if

i) W is adapted to $(\mathcal{F}_t)_{t\in[0,T]}$, with $W_0 = 0$ almost surely and almost all paths continuous; and

ii) for all $s \le t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$ and has law $\mathcal{N}(0, t-s)$.

By Proposition 1.1.34, condition ii) already forces the independence of increments demanded in Definition 1.1.33 iv), so a $(\mathcal{F}_t)_{t\in[0,T]}$-Brownian motion is in particular a Brownian motion; the content of the definition is the *relative* independence, which iv) alone does not give. Conversely, any Brownian motion is a $(\mathcal{F}_t)_{t\in[0,T]}$-Brownian motion for its own natural filtration $(\mathcal{F}_t^W)_{t\in[0,T]}$, augmented as in Convention 1.1.7, and this is the filtration we use unless another is named.[1] The distinction is not idle: Girsanov's theorem (Theorem 1.2.35) changes the measure while keeping the filtration fixed, and its conclusion is exactly that a certain process is a $(\mathcal{F}_t)_{t\in[0,T]}$-Brownian motion under the new measure.

The next definition and proposition highlight some of the structural richness of such processes. In particular, the last property quantifies the *roughness* of sample paths, a key feature of Brownian motion.

Definition 1.1.36 (Quadratic Variation)**.** Let $f : [0,T] \to \mathbb{R}$, and for a partition $\mathcal{P} = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ of $[0,t]$ write

$$\|\mathcal{P}\| = \max_{1\le k\le n} (t_k - t_{k-1}). \tag{1.32}$$

If the limit

$$[f]_t := \lim_{\|\mathcal{P}\|\to 0} \sum_{k=1}^{n} \Big(f(t_k) - f(t_{k-1})\Big)^2 \tag{1.33}$$

exists, it is called the **quadratic variation** of f on $[0,t]$.

Two cautions are in order. First, the limit in (1.33) need not exist for an arbitrary f, which is why the definition is phrased conditionally. Second — and this is the point that will matter — for a *stochastic* process the sums in (1.33) are random variables, and the limit must be specified as a limit of random variables. We shall always understand it in $L^2(\Omega)$, which for our purposes is both the easiest to obtain and strong enough to imply convergence in probability, and hence almost sure convergence along a subsequence.

It is worth pausing on how strange (1.33) is from the point of view of ordinary calculus. If f is continuously differentiable, then $|f(t_k) - f(t_{k-1})| \le C\,(t_k - t_{k-1})$ and the sum is bounded by $C^2\|\mathcal{P}\|\,t \to 0$: smooth functions have zero quadratic variation. The same computation applies to any function of bounded variation that is continuous. A non-zero quadratic variation is therefore a certificate of irregularity, and the following proposition asserts that Brownian motion carries exactly such a certificate.

Proposition 1.1.37. *Let $(W_t)_{t\ge 0}$ be a Brownian motion. Then:*

a) For $t > 0$, $W_t \sim \mathcal{N}(0,t)$. Moreover, for all $s,t \ge 0$,

$$\mathbb{E}[W_s W_t] = \min\{s,t\}. \tag{1.34}$$

b) For $t_0 \ge 0$, the process

$$Y(t) = W(t+t_0) - W(t_0) \tag{1.35}$$

is also a Brownian motion.

c) For any $\lambda > 0$, the process

$$Z(t) = \frac{1}{\sqrt{\lambda}} W(\lambda t) \tag{1.36}$$

is also a Brownian motion.

[1] Analogous considerations hold for a Poisson process; see [20].

d) Brownian motion is both a Markov process and a martingale.

e) The quadratic variation of W over $[0,t]$ exists and equals

$$[W]_t = t \qquad \textit{the limit being taken in } L^2(\Omega). \tag{1.37}$$

f) The process $W_t^2 - t$ is a martingale.

Properties b) and c) are called translation invariance *and* scaling invariance*, respectively.*

Proof. a) By i) we have $W_t = W_t - W_0$ almost surely, so $W_t \sim \mathcal{N}(0,t)$ by iii). For $s < t$, write $W_t = (W_t - W_s) + W_s$ and use that $W_t - W_s$ is independent of W_s with zero mean:

$$\mathbb{E}[W_s W_t] = \mathbb{E}\Big[W_s(W_t - W_s)\Big] + \mathbb{E}[W_s^2] = 0 + s = \min\{s,t\}.$$

b) $Y_0 = 0$ and the paths of Y are continuous, since those of W are. For $s < t$ we have $Y_t - Y_s = W_{t+t_0} - W_{s+t_0} \sim \mathcal{N}(0, t-s)$, and increments of Y over disjoint intervals are increments of W over disjoint intervals, hence independent.

c) Again $Z_0 = 0$ and the paths are continuous. For $s < t$, $Z_t - Z_s = \lambda^{-1/2}(W_{\lambda t} - W_{\lambda s})$ is normal with mean 0 and variance $\lambda^{-1}(\lambda t - \lambda s) = t - s$; independence of increments is inherited from W because $s \mapsto \lambda s$ preserves disjointness of intervals.

d) Integrability is clear from a). For $s < t$, using independence of $W_t - W_s$ from $\mathcal{F}_s$ and $\mathcal{F}_s$-measurability of W_s,

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = \mathbb{E}[W_t - W_s \mid \mathcal{F}_s] + \mathbb{E}[W_s \mid \mathcal{F}_s] = \mathbb{E}[W_t - W_s] + W_s = W_s,$$

so W is a martingale; the same computation with $\mathbb{P}(W_t \in A \mid \mathcal{F}_s)$ in place of $\mathbb{E}[W_t \mid \mathcal{F}_s]$ shows that the conditional law of W_t given $\mathcal{F}_s$ depends on ω only through W_s, which is the Markov property.

e) Let $\mathcal{P} = \{0 = t_0 < \cdots < t_n = t\}$ and set $X_k = (W_{t_k} - W_{t_{k-1}})^2 - (t_k - t_{k-1})$. The X_k are independent with $\mathbb{E}[X_k] = 0$, and since $\mathbb{E}[Z^4] = 3\sigma^4$ for $Z \sim \mathcal{N}(0,\sigma^2)$,

$$\mathbb{E}[X_k^2] = 3(t_k - t_{k-1})^2 - 2(t_k - t_{k-1})^2 + (t_k - t_{k-1})^2 = 2(t_k - t_{k-1})^2.$$

Hence, by independence and orthogonality,

$$\mathbb{E}\Bigg[\bigg(\sum_k (W_{t_k} - W_{t_{k-1}})^2 - t\bigg)^2\Bigg] = \sum_k \mathbb{E}[X_k^2] = 2\sum_k (t_k - t_{k-1})^2 \le 2\|\mathcal{P}\|\, t \longrightarrow 0$$

as $\|\mathcal{P}\| \to 0$, which is (1.37). See also [40, Thm. 4.1.2].

f) Integrability follows from a). For $s < t$, expanding $W_t^2 = (W_t - W_s)^2 + 2W_s(W_t - W_s) + W_s^2$ and conditioning,

$$\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = (t - s) + 2W_s \cdot 0 + W_s^2,$$

so $\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s$. ■

One might wonder whether such a process actually exists, given the strong set of requirements in Definition 1.1.33. The question is not idle: we have demanded independent Gaussian increments *and* continuity of every sample path, and there is no reason a priori why the two should be compatible. A full construction would take us too far afield, but it would be unsatisfying to leave the matter entirely unaddressed, so we indicate the results which settle it — each of which we shall have occasion to use again for its own sake.

The construction proceeds in two stages, and the division of labour between them is worth naming in advance, because it is exactly the division between the finite-dimensional and the pathwise. The first stage produces a process with the right finite-dimensional distributions and says nothing whatever about its paths; the second repairs the paths without disturbing the distributions.

The first stage is the Daniell–Kolmogorov extension theorem. Conditions iii) and iv) of Definition 1.1.33 prescribe, for every finite set of times $t_1 < \cdots < t_n$, the joint law of $(W_{t_1}, \ldots, W_{t_n})$: it is the centred Gaussian law on $\mathbb{R}^n$ with covariance $\min\{t_i, t_j\}$, which is positive semi-definite and therefore legitimate. These laws are *consistent*, in the sense that the law prescribed for a larger set of times marginalises to the law prescribed for a smaller one — an immediate consequence of the Gaussian algebra. The extension theorem then asserts that any consistent family of finite-dimensional distributions is realised by some process on the product space $\mathbb{R}^{[0,\infty)}$; see [39, Thm. 2.2.2] or [20, Ch. 8].

What this stage cannot deliver is condition ii). The set of continuous paths is not measurable in the product σ-algebra — membership in it depends on uncountably many coordinates at once — so continuity is invisible to the finite-dimensional data and must be obtained separately. That is the second stage, and it does so from a moment condition alone.

Theorem 1.1.38 (Kolmogorov's continuity criterion)**.** *Let $X = \{X_t\}_{t\in[0,T]}$ be a stochastic process for which there exist constants $\alpha, \beta, C > 0$ with*

$$\mathbb{E}\Big[\, |X_t - X_s|^\alpha \,\Big] \le C\, |t-s|^{1+\beta}, \qquad s, t \in [0,T]. \tag{1.38}$$

Then X admits a continuous modification $\tilde{X}$. Moreover, the paths of $\tilde{X}$ are almost surely locally γ-Hölder continuous for every $\gamma \in (0, \beta/\alpha)$. ■

Applied to Brownian motion the criterion is immediate. Since $W_t - W_s \sim \mathcal{N}(0, t-s)$, the Gaussian moments give $\mathbb{E}[|W_t - W_s|^{2m}] = c_m |t-s|^m$ for every $m \in \mathbb{N}$, so (1.38) holds with $\alpha = 2m$ and $\beta = m-1$. A continuous modification therefore exists, which is what was needed; and the Hölder conclusion yields exponents $\gamma < (m-1)/2m$, which as $m \to \infty$ approach $1/2$ from below but never reach it. That the supremum is exactly $1/2$, and that it is not attained, is the subject of the next subsection.

Note the word *modification*: what Theorem 1.1.38 produces is a process agreeing with X at each fixed time almost surely, not one agreeing with X along whole paths (Remark 1.1.8). This is not a defect of the theorem but a necessity, since (1.38) constrains only two-dimensional distributions and cannot possibly determine path behaviour on its own. As is customary, we shall always work with the continuous modification and cease to mention the distinction.

That settles existence. But Brownian motion is not merely *a* process meeting the requirements of Definition 1.1.33; it is the process that an enormous class of unrelated constructions converges to. The statement is the following, and it is the promised return to the simple symmetric random walk of Example 1.1.23, where we said we would come back once the jump size and the waiting time between jumps had been adjusted together.

Theorem 1.1.39 (Donsker's invariance principle)**.** *Let $\{\xi_k\}_{k\ge1}$ be i.i.d. with $\mathbb{E}[\xi_1] = 0$ and $\mathbb{V}[\xi_1] = 1$, and let $X_n = \sum_{k=1}^n \xi_k$ be the associated random walk. Define the rescaled, linearly interpolated process*

$$W_t^{(n)} := \frac{1}{\sqrt{n}}\Big(X_{\lfloor nt\rfloor} + (nt - \lfloor nt\rfloor)\, \xi_{\lfloor nt\rfloor+1} \Big), \qquad t \in [0,T]. \tag{1.39}$$

Then $W^{(n)} \Rightarrow W$ as $n \to \infty$, where the convergence is in distribution on the space $C[0,T]$ of continuous functions with the uniform topology, and W is a Brownian motion. ■

The scaling in (1.39) is the whole content of the theorem: space is compressed by $\sqrt{n}$ while time is compressed by n. That is exactly the relation space $\sim \sqrt{\text{time}}$ which Definition 1.1.33 imposes through $W_t - W_s \sim \mathcal{N}(0, t-s)$, and which reappears as the scaling invariance of Proposition 1.1.37(c). The theorem is the exact path-space analogue of the Central Limit Theorem, and it explains why Brownian motion is *universal* in the same sense that the Gaussian distribution is: the limit does not depend on the law of ξ_1 beyond its first two moments, so any accumulation of small, independent, finite-variance shocks produces the same object. This is the honest reason Brownian motion is ubiquitous in modelling, and it is worth keeping in mind that the same universality argument gives no warrant at all for the shocks being independent — a point the empirical evidence of Part II will press hard.

The reflection principle and the running maximum

One consequence of the Markov property is worth recording in full, both because it is the cleanest illustration of what independence of increments buys and because it is the only tool we shall have for pricing a contract that looks at the whole path rather than at its endpoint.

What is needed is the Markov property not at a fixed time but at a *stopping* time.

Theorem 1.1.40 (Strong Markov property)**.** *Let W be a Brownian motion and τ an almost surely finite stopping time. Then the shifted process $\widetilde{W}_s := W_{\tau+s} - W_\tau$, $s \geq 0$, is a Brownian motion independent of $\mathcal{F}_\tau$ (Definition 1.1.27).* ■

The statement genuinely strengthens Proposition 1.1.37(d), where the Markov property was asserted at deterministic times only, and it is not automatic: it holds here because τ may be approximated from above by stopping times taking countably many values, for each of which the claim reduces to the ordinary Markov property, and the continuity of the paths lets one pass to the limit. See [39, Thm. 2.6.16].

Fix $a > 0$ and let $\tau_a := \inf\{t \geq 0 : W_t > a\}$ be the first hitting time of the level a, a stopping time by Example 1.1.13. The *reflection principle* is the observation that, on the event $\{\tau_a \leq t\}$, the post-τ_a increments form a Brownian motion independent of $\mathcal{F}_{\tau_a}$, and that a Brownian motion has the same law as its negative. Reflecting the path in the level a after time τ_a therefore produces a path of the same law, and it exchanges the events $\{W_t > a\}$ and $\{W_t < a\}$ while fixing $\{\tau_a \leq t\}$.

Theorem 1.1.41 (Reflection principle)**.** *Let W be a Brownian motion and write $\overline{W}_t := \max_{0 \leq s \leq t} W_s$ for its* ***running maximum****. Then for every $a \geq 0$,*

$$\mathbb{P}\Big(\overline{W}_t \geq a\Big) \;=\; \mathbb{P}\Big(\tau_a \leq t\Big) \;=\; 2\,\mathbb{P}\Big(W_t > a\Big) \;=\; \mathbb{P}\Big(|W_t| > a\Big), \tag{1.40}$$

so that $\overline{W}_t$ and $|W_t|$ have the same law. Moreover the pair $(W_t, \overline{W}_t)$ has joint density

$$f_{(W_t,\overline{W}_t)}(x,m) \;=\; \frac{2(2m-x)}{\sqrt{2\pi t^3}}\,\exp\left(-\frac{(2m-x)^2}{2t}\right), \qquad m \geq \max\{x, 0\}. \tag{1.41}$$

■

Sketch. The first chain of equalities is the argument just given: $\{\overline{W}_t \geq a\} = \{\tau_a \leq t\}$ because the path is continuous, and conditionally on $\tau_a \leq t$ the reflected path is as likely to finish above a as below, so $\mathbb{P}(\tau_a \leq t) = 2\mathbb{P}(W_t > a)$ — the boundary case $W_t = a$ having probability zero. For (1.41) one applies the same reflection to obtain $\mathbb{P}(W_t \leq x,\ \overline{W}_t \geq m) = \mathbb{P}(W_t \geq 2m - x)$ for $m \geq \max\{x, 0\}$, and differentiates in both arguments; see [39, Sec. 2.6] or [56].

Two things follow that we shall want later. First, $\mathbb{E}[\overline{W}_t] = \mathbb{E}|W_t| = \sqrt{2t/\pi}$, which already says that the expected high-water mark of a driftless price over $[0, t]$ grows like $\sqrt{t}$, on the same scale as the fluctuation itself. Second — and this is the reason the subsection is here — (1.41) is a *joint* law across the whole interval, not a terminal marginal, and it is therefore exactly the object that Breeden–Litzenberger (Proposition 1.4.6) cannot supply. Barrier options, whose payoff depends on whether $\overline{W}_T$ crossed a level, are priced from (1.41) in the Black–Scholes model and are precisely the contracts for which the local volatility model of Subsection 1.5.1, calibrated to terminal marginals alone, gets the answer wrong. We return to this in Chapter 3.

1.1.5 Path regularity

We now examine how irregular the trajectories of Brownian motion actually are. This subsection is short, and every statement in it is classical, but it is the most consequential material in the chapter: the construction of the Itô integral in Section 1.2 is a *response* to what follows, and the theory developed in Part II is a response to what happens when these estimates are pushed slightly further.

We begin with the observation that quadratic variation and total variation are in direct competition.

Definition 1.1.42 (Total variation)**.** The **total variation** of $f : [0, T] \to \mathbb{R}$ on $[0, t]$ is

$$V_1(f)_t := \sup_{\mathcal{P}} \sum_{k=1}^{n} \Big| f(t_k) - f(t_{k-1}) \Big|, \tag{1.42}$$

the supremum being over all partitions $\mathcal{P} = \{0 = t_0 < \cdots < t_n = t\}$. If $V_1(f)_t < \infty$ we say f has **bounded variation** on $[0, t]$.

Bounded variation is exactly the condition under which the Riemann–Stieltjes integral $\int_0^t g\, df$ is defined for every continuous g; equivalently, it is the condition under which f induces a signed measure df. It is therefore the natural first thing to ask of a prospective integrator.

Proposition 1.1.43. *Let W be a Brownian motion. Then, almost surely, $V_1(W)_t = \infty$ for every $t > 0$.*

Proof: Fix $t > 0$ and a sequence of partitions $\mathcal{P}_n$ with $\|\mathcal{P}_n\| \to 0$. On any partition,

$$\sum_k \Big(W_{t_k} - W_{t_{k-1}}\Big)^2 \ \leq\ \Big(\max_k \Big|W_{t_k} - W_{t_{k-1}}\Big|\Big) \cdot \sum_k \Big|W_{t_k} - W_{t_{k-1}}\Big| \ \leq\ \Big(\max_k \Big|W_{t_k} - W_{t_{k-1}}\Big|\Big)\, V_1(W)_t. \tag{1.43}$$

Suppose, for contradiction, that $V_1(W)_t < \infty$ on an event of positive probability. Sample paths of W are continuous, hence uniformly continuous on the compact $[0, t]$, so the maximal increment on the right of (1.43) tends to 0 as $\|\mathcal{P}_n\| \to 0$. The whole right-hand side therefore tends to 0, forcing $[W]_t = 0$ on that event. This contradicts (1.37), which gives $[W]_t = t > 0$ in $L^2(\Omega)$ and hence almost surely along a subsequence. ∎

This is the single most important fact in the chapter, and it is worth stating its consequence without any hedging. **The integral $\int_0^t f\,dW$ cannot be defined pathwise as a Riemann–Stieltjes integral**, for any non-trivial class of integrands, because W does not induce a measure on $[0,t]$ along any path. Every difficulty in Section 1.2 — the restriction to adapted integrands, the insistence on evaluating at the left endpoint, the approximation in $L^2(\Omega)$ rather than pathwise, the appearance of a second-order term in the chain rule — traces back to (1.43). Itô's construction is not one possible approach among several; it is a way of exploiting probabilistic structure to recover an object which deterministic analysis cannot reach.

The same competition can be expressed more finely by asking, not whether the sum of increments converges, but for which exponent p the sum of p-th powers does.

Definition 1.1.44 (p-variation)**.** Let $p \geq 1$. The p**-variation** of $f : [0,T] \to \mathbb{R}$ on $[0,t]$ is

$$V_p(f)_t := \left(\sup_{\mathcal{P}} \sum_{k=1}^{n} \left| f(t_k) - f(t_{k-1}) \right|^p \right)^{1/p}, \tag{1.44}$$

and f is said to have *finite p-variation* on $[0,t]$ if $V_p(f)_t < \infty$. We write $f \in \mathcal{C}^{p\text{-var}}([0,t])$.

For $p = 1$ this is Definition 1.1.42, and for $p = 2$ it is a supremum version of the quadratic variation of Definition 1.1.36. The scale is monotone: if f is continuous and $V_p(f)_t < \infty$, then $V_q(f)_t < \infty$ for every $q > p$, since $|f(t_k) - f(t_{k-1})|^q \leq (\max_k |f(t_k) - f(t_{k-1})|)^{q-p} |f(t_k) - f(t_{k-1})|^p$. Each function thus has a critical exponent below which its p-variation is infinite and above which it is finite, and this exponent is a precise measure of roughness.

Theorem 1.1.45 (Regularity of Brownian paths)**.** *Let W be a Brownian motion. Then, almost surely:*

i) the sample path $t \mapsto W_t$ is locally γ-Hölder continuous for every $\gamma < \frac{1}{2}$, and is locally γ-Hölder for no $\gamma \geq \frac{1}{2}$;

ii) $V_p(W)_t < \infty$ for every $p > 2$ and every t, whereas $V_p(W)_t = \infty$ for every $p \leq 2$;

iii) the sample path is nowhere differentiable: for almost every ω, there is no t at which $t \mapsto W_t(\omega)$ possesses a finite derivative. ■

Part i) was half-proved above via Kolmogorov's criterion; the negative half, like iii), is due to Paley, Wiener and Zygmund, and proofs of all three may be found in [23] or [39]. Part iii) is worth a moment's reflection. A continuous function which is differentiable nowhere is already a celebrated pathology in real analysis, exhibited with some effort by Weierstrass; the theorem says that if one picks a Brownian path at random one obtains such a function with probability one. The pathological case is the generic case.

It is part ii) which will matter most. Let us record it in the form we shall use.

Remark 1.1.46 (The critical exponent)*.* Brownian motion has finite p-variation exactly for $p > 2$. The number 2 is a threshold in a strong sense, and the reason is the following. Young's integral $\int g\,df$ can be constructed *pathwise*, by purely deterministic means and with no probability involved, whenever f has finite p-variation, g has finite q-variation, and

$$\tfrac{1}{p} + \tfrac{1}{q} > 1. \tag{1.45}$$

The threshold 2 is not part of (1.45) but a consequence of it in the case that matters to us. The integrands we shall meet are of the form $g = F(f)$ for smooth F, which inherits the regularity of f itself, so $q = p$ and (1.45) collapses to $2/p > 1$, that is $p < 2$. Brownian motion misses this regime — but only just, and by an amount that no refinement of the estimate can recover, since $V_2(W)_t = \infty$.

The Itô integral of Section 1.2 is precisely the tool that bridges this gap, and it does so by spending probabilistic structure: the integrand must be adapted, the approximation is in $L^2(\Omega)$ rather than pathwise, and the resulting object is defined only up to null sets. That is a real price. It buys us the entire theory of Part I, but it also means the theory is tied to a specific process and a specific filtration, and it is exactly this dependence which becomes an obstacle when the driving signal is rougher than Brownian — when even $p = 2$ is out of reach, and neither Young nor Itô applies. The theory that operates in that regime is the subject of Part II.

[FIGURE HERE] Simulated Brownian path with, alongside it, the partition sums for total variation (diverging) and quadratic variation (converging to t) as the mesh is refined. This is the visual statement of Proposition 1.1.43.

Finally, we record the relation between the pathwise quadratic variation of Definition 1.1.36 and the predictable version produced by the Doob–Meyer decomposition of Subsection 1.2.5, since both notations occur in the literature and we shall use them interchangeably.

Proposition 1.1.47. *Let M be a continuous square-integrable martingale. Then the pathwise quadratic variation $[M]_t$ of Definition 1.1.36 exists as a limit in $L^2(\Omega)$ and coincides, up to indistinguishability, with the compensator $\langle M\rangle_t$ of Theorem 1.2.28. Furthermore, the polarised quantity*

$$\langle M, N\rangle_t := \tfrac{1}{2}\Big(\langle M+N\rangle_t - \langle M\rangle_t - \langle N\rangle_t\Big) = \lim_{\|\mathcal{P}\|\to 0} \sum_k \Big(M_{t_k} - M_{t_{k-1}}\Big)\Big(N_{t_k} - N_{t_{k-1}}\Big) \tag{1.46}$$

*defines the **quadratic covariation** of two continuous square-integrable martingales, is bilinear and symmetric, and satisfies the Kunita–Watanabe inequality*

$$\int_0^t \Big|d\langle M, N\rangle_s\Big| \;\le\; \sqrt{\langle M\rangle_t}\,\sqrt{\langle N\rangle_t} \qquad \textit{almost surely.} \blacksquare \tag{1.47}$$

The distinction between $[M]$ and $\langle M\rangle$ is genuine for processes with jumps, where the two differ by the accumulated squared jumps; for the continuous processes of this dissertation it is empty, and we shall write $\langle M\rangle$ throughout. Quadratic covariation is what gives meaning to the off-diagonal entries $\rho_{ij}\,dt$ of the multiplication table in Subsection 1.2.4, and it is the object that the correlation parameter of a stochastic volatility model actually controls.

1.2 Stochastic Calculus

Stochastic Calculus is a wide area in Mathematics that allows, for example, the incorporation of randomness and uncertainty into differential equations. One of our main aims is to give meaning to expressions of the kind

$$dX_t = a(t, X_t)\,dt + b(t, X_t)\,dW_t, \qquad X_0 \in \mathbb{R}. \tag{1.48}$$

Here dW_t will be specified in the next subsection. Notice that this generalizes the deterministic case: if $b \equiv 0$, then X_t reduces to the solution of the ODE $dY_t = a(t, Y_t)\,dt$.

We shall first construct a rigorous integral interpretation of such expressions, providing concrete meaning to the stochastic term. Afterwards, we will introduce the stochastic chain rule (Itô's Lemma), which differs from the classical one in a somewhat surprising way, and then proceed to study the existence and uniqueness of solutions, and some of their properties.

It must be highlighted that the following dissertation can be expanded up to the most general setting, however we will stick to the most classical presentation of the subject, leaving the most general case a worthy space at the end, but without letting the trees hide the forest.

1.2.1 The Itô Integral

The stochastic differential equation (1.48) can be reformulated in integral form as

$$X_t - X_0 = \int_0^t a(s, X_s)\, ds \;+\; \int_0^t b(s, X_s)\, dW_s. \tag{1.49}$$

For each $\omega \in \Omega$, the first integral on the right-hand side is nothing new. Regarding the second one, we may ask whether there is a sensible way of defining

$$M_t = \int_0^t f(s,\omega)\, dW_s, \quad 0 \le t \le T,$$

so that $\{M_t\}_{t\in[0,T]}$ is a martingale. Another important consideration is that the integrand must be *adapted* to the filtration associated with W_t. These requirements naturally lead to the following definitions.

Definition 1.2.1 (Adapted square-integrable processes)**.** Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ be a filtered probability space.

i) The space of stochastic processes on $[0,T]$, adapted to a filtration $(\mathcal{F}_t)_{t\ge0}$ and such that

$$\int_0^T \mathbb{E}[|f(t)|^2]\, dt < \infty,$$

will be denoted by $L^2_{ad}([0,T]\times\Omega)$. This is a Hilbert space with the inner product

$$\langle f, g\rangle = \int_0^T \mathbb{E}[f(t)g(t)]\, dt.$$

ii) We say $f \in L^2_{ad}([0,T]\times\Omega)$ is a **step stochastic process** if

$$f(t,\omega) = \sum_{i=1}^n \varphi_\omega(t_{i-1})\, \mathbb{1}_{[t_{i-1},t_i)}(t),$$

where $\varphi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$-measurable and has finite variance ($\mathbb{E}[\varphi(t_{i-1})^2] < \infty$).

The construction of the stochastic integral begins by defining it on step processes and extending by density. For such f, take $t_0 = 0$, $t_n = T$, and define[2]

$$I(f) = \int_0^T f(t)\, dW_t := \sum_{i=1}^n \varphi(t_{i-1}) \Big(W(t_i) - W(t_{i-1})\Big). \tag{1.50}$$

This mapping $f \mapsto I(f)$ is linear. The following lemmas are key.

[2]The choice of the left endpoint t_{i-1} ensures that the stochastic integral preserves the martingale property. Variants with different evaluation points lead to different calculi, e.g. Stratonovich calculus, where the classical chain rule holds (see [40]).

Lemma 1.2.2. *If f is a step stochastic process, then*

$$\mathbb{E}[I(f)] = 0, \quad \mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2]\,dt.$$

Proof: Write $\Delta_i W = W(t_i) - W(t_{i-1})$, so that $I(f) = \sum_{i=1}^n \varphi(t_{i-1})\Delta_i W$. Since $\varphi(t_{i-1})$ is $\mathcal{F}_{t_{i-1}}$-measurable and $\Delta_i W$ is independent of $\mathcal{F}_{t_{i-1}}$ with mean zero, conditioning gives

$$\mathbb{E}\Big[\varphi(t_{i-1})\Delta_i W\Big] = \mathbb{E}\Big[\varphi(t_{i-1})\,\mathbb{E}\Big[\Delta_i W \mid \mathcal{F}_{t_{i-1}}\Big]\Big] = 0,$$

and summing yields $\mathbb{E}[I(f)] = 0$. For the second moment, expand the square. If $i < j$ then $\varphi(t_{i-1})\varphi(t_{j-1})\Delta_i W$ is $\mathcal{F}_{t_{j-1}}$-measurable while $\Delta_j W$ is independent of it with mean zero, so the cross terms vanish by the same argument. The diagonal terms give, using independence of $\Delta_i W$ from $\mathcal{F}_{t_{i-1}}$ once more,

$$\mathbb{E}\Big[\varphi(t_{i-1})^2(\Delta_i W)^2\Big] = \mathbb{E}\Big[\varphi(t_{i-1})^2\Big](t_i - t_{i-1}) = \int_{t_{i-1}}^{t_i} \mathbb{E}[f(t)^2]\,dt,$$

since $f \equiv \varphi(t_{i-1})$ on $[t_{i-1}, t_i)$. Summing over i gives $\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2]\,dt$. ■

Note that both computations rest on exactly one structural fact: the integrand is evaluated at the *left* endpoint, which is what makes $\varphi(t_{i-1})$ independent of the increment it multiplies. Evaluating anywhere else destroys both identities.

Lemma 1.2.3 (Density of step processes). *For every $f \in L^2_{ad}([0,T]\times\Omega)$, there exists a sequence $\{f_n\}_{n=1}^\infty$ of step stochastic processes such that*

$$\lim_{n\to\infty} \int_0^T \mathbb{E}[(f(t) - f_n(t))^2]\,dt = 0.$$

That is, step processes are dense in $L^2_{ad}([0,T]\times\Omega)$.

Proof: The argument is given in Appendix B.2; see also [40, Lemma 4.3.3]. ■

We are now ready to extend the definition.

Definition 1.2.4 (Stochastic Integral). Let $f \in L^2_{ad}([0,T]\times\Omega)$. We define its *stochastic integral* over $[0,T]$ as the limit in $L^2(\Omega)$ of $I(f_n)$ for any approximating sequence of step processes $\{f_n\}$ with $f_n \to f$ in $L^2_{ad}([0,T]\times\Omega)$.

The following fundamental result follows.

Theorem 1.2.5 (Itô's isometry). *Let $f \in L^2_{ad}([0,T]\times\Omega)$.*

i) The stochastic integral $I(f)$ is a centered random variable and

$$\mathbb{E}[I(f)^2] = \int_0^T \mathbb{E}[f(t)^2]\,dt.$$

ii) For another $g \in L^2_{ad}([0,T]\times\Omega)$,

$$\mathbb{E}\left[\left(\int_0^T f(t)\,dW_t\right)\left(\int_0^T g(t)\,dW_t\right)\right] = \int_0^T \mathbb{E}[f(t)g(t)]\,dt.$$

Proof: i) Let $\{f_n\}$ be step processes with $f_n \to f$ in $L^2_{ad}([0,T]\times\Omega)$, as furnished by Lemma 1.2.3. By Lemma 1.2.2 applied to $f_n - f_m$,

$$\mathbb{E}\Big[(I(f_n)-I(f_m))^2\Big] = \int_0^T \mathbb{E}\Big[(f_n(t)-f_m(t))^2\Big]\,dt \xrightarrow[n,m\to\infty]{} 0,$$

so $\{I(f_n)\}$ is Cauchy in $L^2(\Omega)$ and its limit $I(f)$ exists and is independent of the approximating sequence. Both $f \mapsto \mathbb{E}[I(f)]$ and $f \mapsto \mathbb{E}[I(f)^2]$ are continuous for the L^2 norms involved, so the identities $\mathbb{E}[I(f_n)] = 0$ and $\mathbb{E}[I(f_n)^2] = \int_0^T \mathbb{E}[f_n^2]\,dt$ pass to the limit.

ii) Apply i) to $f+g$ and expand: $\mathbb{E}[I(f+g)^2] = \mathbb{E}[I(f)^2] + 2\mathbb{E}[I(f)I(g)] + \mathbb{E}[I(g)^2]$ by linearity of I, while the right-hand side expands identically. Cancelling the square terms gives the stated polarisation. ■

In the language of Proposition 1.1.4, item 9, this is precisely the statement that I is an isometry from $L^2_{ad}([0,T]\times\Omega)$ into $L^2(\Omega)$, extended from a dense subspace by continuity — the standard extension of a bounded operator, and the reason the construction works at all.

It is worth noting that if f is deterministic (i.e. does not depend on ω), then

$$I(f) \sim \mathcal{N}\left(0, \int_0^T f(t)^2\,dt\right).$$

For $0 \le t \le T$, we have $\int_0^t \mathbb{E}[f(s)^2]\,ds \le \int_0^T \mathbb{E}[f(s)^2]\,ds$, so $f \in L^2_{ad}([0,t]\times\Omega)$ and the stochastic integral is well defined on $[0,t]$. Consider then the process $X_t = \int_0^t f(s)dW(s)$. We know this process has finite first two moments, and so we can take the conditional expectation of X_t with respect to the given filtration.

Theorem 1.2.6 (Martingale property)**.** *Take $f \in L^2_{ad}([0,T]\times\Omega)$. Then the process given by $X_t = I(f\mathbb{1}_{[0,t]})$ is a martingale with respect to the filtration $\mathcal{F}_t$.*

Proof: Integrability follows from Itô's isometry, since $\mathbb{E}[X_t^2] = \int_0^t \mathbb{E}[f(s)^2]\,ds < \infty$ and $L^2(\Omega) \subset L^1(\Omega)$ on a probability space. Adaptedness is inherited from the approximating step processes.

For the martingale identity, take first f a step process and $s < t$ lying in the partition defining it (refining the partition if necessary, which changes nothing). Then $X_t - X_s = \sum_i \varphi(t_{i-1})\Big(W(t_i) - W(t_{i-1})\Big)$, the sum running over the subintervals of $[s,t]$. For each such i we have $t_{i-1} \ge s$, so conditioning first on $\mathcal{F}_{t_{i-1}}$ and then on $\mathcal{F}_s$ gives

$$\mathbb{E}\Big[\varphi(t_{i-1})(W(t_i)-W(t_{i-1})) \mid \mathcal{F}_s\Big] = \mathbb{E}\Big[\varphi(t_{i-1})\,\mathbb{E}\Big[W(t_i)-W(t_{i-1}) \mid \mathcal{F}_{t_{i-1}}\Big]\,\Big|\,\mathcal{F}_s\Big] = 0,$$

by the tower property (Proposition A.1.1, item 5) and independence of the increment from $\mathcal{F}_{t_{i-1}}$. Hence $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$.

For general $f \in L^2_{ad}$, take step processes $f_n \to f$ and write X^n for the corresponding integrals. Itô's isometry gives $X_t^n \to X_t$ in $L^2(\Omega)$, hence in $L^1(\Omega)$, and conditional expectation is a contraction on L^1 (Proposition A.1.1, item 7), so $\mathbb{E}[X_t^n \mid \mathcal{F}_s] \to \mathbb{E}[X_t \mid \mathcal{F}_s]$ in $L^1(\Omega)$. The left-hand side equals $X_s^n \to X_s$, and the martingale identity passes to the limit. ■

The Markov property also holds, but we will return to that when we see these processes as diffusions in Subsection 1.2.6.

We shall now address the continuity of the process X_t. We must keep in mind that this integral is not defined for each ω in a Riemann sense. The following result is therefore not as elementary as in real analysis.

Theorem 1.2.7 (Continuity property)**.** *Suppose* $f \in L^2_{ad}([0,T]\times\Omega)$. *Then the process* $X_t = \int_0^t f(s)dW(s)$ *admits a continuous modification, with which we shall always work.*

Proof: The argument is given in Appendix B.3; see also [40, Thm. 4.6.2]. ■

This is the payoff of Proposition 1.1.30 announced earlier: the maximal inequality is exactly what upgrades convergence at each fixed time to uniform convergence along a subsequence, and hence L^2 convergence to a pathwise statement.

Finally, a result that actually links Riemann sums with stochastic integrals will be helpful in the next section.

Theorem 1.2.8. *Suppose* $f \in L^2_{ad}([0,T]\times\Omega)$ *and assume that* $\mathbb{E}[f(t)f(s)]$ *is a continuous function of* t *and* s. *Then, for partitions of* $[0,T]$, *given by* $\mathcal{P} = \{0 = t_0 < t_1 < \cdots < t_n = T\}$,

$$\int_0^T f(t)\,dW(t) = \lim_{\|\mathcal{P}\|\to 0} \sum_{i=1}^{n} f(t_{i-1})\Big(W(t_i) - W(t_{i-1})\Big) \tag{1.51}$$

in $L^2(\Omega)$, *where the limit is taken as in Definition 1.1.36.*

Proof: The argument is given in Appendix B.4; see also [40, Thm. 4.7.1]. ■

The continuity hypothesis on $\mathbb{E}[f(t)f(s)]$ is what makes this work and cannot simply be dropped: without it the left-endpoint approximation need not converge, and the Riemann-sum interpretation of the Itô integral fails.

Theorem 1.2.8 also isolates the one genuinely arbitrary choice in the whole construction, and it is worth naming the alternative, because Chapter 2 will show that the two are on an equal footing.

Definition 1.2.9 (Stratonovich integral)**.** The **Stratonovich integral** is defined, under the hypotheses of Theorem 1.2.8, as the limit of the same Riemann sums with the integrand evaluated at the *midpoint* of each subinterval rather than at the left endpoint:

$$\int_0^T f(t)\circ dW(t) := \lim_{\|\mathcal{P}\|\to 0} \sum_{i=1}^{n} f\left(\tfrac{t_{i-1}+t_i}{2}\right)\Big(W(t_i) - W(t_{i-1})\Big), \tag{1.52}$$

the limit again being taken in $L^2(\Omega)$.

For integrands of the form $f = F(W)$ with F smooth the two integrals both exist and differ by a computable amount,

$$\int_0^T F(W_t)\circ dW_t = \int_0^T F(W_t)\,dW_t + \tfrac{1}{2}\int_0^T F'(W_t)\,dt, \tag{1.53}$$

the correction being one half of the quadratic covariation $\langle F(W), W\rangle_T$. The Itô integral is the one we shall use throughout Part I, for the reason given in the footnote above: it alone preserves the martingale property, because it alone evaluates the integrand before the increment it multiplies. The Stratonovich integral buys something else in exchange, namely the classical chain rule with no second-order term, which makes it the natural choice in differential geometry and in the Wong–Zakai approximation results of Chapter 2. Neither is more correct than the other; they answer different questions, and (1.53) is the dictionary. Chapter 2 will recover this dictionary as a special case of a general formula, and will identify the choice between the two as a choice of *lift* rather than a choice of convention.

More generally, we can consider a larger family of integrands. Let $\mathcal{L}(\Omega, L^2[0,T])$ be the family of processes f such that

- $f(t)$ is $\mathcal{F}_t$−adapted.
- $\int_0^T |f(t)|^2 \, dt < \infty$ almost surely.

Notice how we dropped the condition of having finite expectation, so that we have indeed a larger class of processes. Consider for example the process $f(t) = e^{W(t)^2}$. It follows that

$$\mathbb{E}[f(t)^2] = \begin{cases} \dfrac{1}{\sqrt{1-4t}} & \text{if } 0 \le t < 1/4 \\ \infty & \text{if } t \ge 1/4. \end{cases} \tag{1.54}$$

Hence $\int_0^1 e^{W(t)^2} \, dW_t$ does not make sense with the previous construction. It can be shown, however, that these processes are still pretty worth the attention. Beyond the scope of our purposes, it can be shown that:

Theorem 1.2.10 (Local Martingale Property)**.** *Let $f \in \mathcal{L}(\Omega, L^2[0,T])$ and $t \in [0,T]$. Then*

$$X_t = \int_0^t f(s) \, dW_s$$

is a local martingale. Furthermore, it has a continuous realization. ■

1.2.2 Itô's Chain Rule

One of the most shocking facts about this new integration is a sophisticated version of the Chain Rule, that generalizes the familiar expression $d(f(g))(x) = f'(g(x))dg(x)$. The key underlying fact for this new integration rule is the non-zero quadratic variation of Brownian motion. This fantastic new calculus allows us to find some new relations and solve new equations that naturally generalize the already rich world of differential equations.

Definition 1.2.11 (Itô process)**.** Given an initial random variable X_0, an Itô process is a stochastic process on $[0,T]$ of the form

$$X_t = X_0 + \int_0^t g(s) \, ds + \int_0^t f(s) \, dW_s \iff dX_t = g(s) \, ds + f(s) \, dW_s, \quad X(0) = X_0$$

where we emphasize the equivalent formulations, since the latter is just a way of writing the former. We impose X_0 being $\mathcal{F}_0$-measurable, $g \in \mathcal{L}(\Omega, L^1[0,T])$ and $f \in \mathcal{L}(\Omega, L^2[0,T])$. Note how the first integral is just a Lebesgue integral for each ω, that is, for each path of W. The second is a proper Itô integral.

We directly state the more general form of Itô's new calculus rule, in the convenient *differential formulation.* A complete proof is beyond the scope of this Thesis. However, such a renowned Theorem has been addressed by several authors, for example in [40].

Theorem 1.2.12 (Itô's Rule). *Let $dX_t = \mu(t)\,dt + \sigma(t)\,dW_t$ be an Itô process, and let $V(t,x)$ be a continuous function on $[0,T]\times\mathbb{R}$ with continuous partial derivatives $\frac{\partial V}{\partial t}$, $\frac{\partial V}{\partial x}$ and $\frac{\partial^2 V}{\partial x^2}$. Then $V(t,X_t)$ is also an Itô process, and*

$$dV(t,X_t) = \left(\frac{\partial V}{\partial t}(t,X_t) + \mu(t)\frac{\partial V}{\partial x}(t,X_t) + \frac{1}{2}\sigma^2(t)\frac{\partial^2 V}{\partial x^2}(t,X_t)\right)dt + \sigma(t)\frac{\partial V}{\partial x}(t,X_t)\,dW_t. \quad (1.55)$$

■

A convenient way to derive Equation (1.55) is through a symbolic derivation using the Taylor expansion to first order in dt and second order in dX_t, together with the *Itô table*:

$\times$	dW_t	dt
dW_t	dt	0
dt	0	0

Applying the Taylor expansion to $V(t,X_t)$ gives

$$dV(t,X_t) = \frac{\partial V}{\partial t}(t,X_t)\,dt + \frac{\partial V}{\partial x}(t,X_t)\,dX_t + \frac{1}{2}\frac{\partial^2 V}{\partial x^2}(t,X_t)\,(dX_t)^2.$$

Using the Itô table, we have $(dX_t)^2 = \sigma(t)^2\,dt$, so using $dX_t = \mu(t)\,dt + \sigma(t)\,dW_t$ yields

$$dV(t,X_t) = \frac{\partial V}{\partial x}(t,X_t)\sigma(t)\,dW_t + \left[\frac{\partial V}{\partial t}(t,X_t) + \mu(t)\frac{\partial V}{\partial x}(t,X_t) + \frac{1}{2}\sigma(t)^2\frac{\partial^2 V}{\partial x^2}(t,X_t)\right]dt,$$

which is precisely Equation (1.55).

We stress that the computation just performed is a *mnemonic*, not a proof: the symbols dW_t and $(dW_t)^2$ have been manipulated as though they were infinitesimals, which they are not. What makes the manipulation legitimate is the content of (1.37) — the second-order term survives precisely because W has non-vanishing quadratic variation — and a genuine proof proceeds by Taylor-expanding along a partition and controlling the remainder in $L^2(\Omega)$. We refer to [40] for the details. The value of the Itô table is that it makes the bookkeeping automatic once one accepts that $\langle W\rangle_t = t$, and we shall use it in that spirit throughout.

Just like ordinary Chain Rule, this has opened plenty of possibilities in order to calculate some stochastic integrals in terms of others.

Example 1.2.13. We can use this result to easily obtain a simpler form for the process $W_t\,dW_t$. We can simply consider $g(x) = x^2$ and define $H_t = g(W_t)$ to obtain

$$dH_t = 2W_t\,dW_t + \frac{2}{2}\,(dW_t)^2 = dt + 2W_t\,dW_t \quad (1.56)$$

so that $\int_0^t W_s\,dW_s = \frac{1}{2}\,(W_t^2 - t)$, which is a much simpler expression, and clearly a martingale.

Example 1.2.14. In Finance, the Black-Scholes model relies on the *log-normal* equation, given by $dS_t = rS_t\,dt + \sigma S_t\,dW_t$, where $r,\sigma\in\mathbb{R}$ and $\sigma > 0$. It can be solved by taking $H_t = \ln(S_t)$, so that

$$dH_t = \frac{1}{S_t}dS_t - \frac{1}{2S_t^2}(dS_t)^2 = r\,dt + \sigma\,dW_t - \frac{1}{2}\sigma^2\,dt \quad (1.57)$$

obtaining $H_t = H_0 + (r-\sigma^2/2)\,t + \sigma W_t \implies S_t = S_0 e^{(r-\sigma^2/2)\,t+\sigma W_t}$.

Example 1.2.15 (Lamperti's Transform)**.** Consider a general process $dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)\, dW_t$ and assume that σ is *strictly positive* on the range of X — without which the integrand below is not defined — and differentiable once with respect to time and once with respect to x, which is what is needed for $\Gamma \in \mathcal{C}^{1,2}$ and hence for Itô's rule to apply. Define the function

$$\Gamma(t, x) = \int_{X_0}^{x} \frac{1}{\sigma(t, s)}\, ds \tag{1.58}$$

and consider the process $H_t = \Gamma(t, X_t)$. After a few calculations, we arrive to

$$\begin{aligned} dH_t &= \left(\frac{\partial \Gamma}{\partial t} + \mu(t, X_t) \frac{\partial \Gamma}{\partial x} + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 \Gamma}{\partial x^2} \right) dt + \underbrace{\sigma(t, X_t) \frac{\partial \Gamma}{\partial x}}_{=1}\, dW_t \\ &= \left(\frac{\partial \Gamma}{\partial t} + \frac{\mu(t, X_t)}{\sigma(t, X_t)} - \frac{1}{2} \frac{\partial \sigma}{\partial x}(t, X_t) \right) dt + dW_t \end{aligned} \tag{1.59}$$

so that all the complexity of the original process is contained in the drift. This transformation is convenient because it creates a process of constant volatility, which enables much faster and accurate numerical methods, such as derivative pricing on recombining trees; we return to this in Chapter 3.

1.2.3 Stochastic Differential Equations

The main aim of this section is to provide necessary and sufficient conditions for the existence of Itô processes. We will follow a direct approach and state the existence and uniqueness theorem. In addition, notice how we will be working with the differential form of the equations, however realising the integral sense they have been provided with. We need a precise definition for *stochastic differential equation* or *SDE* and thus some preliminary results.

Given an $\mathcal{F}_0$-measurable random variable X_0 of finite variance, consider the following formal equation on $[0, T]$:

$$dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)\, dW_t \qquad X(0) = X_0. \tag{1.60}$$

Definition 1.2.16 (Solution of an SDE)**.** A stochastic process X_t, $0 \leq t \leq T$, is called a solution of (1.60) if it satisfies:

- $\sigma(t, X_t) \in \mathcal{L}(\Omega, L^2[0, T])$ so its Itô integral makes sense for each $t \in [0, T]$.
- Almost all sample paths of $\mu(t, X_t)$ belong to $L^1[0, T]$.
- Equation 1.60 holds almost surely.

We need to establish some growth conditions on μ and σ for the equation to make sense on, at least, an open interval, just like in ordinary differential equations. Let us define the following conditions:

Definition 1.2.17. Let $g(t, x)$ a measurable function on $[0, T] \times \mathbb{R}$.

- It is said to satisfy a Lipschitz condition in x if there exists $K > 0$ such that $|g(t, x) - g(t, y)| \leq K|x - y|$ for all $0 \leq t \leq T$, $x, y \in \mathbb{R}$.

- It is said to satisfy a linear growth condition in x if there exists $K > 0$ such that $|g(t,x)| \leq K(1+|x|)$ for all $0 \leq t \leq T$, $x \in \mathbb{R}$.

Now, notice that for $x \geq 0$, it holds that $1+x^2 \leq (1+x)^2 \leq 2(1+x^2)$ so that the linear growth condition can be restated in terms of a new constant M such that $|g(t,x)| \leq M\sqrt{1+x^2}$.

The proof of existence and uniqueness rests on a Picard iteration, exactly as for ordinary differential equations, and the comparison step is furnished by Grönwall's inequality (Lemma A.4.1 in Appendix A); since the argument is deterministic and entirely standard we do not reproduce it here.

Theorem 1.2.18. *Let μ and σ be measurable functions on $[0,T]\times\mathbb{R}$ satisfying both the Lipschitz and the linear growth conditions in x of Definition 1.2.17, and let the initial condition X_0 be $\mathcal{F}_0$-measurable with $\mathbb{E}[X_0^2] < \infty$. Then the stochastic differential equation 1.60 admits a solution, which is unique in the following sense: any two solutions are indistinguishable. Moreover the solution has continuous sample paths.*

Proof: We shall only prove the uniqueness part, the existence being a Picard iteration as indicated above. Let X and Y be two continuous solutions and put $Z_t = X_t - Y_t$, so that

$$Z_t = \int_0^t \Big[\sigma(s,X_s) - \sigma(s,Y_s)\Big] dW_s + \int_0^t \Big[\mu(s,X_s) - \mu(s,Y_s)\Big] ds.$$

We cannot yet take expectations in this identity. By Definition 1.2.16 the integrand of the stochastic term lies only in the pathwise class $\mathcal{L}(\Omega, L^2[0,T])$, on which the Itô integral is constructed by localisation and is a local martingale rather than a martingale; Itô's isometry, a statement about $L^2_{ad}([0,T]\times\Omega)$, is therefore not available. Nor do we know that $\mathbb{E}[Z_t^2]$ is finite, and an inequality between infinite quantities would tell us nothing. Both obstructions are removed by the same device. Define, for $n \in \mathbb{N}$,

$$\tau_n := \inf\Big\{t \in [0,T] : |X_t| \geq n \quad\text{or}\quad |Y_t| \geq n\Big\} \wedge T, \tag{1.61}$$

which is a stopping time by Proposition 1.1.11, since X and Y are continuous and adapted. As their paths are continuous, they are bounded on $[0,T]$ for almost every ω, whence $\tau_n \uparrow T$ almost surely.

Fix n and stop the identity above at τ_n. On $\{s \leq \tau_n\}$ the Lipschitz condition gives $|\sigma(s,X_s) - \sigma(s,Y_s)| \leq K|Z_s| \leq 2Kn$, so the stopped integrand $\mathbb{1}_{[0,\tau_n]}(s)\Big[\sigma(s,X_s) - \sigma(s,Y_s)\Big]$ is bounded, adapted, and hence genuinely an element of $L^2_{ad}([0,T]\times\Omega)$. Itô's isometry now applies to it, and together with the Lipschitz condition yields

$$\mathbb{E}\left[\left(\int_0^{t\wedge\tau_n} \Big[\sigma(s,X_s) - \sigma(s,Y_s)\Big] dW_s\right)^2\right] = \int_0^t \mathbb{E}\Big[(\sigma(s,X_s) - \sigma(s,Y_s))^2\,\mathbb{1}_{\{s\leq\tau_n\}}\Big] ds \leq K^2 \int_0^t \mathbb{E}\Big[Z_s^2\,\mathbb{1}_{\{s\leq\tau_n\}}\Big] ds.$$

For the Lebesgue term, the Cauchy–Schwarz inequality followed by the Lipschitz condition gives, pathwise,

$$\left(\int_0^{t\wedge\tau_n} \Big[\mu(s,X_s) - \mu(s,Y_s)\Big] ds\right)^2 \leq T \int_0^t \Big(\mu(s,X_s) - \mu(s,Y_s)\Big)^2 \mathbb{1}_{\{s\leq\tau_n\}}\, ds \leq TK^2 \int_0^t Z_s^2\,\mathbb{1}_{\{s\leq\tau_n\}}\, ds.$$

Write $u_n(t) := \mathbb{E}\Big[Z^2_{t\wedge\tau_n}\Big]$. Since $|Z_{t\wedge\tau_n}| \leq 2n$ we have $u_n(t) \leq 4n^2 < \infty$ for every t, which is the finiteness the argument was missing. Moreover $Z_s^2\,\mathbb{1}_{\{s\leq\tau_n\}} \leq Z^2_{s\wedge\tau_n}$ pointwise, so combining the two estimates through $(a+b)^2 \leq 2(a^2+b^2)$ and setting $\lambda := 2K^2(1+T)$ gives

$$u_n(t) \;\leq\; \lambda \int_0^t u_n(s)\, ds, \qquad 0 \leq t \leq T.$$

This is a Grönwall inequality with $g \equiv 0$ applied to a finite, non-negative function, so Lemma A.4.1 of Appendix A forces $u_n \equiv 0$. Hence $Z_{t\wedge\tau_n} = 0$ almost surely for each t and each n; letting $n \to \infty$ and using $\tau_n \uparrow T$ together with the continuity of Z gives $Z_t = 0$ almost surely for each fixed $t \in [0,T]$.

It remains to upgrade this to indistinguishability, and the step is not vacuous: the null set so far depends on t (Remark 1.1.8). Let $\{r_n\}$ enumerate the rationals of $[0,T]$ and let Ω_0 be the intersection of the full-measure events $\{Z_{r_n} = 0\}$ with the full-measure event on which Z has continuous paths. Then $\mathbb{P}(\Omega_0) = 1$, and for $\omega \in \Omega_0$ the function $t \mapsto Z_t(\omega)$ is continuous and vanishes on a dense set, hence vanishes identically. Thus X and Y are indistinguishable. ■

The localisation in (1.61) is worth pausing over, since it is the argument Example 1.1.12 was introduced to prepare. The pattern is invariably the same: one cannot integrate or take expectations in the class where the object naturally lives, so one stops the process at the first time it leaves a bounded region, works in the well-behaved class there, and lets the region exhaust the space. Without it the proof above would appeal to an isometry that does not hold for these integrands and to a Grönwall inequality that could be vacuous.

Three comments on the hypotheses, none of which is pedantry.

First, both conditions are needed, and they do different jobs: the Lipschitz condition delivers uniqueness, whereas the linear growth condition prevents the solution from exploding in finite time — exactly as $\dot{y} = y^2$ does in the deterministic theory. If μ and σ are globally Lipschitz in x uniformly in t, and $|\mu(t,0)|$ and $|\sigma(t,0)|$ are bounded on $[0,T]$, then linear growth follows automatically, so the second hypothesis is often not stated separately; we state it because the multidimensional version, Theorem 1.2.27, does.

Second, the square-integrability of the initial condition is not decoration. The whole construction of the Itô integral took place in L^2, and the Picard iterates are estimated there; without $\mathbb{E}[X_0^2] < \infty$ the very first iterate need not lie in the space where the argument is conducted.

Third, and most consequentially for what follows, the word "unique" conceals a distinction that this formulation is too coarse to see. We have asserted *pathwise* uniqueness: given the Brownian motion, the solution is determined. One may instead ask only that the *law* of the solution be determined, which is a weaker requirement and corresponds to the notion of a *weak* solution — one where the driving Brownian motion is part of the output rather than the input. The two notions genuinely differ, and the gap between them is not a curiosity: the Lipschitz hypothesis above is violated by one of the most widely used models in this dissertation, the square-root variance process of Subsection 1.5.2, whose diffusion coefficient $\xi\sqrt{v}$ is only Hölder continuous of order $\frac{1}{2}$ at the origin. Theorem 1.2.18 as stated therefore does *not* cover the Heston model. We now repair this.

Beyond Lipschitz: strong and weak solutions

Let us first make the distinction precise, since it is systematically blurred in the applied literature and the blurring causes real confusion.

Definition 1.2.19 (Strong and weak solutions)**.** Consider the equation $dX_t = \mu(t,X_t)\,dt + \sigma(t,X_t)\,dW_t$ with a given $\mathcal{F}_0$-measurable X_0.

i) A **strong solution** is a process X, adapted to the augmented filtration generated by a *given* Brownian motion W and the initial condition, satisfying the equation almost surely.

Pathwise uniqueness holds if any two strong solutions on the same space, driven by the same W, are indistinguishable.

ii) A **weak solution** is a pair (X, W) together with a filtered probability space, such that W is a Brownian motion for that filtration and the equation holds. *Uniqueness in law* holds if any two weak solutions induce the same law on $C[0, T]$.

The distinction is one of information. For a strong solution the noise is handed to us and the trajectory is a measurable functional of it — one may, at least in principle, simulate the solution from a given stream of random numbers. For a weak solution we are permitted to construct the noise ourselves, and all we control is the distribution of the result. For pricing this is often enough, since prices are expectations and expectations depend only on the law; for hedging, and for any pathwise statement, it is not.

The relation between the four notions is settled by the Yamada–Watanabe theorem: pathwise uniqueness together with existence of a weak solution implies existence of a strong solution and uniqueness in law. What we require is the companion criterion, which relaxes the Lipschitz condition in exactly the direction needed.

Theorem 1.2.20 (Yamada–Watanabe)**.** *Consider the one-dimensional equation* $dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)\, dW_t$. *Suppose there exist a strictly increasing function* $\eta : [0, \infty) \to [0, \infty)$ *with* $\eta(0) = 0$ *and*

$$\int_{0+} \frac{du}{\eta(u)^2} = \infty, \tag{1.62}$$

and a strictly increasing concave ϱ *with* $\varrho(0) = 0$ *and* $\int_{0+} du/\varrho(u) = \infty$, *such that for all* t *and all* x, y

$$\Big|\sigma(t, x) - \sigma(t, y)\Big| \le \eta\Big(|x - y|\Big), \qquad \Big|\mu(t, x) - \mu(t, y)\Big| \le \varrho\Big(|x - y|\Big).$$

Then pathwise uniqueness holds for the equation [59]. ■

The modulus for the diffusion coefficient is written η rather than the h of the original paper, since h denotes the terminal payoff function in Theorem 1.3.7. The point of (1.62) is that it admits $\eta(u) = u^\gamma$ for every $\gamma \ge \frac{1}{2}$, since $\int_{0+} u^{-2\gamma} du$ diverges precisely when $2\gamma \ge 1$. Hölder continuity of order $\frac{1}{2}$ is thus permitted for the diffusion coefficient — and only just, the exponent $\frac{1}{2}$ being critical. The drift is treated separately and more leniently, needing only an Osgood-type modulus.

Example 1.2.21 (The square-root process)**.** Consider the process which will serve as the variance in the Heston model of Subsection 1.5.2,

$$dv_t = \kappa(\theta - v_t)\, dt + \xi\sqrt{v_t}\, dW_t, \qquad v_0 > 0, \tag{1.63}$$

with $\kappa, \theta, \xi > 0$. The drift is affine, hence Lipschitz. The diffusion coefficient $\sigma(v) = \xi\sqrt{v}$ is not Lipschitz at the origin, but it satisfies $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$, so Theorem 1.2.20 applies with $\eta(u) = \xi\sqrt{u}$, and pathwise uniqueness holds. A non-negative solution exists, and (1.63) is therefore well posed — but by Yamada–Watanabe, not by Theorem 1.2.18.

Well-posedness is not the end of the story for (1.63), because we also need the solution to stay strictly positive: it is to be a variance, and $\sqrt{v_t}$ must make sense. Note that the diffusion coefficient vanishes at $v = 0$ while the drift there equals $\kappa\theta > 0$, so the origin is a point where the noise switches off and the drift pushes upward. Whether that upward push is strong enough is a quantitative question with a sharp answer.

Proposition 1.2.22 (Feller condition)**.** *Let v solve* (1.63) *with $v_0 > 0$. If*

$$2\kappa\theta \geq \xi^2, \tag{1.64}$$

then $\mathbb{P}(v_t > 0$ for all $t \in [0,T]) = 1$. If $2\kappa\theta < \xi^2$, the process reaches 0 with positive probability, though it is instantaneously reflecting and does not become negative [24]. ■

Condition (1.64) has a transparent reading: mean reversion must be strong enough, and the volatility of volatility small enough, that the drift dominates the noise near the origin. It is worth knowing that in practice it is very often *violated* by calibrated parameters — fitting a steep short-maturity skew tends to demand a large ξ — and that this is a well-known source of difficulty in numerical work, since a naive Euler discretisation of (1.63) will produce negative values and then fail on $\sqrt{v_t}$. Practitioners resort to truncation or reflection schemes, or to exact simulation of the non-central χ^2 transition law. We flag the point here because it is a genuine instance of the theory in this section constraining what can be done in Chapter 3, rather than a technicality to be waved through.

1.2.4 The multidimensional case

For the time being, we have only addressed processes in dimension 1. Almost every result already covered can be restated for multidimensional processes or has a direct analogue.

By a multidimensional Brownian motion on $\mathbb{R}^n$ we mean a stochastic process of the form $\mathbf{W}(t) = (W_1(t), W_2(t), \dots, W_n(t))$ formed by n independent Brownian motions on $\mathbb{R}$. Similarly, we can set them to start at $x_0 \in \mathbb{R}^n$ just by a simple translation. It is a martingale and a Markov process that, starting at $\vec{0}$, has a transition density function given by $p_t(x) = (2\pi t)^{-n/2} e^{-\|x\|^2/2t}$.

There are certain properties of these processes that dramatically change with dimension. For example, for $n = 1$, it returns to 0 infinitely often, and for $n = 2$ enters arbitrarily small neighborhoods of 0 infinitely often with probability 1, however the probability that it reaches $\vec{0}$ is 0. For $n \geq 3$, the process is much more sparse, satisfying $\lim_{t\to\infty}\|\mathbf{W}(t)\| = \infty$ almost surely.

In terms of fractal dimensions, the *range* $\mathbf{W}([0,t])$ has Hausdorff dimension 2 for $n \geq 2$, and dimension 1 for $n = 1$ (where it is simply an interval). The *graph* $\{(s, \mathbf{W}(s)) : s \in [0,t]\} \subset \mathbb{R}^{1+n}$ behaves differently: it has dimension $3/2$ for $n = 1$, a classical result of Taylor [57], and dimension 2 for $n \geq 2$. These fascinating properties of Brownian motion can be found in [23].

The exponent $3/2$ in the one-dimensional graph is not an isolated curiosity. A graph of dimension $3/2$ is exactly what one obtains from a function which is α-Hölder for every $\alpha < 1/2$ and for no $\alpha \geq 1/2$, and that critical exponent $1/2$ — together with what happens on either side of it — will return in Part II as the organising theme of the whole dissertation.

We can therefore define a multidimensional stochastic integral (expressed in differential form) as

$$dX_t = \mu\, dt + \sum_{i=1}^{n} \sigma_i\, dW_t^i \tag{1.65}$$

where each integral must be considered separately. Let $W_1(t), W_2(t), \dots, W_m(t)$ be independent Brownian motions. Consider n Itô processes $X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(n)}$ given by

$$X_t^{(i)} = X_0^{(i)} + \int_0^t \mu_i(s)\, ds + \sum_{j=1}^{m} \int_0^t \sigma_{ij}(s)\, dW_j(s), \quad 1 \leq i \leq n, \tag{1.66}$$

where $\mu_i \in \mathcal{L}(\Omega, L^1[0,T])$ and $\sigma_{ij} \in \mathcal{L}(\Omega, L^2[0,T])$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$. Collecting $W = (W_1, \dots, W_m)^\top$ and $X_t = (X_t^{(1)}, \dots, X_t^{(n)})^\top$ into column vectors, $\mu = (\mu_1, \dots, \mu_n)^\top$ likewise, and $\sigma = [\sigma_{ij}]$ into an $n \times m$ matrix, equation (1.66) becomes

$$X_t = X_0 + \int_0^t \mu(s)\, ds + \int_0^t \sigma(s)\, dW(s), \qquad 0 \leq t \leq T,$$

the integrals being taken componentwise. Growth conditions on μ_i and σ_{ij} must again be imposed for the solution to make sense. Itô's formula holds as before, now accounting for the crossed second derivatives; since every subsequent computation in this dissertation — the Itô product rule, the Black–Scholes derivation of Subsection 1.4.1, the Heston system of Subsection 1.5.2 and the proof of Girsanov's theorem in Appendix B.6 — rests on it, we state it explicitly.

Theorem 1.2.23 (Multidimensional Itô formula)**.** *Let X be the n-dimensional Itô process of* (1.66)*, driven by an m-dimensional Brownian motion with independent components, and let $\theta \in \mathcal{C}^{1,2}\Big([0,T] \times \mathbb{R}^n, \mathbb{R}\Big)$. Then, almost surely and for every $t \in [0,T]$,*

$$d\theta(t, X_t) = \frac{\partial \theta}{\partial t}\, dt + \sum_{l=1}^{n} \frac{\partial \theta}{\partial x_l}\, dX_t^{(l)} + \frac{1}{2} \sum_{l,h=1}^{n} \frac{\partial^2 \theta}{\partial x_l \partial x_h}\, d\Big\langle X^{(l)}, X^{(h)} \Big\rangle_t, \tag{1.67}$$

where the cross-variations are

$$d\Big\langle X^{(l)}, X^{(h)} \Big\rangle_t = \sum_{j=1}^{m} \sigma_{lj}(t) \sigma_{hj}(t)\, dt = \Big(\sigma \sigma^\top\Big)_{lh} dt. \tag{1.68}$$

If instead the driving Brownian motion has correlation matrix $P = (\rho_{jk})$ in the sense of Definition 1.2.24 below, (1.67) *is unchanged and* (1.68) *becomes*

$$d\Big\langle X^{(l)}, X^{(h)} \Big\rangle_t = \sum_{j,k=1}^{m} \sigma_{lj}(t)\, \rho_{jk}\, \sigma_{hk}(t)\, dt = \Big(\sigma P \sigma^\top\Big)_{lh} dt. \tag{1.69}$$

■

The proof is the one-dimensional argument of Theorem 1.2.12 applied coordinatewise, with the multiplication table of Table 1.1 supplying the second-order terms; see [40, Ch. 7] or [49, Ch. 4]. Taking $\theta(x_1, x_2) = x_1 x_2$ and $n = 2$ recovers the Itô product rule.

The correlated form (1.69) is stated because it, and not (1.68), is what the Heston system (1.138) requires: that model is specified with $d\langle W^S, W^v \rangle_t = \rho\, dt$, so its drivers are correlated by construction. Nothing new is involved — one may equally decorrelate first, writing $\widetilde{\mathbf{W}} = L\mathbf{W}$ with $LL^\top = P$ as in Definition 1.2.24, and then apply (1.68) to σL — but keeping the general form to hand avoids performing that substitution silently every time the table is used.

Before writing the corresponding multiplication table we must introduce an object which the table presupposes and which the definition above does not provide. So far, a multidimensional Brownian motion has *independent* components. In modelling one almost always wants the components to be correlated — a falling equity price accompanied by rising volatility, two rates moving together, an exchange rate and the foreign interest rate — and the following makes that precise.

Definition 1.2.24 (Correlated Brownian motion)**.** Let $P = (\rho_{ij}) \in \mathbb{R}^{n \times n}$ be symmetric and positive semi-definite with $\rho_{ii} = 1$ for all i. An n-dimensional process $\widetilde{\mathbf{W}} = (\widetilde{W}^1, \dots, \widetilde{W}^n)$ is a **Brownian motion with correlation matrix** P if each $\widetilde{W}^i$ is a one-dimensional Brownian motion and

$$\Big\langle \widetilde{W}^i, \widetilde{W}^j \Big\rangle_t = \rho_{ij}\, t, \qquad 1 \leq i, j \leq n, \quad t \in [0,T]. \tag{1.70}$$

Such a process always exists, and is obtained from an independent one by a linear map. Since P is symmetric positive semi-definite it admits a factorisation $P = LL^\top$ — the Cholesky factorisation when P is positive definite, and a square root more generally. Setting $\widetilde{\mathbf{W}} := L\,\mathbf{W}$ for an independent $\mathbf{W}$, bilinearity of the quadratic covariation (1.46) gives

$$\left\langle \widetilde{W}^i, \widetilde{W}^j \right\rangle_t = \sum_{k,l} L_{ik} L_{jl} \left\langle W^k, W^l \right\rangle_t = \sum_k L_{ik} L_{jk}\, t = (LL^\top)_{ij}\, t = \rho_{ij}\, t,$$

and each component is a continuous local martingale with $\langle \widetilde{W}^i \rangle_t = t$, hence a Brownian motion by Theorem 1.2.29.

The positive semi-definiteness of P is a genuine constraint and not a formality. It is what makes the factorisation possible, and it is a real restriction as soon as $n \geq 3$: one cannot independently choose ρ_{12}, ρ_{13} and ρ_{23}, since a strongly positive correlation between the first two pairs forbids a strongly negative one between the third. This is a familiar nuisance in practice, where correlation matrices estimated entrywise from market data frequently fail to be positive semi-definite and must be repaired before they can be used.

With this in hand, the Itô multiplication table extends as follows.

$\times$	dt	dW_t^1	dW_t^2	$\cdots$	dW_t^n
dt	0	0	0	$\cdots$	0
dW_t^1	0	dt	$\rho_{12}\, dt$	$\cdots$	$\rho_{1n}\, dt$
dW_t^2	0	$\rho_{21}\, dt$	dt	$\cdots$	$\rho_{2n}\, dt$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
dW_t^n	0	$\rho_{n1}\, dt$	$\rho_{n2}\, dt$	$\cdots$	dt

Table 1.1: Itô multiplication table for correlated n-dimensional Brownian motion.

The entries are not a new convention but a restatement of (1.70): the symbol $dW_t^i\, dW_t^j$ abbreviates $d\langle W^i, W^j \rangle_t = \rho_{ij}\, dt$. The independent case is recovered by taking $P = \mathbb{I}_{n\times n}$.

Example 1.2.25. Let $f : \mathbb{R}^n \to \mathbb{R}$ a $\mathcal{C}^2$ function and let $\nabla f = (\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n})$ and $\Delta f = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}$ to obtain

$$f(\mathbf{W}(t)) = f(\mathbf{W}(0)) + \int_0^t \nabla f(\mathbf{W}(s)) \cdot d\mathbf{W}_s + \frac{1}{2} \int_0^t \Delta f(\mathbf{W}(s))\, ds.$$

This equation exhibits the deep connection between Brownian motion and diffusions, showing how *one half of the Laplace operator* is the generator of Brownian motion; see Subsection 1.2.6, where the same $\frac{1}{2}$ reappears from the semigroup side. The factor is not decorative: it is the quadratic variation $\langle W^i \rangle_t = t$ entering through the second-order term of Itô's rule, and it is the reason the heat equation associated with Brownian motion is $\partial_t u = \frac{1}{2}\Delta u$ rather than $\partial_t u = \Delta u$.

Example 1.2.26. An important example is given by $\theta(x, y) = xy$. Clearly, $\partial\theta/\partial x = y$ and also $\partial\theta/\partial y = x$. Therefore, if we have processes X_t and Z_t, we obtain

$$d(X_t\, Z_t) = Z_t\, dX_t + X_t\, dZ_t + \frac{1}{2} dX_t\, dZ_t + \frac{1}{2} dX_t\, dZ_t = Z_t\, dX_t + X_t\, dZ_t + dX_t\, dZ_t \tag{1.71}$$

what is usually known as Itô product formula, understanding $dX_t\, dZ_t$ as in table 1.1.

The following generalization holds.

Theorem 1.2.27. *Let $\sigma(t,x)$ be an $n \times m$ matrix-valued function and $\mu(t,x)$ an $\mathbb{R}^n$-valued function. Assume there exists $C > 0$ such that for all $t \in [0,T]$ and $x, y \in \mathbb{R}^n$,*

$$\|\sigma(t,x) - \sigma(t,y)\| \leq C|x-y| \quad , \quad |\mu(t,x) - \mu(t,y)| \leq C|x-y|$$

and

$$\|\sigma(t,x)\|^2 \leq C(1+|x|^2) \quad , \quad |\mu(t,x)|^2 \leq C(1+|x|^2)$$

where $\|\sigma\|^2 = \sum_i \sum_j \sigma_{ij}^2$. For any $\mathcal{F}_0$-measurable random vector X_0 of finite variance, the SDE $dX_t = \mu(t,X_t)\,dt + \sigma(t,X_t)\,dW_t$, $X_0 = \xi$ has a unique continuous solution. ■

1.2.5 Probability measure changes

Another important topic we must cover has to do with probability changes. Brownian motion depends crucially on the underlying probability $\mathbb{P}$. Given a process X_t, it seems legit to ask whether there is a probability $\mathbb{Q}$ so that X_t is a Brownian motion with respect to $\mathbb{Q}$. We assume the σ-algebra and the filtration remain unchanged.

The Doob–Meyer theorem below is stated in the language of *predictable* processes, and one word is owed on why the notion appears at all, since for the processes studied here it will make no difference whatsoever. Writing $\mathbb{L}$ for the adapted processes with almost all sample paths left-continuous, the **predictable** σ-algebra is the smallest σ-algebra on $[0,T] \times \Omega$ making every element of $\mathbb{L}$ measurable, and a process is predictable if it is measurable with respect to it.

When the integrator has continuous sample paths — as Brownian motion does, and as every model in Part I does — predictable and merely adapted integrands give rise to the same stochastic integrals, which is why the construction of Subsection 1.2.1 could proceed without the notion. The distinction becomes essential the moment the integrator is allowed to jump, and for a reason about information rather than technique: if the integrand may "see" a jump as it happens, one can bet on the jump at the instant it occurs and the resulting integral fails to be a martingale. Predictability forbids exactly this, and its financial reading is that one may not trade on information one does not yet have. This is not idle for genuinely discontinuous underlyings — electricity spot prices, several commodities, and credit-sensitive instruments where default is by nature a jump — but we shall not need it again.

Theorem 1.2.28 (Doob–Meyer Decomposition Theorem)**.** *Let $M(t)$, $t \in [0,T]$ be a càdlàg, square integrable martingale. Then there exists a unique decomposition of the form*

$$M_t^2 = R(t) + \langle M \rangle_t \tag{1.72}$$

where $R(t)$ is càdlàg martingale and $\langle M \rangle_t$ is a predictable, right continuous and increasing process such that $\langle M \rangle_0 = 0$ and $\mathbb{E}\Big[\langle M \rangle_t\Big] < \infty$ for all $t \in [0,T]$. We call $\langle M \rangle_t$ the ***compensator*** *of M_t^2 and we say it is the predictable quadratic variation of M_t.* ■

The theorem holds in a way more general form for submartingales, but if M_t is a square integrable martingale, then, by Jensen's inequality, M_t^2 is a submartingale and it suffices for our purposes. For a proof of the Doob–Meyer decomposition theorem, see [40] or [56].

Theorem 1.2.29 (Lévy's Characterisation Theorem)**.** *Let X be a continuous local martingale on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ with $X_0 = 0$ and $\langle X \rangle_t = t$ almost surely for every $t \in [0,T]$. Then X is an $(\mathcal{F}_t)_{t\in[0,T]}$-Brownian motion under $\mathbb{P}$.*

More generally, let $X = (X^1, \ldots, X^n)$ be a vector of continuous local martingales with $X_0 = 0$ and

$$\langle X^i, X^j \rangle_t = \delta_{ij}\, t \qquad \textit{almost surely, for all } i, j \textit{ and all } t,$$

where δ_{ij} is the Kronecker delta. Then X is an n-dimensional $(\mathcal{F}_t)_{t\in[0,T]}$-Brownian motion with independent components.

Proof: The argument is given in Appendix B.5; see also [40, Thm. 8.4.2]. ■

The statement deserves to be read slowly, because it is considerably stronger than it looks. Being a Brownian motion is, by Definition 1.1.33, a statement about the *joint law* of the whole family $\{X_t\}$: independent, stationary, Gaussian increments. Lévy's theorem says that all of this is already implied by two facts about second moments — that X is a local martingale and that its quadratic variation is exactly t. Gaussianity is not assumed anywhere; it is a conclusion.

The multidimensional form, with the condition $\langle X^i, X^j\rangle_t = \delta_{ij}t$, is the one we shall actually need. It is what will allow us, in the proof of Girsanov's theorem, to recognise a process as a Brownian motion under a *new* measure without recomputing any distributions — one need only check that quadratic variation is preserved, which it is, since quadratic variation is a pathwise notion and equivalent measures agree on which paths occur.

Definition 1.2.30 (Exponential Process)**.** Let $\alpha \in \mathcal{L}(\Omega, L^2[0,T])$. We define its exponential process by

$$\mathcal{E}_\alpha(t) = e^{\int_0^t \alpha(s)dW(s) - \frac{1}{2}\int_0^t \alpha^2(s)ds} \tag{1.73}$$

which is precisely the solution of $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)dW(t)$, $\mathcal{E}_\alpha(0) = 1$ so that $\mathcal{E}_\alpha(t)$ is a local martingale.

This exponential process is important because it leads to the construction of a new probability $\mathbb{Q}$ for which $\mathcal{E}_\alpha(t)$ is actually a martingale.

Theorem 1.2.31. *Let $\alpha \in \mathcal{L}(\Omega, L^2[0,T])$. Then $\mathcal{E}_\alpha(t)$ is a martingale if, and only if, $\mathbb{E}[\mathcal{E}_\alpha(t)] = 1$ for all $t \in [0,T]$.*

Proof: Necessity is immediate: if $\mathcal{E}_\alpha$ is a martingale then $\mathbb{E}[\mathcal{E}_\alpha(t)] = \mathbb{E}\Big[\mathbb{E}[\mathcal{E}_\alpha(t) \mid \mathcal{F}_0]\Big] = \mathbb{E}[\mathcal{E}_\alpha(0)] = 1$ for every t.

For sufficiency, recall from the definition that $\mathcal{E}_\alpha(t) = 1 + \int_0^t \alpha(s)\mathcal{E}_\alpha(s)\,dW_s$, so $\mathcal{E}_\alpha$ is a non-negative local martingale, and let (τ_n) be a localising sequence. A non-negative local martingale is always a supermartingale — apply the conditional Fatou lemma (Proposition A.1.2) to $\mathcal{E}_\alpha(t\wedge \tau_n) \to \mathcal{E}_\alpha(t)$ — so for $s < t$,

$$\mathbb{E}\Big[\mathcal{E}_\alpha(t) \mid \mathcal{F}_s\Big] \le \mathcal{E}_\alpha(s) \qquad \text{almost surely.}$$

Both sides are non-negative and, by hypothesis, have the same expectation, namely 1. A non-negative random variable dominated by another of equal expectation is almost surely equal to it, so the inequality is an equality and $\mathcal{E}_\alpha$ is a martingale. See [40, Thm. 8.7.3]. ■

The previous result is tautological and hard to check in practice, so we need a more practical condition. The following result is known as Novikov's condition.

Theorem 1.2.32 (Novikov's condition)**.** *Let $\alpha \in \mathcal{L}(\Omega, L^2[0,T])$. If $\mathbb{E}\left[e^{\frac{1}{2}\int_0^T \alpha^2(s)ds}\right] < \infty$, then $\mathcal{E}_\alpha(t)$ is a martingale. A proof may be found in [40, Sec. 8.7].* ■

This is a much more practical way to check whether $\mathcal{E}_\alpha(t)$ is a martingale, and therefore whether we can construct a new probability $\mathbb{Q}$ such that $d\mathbb{Q} = \mathcal{E}_\alpha(T)\,d\mathbb{P}$. For the following Theorem, we need to make use of two auxiliary lemmas.

Lemma 1.2.33 (Conditional expectation under a change of measure)**.** *Let γ be a nonnegative integrable random variable, so that $d\mathbb{Q} = \gamma\,d\mathbb{P}$ defines a new probability measure (see 1.2). Then, for any σ-algebra $\mathcal{G} \subset \mathcal{F}$ and Y such that $\mathbb{E}_\mathbb{Q}[Y] < \infty$, it follows that*

$$\mathbb{E}_\mathbb{Q}[Y \mid \mathcal{G}] = \frac{\mathbb{E}_\mathbb{P}[\gamma Y \mid \mathcal{G}]}{\mathbb{E}_\mathbb{P}[\gamma \mid \mathcal{G}]}, \ \mathbb{Q}\text{-almost surely.} \tag{1.74}$$

Proof: Integrability holds because $\mathbb{E}_{\mathbb{P}}[|Y|\gamma] = \mathbb{E}_{\mathbb{Q}}[|Y|] < \infty$, so $\mathbb{E}_{\mathbb{P}}[\gamma Y \mid \mathcal{G}]$ is defined. Fix $G \in \mathcal{G}$. By the defining property of conditional expectation under $\mathbb{P}$ and the definition of $\mathbb{Q}$,

$$\int_G \mathbb{E}_{\mathbb{P}}[\gamma Y \mid \mathcal{G}]\, d\mathbb{P} = \int_G \gamma Y\, d\mathbb{P} = \int_G Y\, d\mathbb{Q} = \int_G \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}]\, d\mathbb{Q}.$$

On the other hand, writing $d\mathbb{Q} = \gamma\, d\mathbb{P}$ and using that $\mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}]$ is $\mathcal{G}$-measurable, so that it may be taken out of a conditional expectation given $\mathcal{G}$ (Proposition A.1.1, item 4),

$$\int_G \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}]\, d\mathbb{Q} = \int_G \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}]\, \gamma\, d\mathbb{P} = \int_G \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}]\, \mathbb{E}_{\mathbb{P}}[\gamma \mid \mathcal{G}]\, d\mathbb{P}.$$

The two displays agree for every $G \in \mathcal{G}$, and both integrands are $\mathcal{G}$-measurable, so they coincide almost surely. Dividing by $\mathbb{E}_{\mathbb{P}}[\gamma \mid \mathcal{G}]$, which is strictly positive $\mathbb{Q}$-almost surely, gives the result [40, Lemma 8.9.2]. ■

Then, if $\alpha \in \mathcal{L}(\Omega, L^2[0,T])$ satisfies the condition on Theorem 1.2.31, it follows that $\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$-martingale. Let $\mathbb{Q}$ be defined by $d\mathbb{Q} = \mathcal{E}_\alpha(T)\, d\mathbb{P}$, that is, $\mathbb{Q}(A) = \int_A \mathcal{E}_\alpha(T)\, d\mathbb{P}$ for every $A \in \mathcal{F}$.

Lemma 1.2.34 (Martingale transfer)**.** *Let $\alpha \in \mathcal{L}(\Omega, L^2[0,T])$. Then a $\{\mathcal{F}_t\}$-adapted process X_t, $t \in [0,T]$ is a $\mathbb{Q}$-martingale if, and only if, $X(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$-martingale.*

Proof: Apply Lemma 1.2.33 with $\gamma = \mathcal{E}_\alpha(T)$ and $\mathcal{G} = \mathcal{F}_s$. Since $\mathcal{E}_\alpha$ is a $\mathbb{P}$-martingale, $\mathbb{E}_{\mathbb{P}}[\mathcal{E}_\alpha(T) \mid \mathcal{F}_s] = \mathcal{E}_\alpha(s)$, so for $s \le t$

$$\mathbb{E}_{\mathbb{Q}}\Big[X(t) \mid \mathcal{F}_s\Big] = \frac{\mathbb{E}_{\mathbb{P}}\Big[X(t)\mathcal{E}_\alpha(T) \mid \mathcal{F}_s\Big]}{\mathcal{E}_\alpha(s)}. \tag{1.75}$$

The numerator simplifies by conditioning on $\mathcal{F}_t$ first and using that $X(t)$ is $\mathcal{F}_t$-measurable:

$$\mathbb{E}_{\mathbb{P}}\Big[X(t)\mathcal{E}_\alpha(T) \mid \mathcal{F}_s\Big] = \mathbb{E}_{\mathbb{P}}\Big[X(t)\, \mathbb{E}_{\mathbb{P}}[\mathcal{E}_\alpha(T) \mid \mathcal{F}_t] \,\Big|\, \mathcal{F}_s\Big] = \mathbb{E}_{\mathbb{P}}\Big[X(t)\mathcal{E}_\alpha(t) \mid \mathcal{F}_s\Big].$$

Hence $\mathbb{E}_{\mathbb{Q}}[X(t) \mid \mathcal{F}_s] = X(s)$ if and only if $\mathbb{E}_{\mathbb{P}}[X(t)\mathcal{E}_\alpha(t) \mid \mathcal{F}_s] = X(s)\mathcal{E}_\alpha(s)$, which is the assertion [40, Lemma 8.9.3]. ■

Finally, the following key result holds from Lemmas 1.2.33 and 1.2.34.

Theorem 1.2.35 (Girsanov's Theorem)**.** *Let $\alpha \in \mathcal{L}(\Omega, L^2[0,T])$ such that $\mathcal{E}_\alpha$ is a martingale. Then the process*

$$Z(t) = W(t) - \int_0^t \alpha(s)\, ds, \quad t \in [0,T] \tag{1.76}$$

is a Brownian motion with respect to the measure $d\mathbb{Q} = \mathcal{E}_\alpha(T)d\mathbb{P}$.

Proof: The argument is given in Appendix B.6; see also [40, Thm. 8.9.4]. ■

Girsanov's theorem is the formal statement of something we shall use constantly: *changing the measure changes the drift and leaves the volatility alone.* The Brownian motion Z under $\mathbb{Q}$ is the old W with a drift subtracted, and nothing else has moved. The reason is already contained in Theorem 1.2.29: to verify that Z is a $\mathbb{Q}$-Brownian motion one need only check that it is a continuous local martingale with $\langle Z \rangle_t = t$, and quadratic variation is a pathwise limit, so it is unaffected by passing to an equivalent measure. Two equivalent measures disagree about how likely a path is, never about which paths exist. This observation is the mathematical content of the statement — which we shall meet again in Section 1.4 — that an option's price does not depend on the expected return of the underlying, while it depends critically on its volatility.
For the multi-factor models of Section 1.5 we require the vector-valued form.

Theorem 1.2.36 (Girsanov's Theorem, multidimensional)**.** *Let* $\mathbf{W} = (W^1, \dots, W^n)$ *be an* n*-dimensional Brownian motion and let* $\alpha = (\alpha^1, \dots, \alpha^n)$ *be an adapted process with* $\int_0^T \|\alpha(s)\|^2\, ds < \infty$ *almost surely. Define*

$$\mathcal{E}_\alpha(t) = \exp\left(\int_0^t \alpha(s)^\top\, d\mathbf{W}(s) - \frac{1}{2}\int_0^t \|\alpha(s)\|^2\, ds\right), \tag{1.77}$$

and suppose Novikov's condition holds in the form $\mathbb{E}\Big[\exp\Big(\frac{1}{2}\int_0^T \|\alpha(s)\|^2 ds\Big)\Big] < \infty$*. Then* $\mathcal{E}_\alpha$ *is a martingale,* $d\mathbb{Q} = \mathcal{E}_\alpha(T)\, d\mathbb{P}$ *defines a probability measure equivalent to* $\mathbb{P}$*, and*

$$\mathbf{Z}(t) = \mathbf{W}(t) - \int_0^t \alpha(s)\, ds, \qquad t \in [0, T], \tag{1.78}$$

is an n*-dimensional* $\mathbb{Q}$*-Brownian motion with independent components.* ■

That the components of $\mathbf{Z}$ remain independent under $\mathbb{Q}$ deserves emphasis, since it is not obvious and is exactly what the multidimensional Lévy characterisation delivers: the cross-variations $\langle Z^i, Z^j\rangle_t = \delta_{ij} t$ are pathwise quantities and survive the change of measure untouched. Note also that this is a statement about *independent* driving factors; when a model is specified with correlated Brownian motions, as the Heston model of Subsection 1.5.2 is, one applies the theorem after decorrelating, in the manner of Definition 1.2.24 above.

The Martingale Representation Theorem

Girsanov's theorem tells us which measures are available. The following tells us which *payoffs* are attainable, and it is the deepest result of this subsection. Together the two constitute the entire mathematical content of the fundamental theorems of asset pricing in Section 1.3.

Theorem 1.2.37 (Itô's Martingale Representation Theorem)**.** *Let* $\mathbf{W}$ *be an* n*-dimensional Brownian motion on* $(\Omega, \mathcal{F}, \mathbb{P})$ *and let* $(\mathcal{F}_t^W)_{t\in[0,T]}$ *be its augmented natural filtration. Let* M *be a square-integrable* $\mathbb{P}$*-martingale with respect to that filtration. Then there exists a unique adapted process* $\varphi = (\varphi^1, \dots, \varphi^n) \in L^2_{\mathrm{ad}}([0,T] \times \Omega)$ *such that*

$$M_t = M_0 + \int_0^t \varphi(s)^\top\, d\mathbf{W}(s), \qquad t \in [0, T], \tag{1.79}$$

almost surely. If M *is only a local martingale, the same holds with* φ *in the larger class* $\mathcal{L}(\Omega, L^2[0,T])$*.* ■

The theorem should be read as a converse to Theorem 1.2.6. There we showed that every Itô integral is a martingale; here we learn that, provided the filtration is the one generated by $\mathbf{W}$ and nothing more, *every* martingale is an Itô integral. In the language of Proposition 1.1.4, item 9, the stochastic integrals do not merely sit inside the space of martingales: they exhaust it.

Two aspects of the hypothesis deserve to be underlined, because both will be load-bearing.

The first is that the filtration must be generated by the Brownian motion itself. If $(\mathcal{F}_t)_{t\in[0,T]}$ carries additional information — another independent Brownian motion, a Poisson process, an independent random variable revealed at time T — the conclusion fails outright, and it fails for an obvious reason: a martingale driven by that extra randomness cannot be written as an integral against $\mathbf{W}$, because the integral would be measurable with respect to the wrong σ-algebra. The theorem is a statement about there being *no more randomness in the model than the driving Brownian motion supplies.*

The second is that φ is not constructed. The proof is an orthogonality argument: one shows that the stochastic integrals form a closed subspace of $L^2(\Omega, \mathcal{F}_T)$ whose orthogonal complement contains only constants, using the fact that exponential martingales $\mathcal{E}_\alpha(T)$ span a dense set. Existence is therefore obtained without any formula for φ, which is worth knowing in advance: the theorem will guarantee that a replicating strategy exists while saying nothing whatever about how to compute it. Recovering φ in concrete cases is precisely the delta-hedging computation of Section 1.4, and in models where no closed form is available it is a numerical problem.

We can already state the consequence for which we shall need it, though the vocabulary is only introduced in the next section.

Remark 1.2.38 (Why this matters for pricing). Let H_T be a square-integrable contingent claim and let $\mathbb{Q}$ be an equivalent martingale measure. The discounted price process $M_t := \mathbb{E}^{\mathbb{Q}}[H_T/S_T^0 \mid \mathcal{F}_t]$ is by construction a $\mathbb{Q}$-martingale — this is the tower property, nothing more. If the filtration is generated by the Brownian motion driving the traded assets, Theorem 1.2.37 applied under $\mathbb{Q}$ produces φ with $M_t = M_0 + \int_0^t \varphi^\top \, d\mathbf{Z}$, and φ is — after rewriting the integral in terms of the discounted asset prices — a self-financing strategy replicating H_T exactly.

One hypothesis is being used silently here and is worth naming, since it is a natural examiner's question. Theorem 1.2.37 is applied under $\mathbb{Q}$ to the $\mathbb{Q}$-Brownian motion $\mathbf{Z}$, so what it requires is that the filtration be generated by $\mathbf{Z}$ — whereas the filtration we have is the one generated by $\mathbf{W}$. The two coincide: by Theorem 1.2.36, $\mathbf{Z}_t = \mathbf{W}_t - \int_0^t \alpha \, ds$ with α adapted, so each is a measurable functional of the other and, after the augmentation of Convention 1.1.7, $(\mathcal{F}_t^W)_{t\in[0,T]} = (\mathcal{F}_t^{\mathbf{Z}})_{t\in[0,T]}$. Girsanov changes the measure and leaves the information alone, which is exactly what makes the two theorems composable.

Every claim is thus attainable, so the market is *complete*, and the price of the claim is not merely a plausible expectation but the unique cost of a portfolio that reproduces it in every state of the world. This is the substance behind the Second Fundamental Theorem (Theorem 1.3.6), and it is the reason the Black–Scholes model is complete (Section 1.4) while the stochastic volatility models of Subsection 1.5.2 are not: in the latter, the filtration is generated by *two* Brownian motions while only *one* asset is traded, the hypothesis of Theorem 1.2.37 fails, and the volatility risk cannot be hedged away.

One word on function spaces, since two appear. Completeness will be defined below by quantifying over $L^\infty(\mathcal{F}_T)$, which is the Delbaen–Schachermayer convention and the right one for the Fundamental Theorems: boundedness is what makes the separating-hyperplane argument work in a dual pair. The Martingale Representation Theorem, by contrast, is an L^2 statement, because it is ultimately a statement about an orthogonal projection in a Hilbert space — Proposition 1.1.4, item 9. The two are reconciled by the inclusion $L^\infty(\mathcal{F}_T) \subset L^2(\mathcal{F}_T)$, valid because $\mathbb{P}$ is a finite measure: every bounded claim is square-integrable, so the representation above applies to every claim the completeness definition quantifies over. We use whichever space the argument at hand requires, and the inclusion is the only thing needed to pass between them.

1.2.6 Itô Diffusions

Recall that, for a Markov process, the *transition probability* $P_{s,x}(t, A) = \mathbb{P}(X_t \in A \mid X_s = x)$, for A measurable, $0 \le s \le t \le T$ and $x \in \mathbb{R}^n$. The Chapman-Kolmogorov equation states that

$$P_{s,x}(t, A) = \int_{\mathbb{R}^n} P_{u,z}(t, A) \, P_{s,x}(u, dz).$$

Definition 1.2.39 (Diffusion Process)**.** An $\mathbb{R}^n$-valued process X_t, $t \in [0, T]$ is called a **diffusion process** if its transition probabilities $\{P_{s,x}(t, \cdot)\}$ satisfy the following three conditions for every $t \in [0, T]$, every $x \in \mathbb{R}^n$ and every $\delta > 0$:

$$\text{i)} \ \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\|>\delta\}} P_{t,x}(t+\varepsilon, dy) = 0$$

$$\text{ii)} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \le \delta\}} (y-x)\, P_{t,x}(t+\varepsilon, dy) = \mu(t,x) \in \mathbb{R}^n \text{ exists,}$$

$$\text{iii)} \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \int_{\{\|y-x\| \le \delta\}} (y-x)(y-x)^\top\, P_{t,x}(t+\varepsilon, dy) = \Sigma(t,x) \in \mathbb{R}^{n\times n} \text{ exists,}$$

where the vectors in $\mathbb{R}^n$ are implicitly taken as columns, so that $(y-x)(y-x)^\top$ yields a symmetric positive semi-definite matrix. μ and Σ are called the *drift* and the *diffusion* coefficient, respectively.

The restriction of the domain of integration to a ball around x is essential and not a technicality: the integral of a probability measure over the whole of $\mathbb{R}^n$ is 1, so dropping the constraint in i) would make the limit $+\infty$ rather than 0. Read correctly, the three conditions say that in a short time ε the process moves a distance of order $\sqrt{\varepsilon}$ with overwhelming probability, that the mean of that displacement is of order ε, and that its covariance is of order ε as well. When conditions ii) and iii) hold, the truncation at level δ turns out not to affect the limits, so we may and shall write them informally as

$$\mu(t,x) = \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \mathbb{E}\left[X_{t+\varepsilon} - x \mid X_t = x\right], \qquad \Sigma(t,x) = \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \mathbb{E}\left[(X_{t+\varepsilon} - x)(X_{t+\varepsilon} - x)^\top \mid X_t = x\right].$$

Note how condition i) forbids the process from having jumps: it is precisely the statement that a displacement of size δ in time ε has probability $o(\varepsilon)$. Therefore the Poisson process, for which such a displacement has probability $\lambda\varepsilon + o(\varepsilon)$, is not a diffusion — which is the promise made after Example 1.1.23, that the Poisson process would reappear as the canonical process that is *not* one.

Example 1.2.40. Take an $\mathbb{R}^n$-valued Brownian motion W_t. Note that $\|W_t\|^2$ is a sum of n independent squares of normal variables, so that $\mathbb{E}\|W_\varepsilon\|^4 = n(n+2)\varepsilon^2$. Hence, for any $c > 0$, Markov's inequality applied to the fourth power gives

$$\mathbb{P}\Big(\|W_{t+\varepsilon} - x\| \ge c \,\Big|\, W_t = x\Big) = \mathbb{P}\Big(\|W_{t+\varepsilon} - W_t\| \ge c\Big) \le c^{-4}\, \mathbb{E}\|W_\varepsilon\|^4 = c^{-4} n(n+2)\varepsilon^2,$$

which is $o(\varepsilon)$. Condition i) of Definition 1.2.39 is therefore satisfied, and a direct computation of the remaining two limits gives $\mu(t,x) = 0$ and $\Sigma(t,x) = \mathbb{I}_{n\times n}$.

The following result finally connects the previous concepts and gives an unambiguous meaning to μ.

Theorem 1.2.41. *Let $\sigma(t,x)$ and $\mu(t,x)$ be functions as in Theorem 1.2.27. Assume they are also continuous on $[0,T] \times \mathbb{R}^n$. The solution to the SDE there defined is a diffusion process where $\mu(t,x)$ is identified and $\Sigma(t,x) = \sigma(t,x)\,\sigma(t,x)^\top$.* ■

The proof of this Theorem is beyond the scope of our purposes and can be found in [40].
We have reached a critical point where diffusions, SDEs and semigroups intersect. These seemingly unrelated mathematical objects give rise to a set of (parabolic) partial differential equations (*PDEs*) that are crucial for computational purposes and pricing financial derivatives. In this context, powerful mathematical tools can be applied.
The first of the approaches, where we consider a semigroup of operators, is included for completeness purposes, showing how its infinitesimal generators may be used to construct diffusion processes. It is also included to illustrate how Functional Analysis arises in this context.

1. Semigroup approach

We restrict to the **stationary case**, in which μ and Σ depend on x alone, and use standard Banach-space language; see [10].

A family $\{T_t\}_{t\geq 0}$ of linear operators on a Banach space $\mathbb{B}$ is a **contraction semigroup** if $T_0 = \mathrm{I}$ and $T_{t+s} = T_t T_s$ (semigroup property), $T_t v \to v$ as $t \downarrow 0$ for every v (strong continuity), and $\|T_t v\| \leq \|v\|$ (contraction). Its **infinitesimal generator** is

$$\mathcal{A}(v) = \lim_{t\downarrow 0} \frac{1}{t}\Big(T_t(v) - v\Big), \tag{1.80}$$

defined on those v for which the limit exists strongly; it is typically unbounded, as already for the translation semigroup $T_t f(x) = f(x+t)$ on $C_0[0,\infty)$, whose generator is $\mathcal{A}f = f'$. The choice of space is not incidental — strong continuity fails on the larger $C_b[0,\infty)$ — but nothing below turns on it, and we take C_0 throughout.

Given a stationary diffusion X_t driven by μ and Σ, one sets $T_t(f)(x) = \int_{\mathbb{R}^n} f(y)\, P_{0,x}(t, dy)$. Under mild regularity on the transition probabilities — essentially that mass neither escapes to infinity in finite time nor moves a macroscopic distance instantaneously, and that $T_t f$ is continuous — the family $\{T_t\}$ is a contraction semigroup on $C_0(\mathbb{R}^n)$, the continuous functions vanishing at infinity, and its generator is the second-order elliptic operator

$$\mathcal{A}(f)(x) = \frac{1}{2}\sum_{i,j=1}^{n} \Sigma_{ij}(x)\frac{\partial^2 f}{\partial x_i \partial x_j} + \sum_{i=1}^{n} \mu_i(x)\frac{\partial f}{\partial x_i}. \tag{1.81}$$

For Brownian motion, where $\Sigma = \mathbb{I}_{n\times n}$ and $\mu = 0$, this is $\mathcal{A}f = \frac{1}{2}\Delta f$.

We shall not develop the semigroup picture further, having obtained from it what we needed: the identification of the generator, and the confirmation that the $\frac{1}{2}$ appearing in Itô's rule is the same $\frac{1}{2}$ appearing in $\frac{1}{2}\Delta$. It is worth recording, however, what the semigroup property actually encodes, because it is the first of the three approaches to fail in Part II. The identity $P_{t+s} = P_t \circ P_s$ is a restatement of the Markov property: the evolution over $t+s$ can be decomposed into evolution over s followed by evolution over t, because at the intermediate instant the present state is a sufficient statistic for the future. When the driving noise has memory — as fractional Brownian motion does — this decomposition is unavailable, the family (P_t) is no longer a semigroup, and the generator ceases to exist as a local operator. Neither the PDE approach nor the semigroup approach survives intact; only the pathwise, integral formulation does, which is a large part of why the theory of Part II is built the way it is.

2. Partial Differential Equations Approach

Let $P_{s,x}(t,\cdot)$ be the transition probabilities of a diffusion X_t driven by μ and Σ, and let $F_{s,x}$ be the corresponding distribution function. The argument is a Taylor expansion inside the Chapman–Kolmogorov equation

$$F_{s,x}(t,y) = \int_{\mathbb{R}^n} F_{u,z}(t,y)\, dF_{s,x}(u,z),$$

and runs as follows. Fix t and y, put $u = s+\varepsilon$, and expand the integrand to second order in z about x. The zeroth-order term gives $F_{s+\varepsilon,x}$; the first-order term contributes $\nabla_x F_{s+\varepsilon,x} \cdot \mu(s,x)\,\varepsilon$, since by Definition 1.2.39(ii) the mean displacement over $[s, s+\varepsilon]$ is $\mu(s,x)\varepsilon + o(\varepsilon)$; and the second-order term contributes $\frac{1}{2}\operatorname{Tr}\Big(H_x F_{s+\varepsilon,x}\,\Sigma(s,x)\Big)\varepsilon$ by Definition 1.2.39(iii). Rearranging, dividing by ε and letting $\varepsilon \to 0$ yields

$$-\frac{\partial F_{s,x}}{\partial s} = \sum_{i=1}^{n} \mu_i \frac{\partial F_{s,x}}{\partial x_i} + \frac{1}{2}\sum_{i,j=1}^{n} \Sigma_{ij}\frac{\partial^2 F_{s,x}}{\partial x_i \partial x_j}, \tag{1.82}$$

together with the terminal condition

$$\lim_{s\uparrow t} F_{s,x}(t,y) = \mathbb{1}_{\{x\leq y\}}. \tag{1.83}$$

Equation (1.82) is called the **Kolmogorov backward equation**, because the variable s is the starting time, effectively moving backwards from the terminal time t. However, we often aim for a formula that enables *forward* calculations. Directly linked to this equation, the Feynman-Kac formula states that the solution to the Kolmogorov backward equation can be represented as an expected value of a functional of the diffusion process X_t. This will be stated in the next section.

Notice that the Kolmogorov backward equation derived in Equation (1.82) is effectively a special case of the Feynman-Kac formula where the interest rate $r(t,x) = 0$ and the terminal condition is the indicator function $h(x) = \mathbb{1}_{\{x \leq y\}}$. In the context of quantitative finance, the Feynman-Kac theorem is a cornerstone result: it establishes the fundamental bridge between the risk-neutral pricing of a derivative (computed as a discounted expected payoff under a martingale measure) and the deterministic PDE that the derivative's price must satisfy to preclude arbitrage.
We shall stay in the one-dimensional case for simplicity now. Assume the transition probabilities of a Markov process X_t possess density functions, that is,

$$P_{s,x}(t, dy) = p_{s,x}(t,y)\, dy, \tag{1.84}$$

so that the Chapman-Kolmogorov equation takes the form

$$p_{s,x}(t,y) = \int_{\mathbb{R}} p_{u,z}(t,y) p_{s,x}(u,z)\, dz. \tag{1.85}$$

By using this identity and some auxiliary test functions (integration by parts), it can be proven that

$$\frac{\partial p_{s_0,x_0}(t,y)}{\partial t} = -\frac{\partial(\mu(t,y)p_{s_0,x_0}(t,y))}{\partial y} + \frac{1}{2}\frac{\partial^2(\Sigma(t,y)p_{s_0,x_0}(t,y))}{\partial y^2} \tag{1.86}$$

which, together with the initial condition

$$\lim_{t \downarrow s_0} p_{s_0,x_0}(t,y) = \delta_{x_0}(y) \tag{1.87}$$

is known as either the **Kolmogorov forward equation** or the **Fokker-Planck equation.** This equation is much more useful in a financial context because it allows calculations *forward* in time, i.e., evolving from known current market variables.

Theorem 1.2.42. *Let Σ and μ satisfy the linear growth and Lipschitz conditions in x. Assume, additionally, that there exists $c > 0$ such that $w^\top \Sigma(t,y) w \geq c\|w\|^2$ for all $w \in \mathbb{R}^n$ (uniform ellipticity). Then:*

a) The Kolmogorov backward equation has a unique solution. Moreover, there exists a continuous Markov process X_t whose transition probabilities satisfy $P_{s,x}(t_0, dy_0) = dF_{s,x}(t_0, y_0)$.

b) Assume additionally that the derivatives of μ and Σ up to second order satisfy the Lipschitz and linear growth conditions. Then the Kolmogorov forward equation has a unique solution.

■

In terms of the infinitesimal generator, we can summarize the previous discussion in the following table:

Forward Equation	**Backward Equation**
$\partial\rho/\partial t = \mathcal{A}^*\rho$	$-\partial u/\partial s = \mathcal{A}u$

where $\mathcal{A}^*$ represents the adjoint of the generator $\mathcal{A}$; see [39] for the functional-analytic details. The duality is worth reading off the table: the backward equation propagates a *payoff* backwards from maturity and is the natural object for pricing a single contract, whereas the forward equation propagates a *density* forwards from today and is the natural object for calibration, since it evolves from known current market data. Dupire's formula in Subsection 1.5.1 is precisely an application of the second.

3. Stochastic Differential Equations Approach

In this case, take μ, Σ and σ such that $\sigma\sigma^\top = \Sigma$. Notice that σU is also a valid solution for any matrix U such that $UU^\top = \mathbb{I}_{n\times n}$ (orthogonal matrix). Therefore, the choice of σ is unique up to multiplication by an orthogonal matrix. Since $UW(t)$ is also a Brownian motion, the transition probabilities of X_t do not depend on the particular choice of σ within its class. We call this class of matrices the *square root* of Σ.
Now, suppose Σ and μ satisfy the linear growth and Lipschitz conditions. It is clear that any square root of Σ satisfies the growth condition. Moreover, if Σ is uniformly elliptic (bounded away from zero), σ also satisfies the Lipschitz conditions. In the 1D case, this can be seen via the inequality:

$$|\sigma(t,x) - \sigma(t,y)| = \frac{|\Sigma(t,x) - \Sigma(t,y)|}{|\sigma(t,x)| + |\sigma(t,y)|} \leq \frac{|\Sigma(t,x) - \Sigma(t,y)|}{2\sqrt{c}}. \tag{1.88}$$

Having this, we can directly state the main theorem corresponding to the SDE approach.

Theorem 1.2.43. *The unique solution to*

$$dX_t = \mu(t,X_t)\,dt + \sigma(t,X_t)\,dW_t, \quad X_{t_0}\ \mathcal{F}_{t_0}\text{-measurable} \tag{1.89}$$

is a diffusion process, whose transition distribution function $F_{s,x}(t,y)$ satisfies the Kolmogorov backward equation. If the additional conditions on part b) of the previous Theorem are satisfied, then X_t has density functions $p_{s,x}(t,y)$, satisfying the corresponding Kolmogorov forward equation. ■

Example 1.2.44. We illustrate the previous discussion with the example that will occupy the rest of this chapter. Take $n = 1$ and consider geometric Brownian motion, $dX_t = rX_t\,dt + \sigma X_t\,dW_t$ on $(0,\infty)$, so that $\mu(x) = rx$ and $\Sigma(x) = \sigma^2x^2$. By the theorem above its generator is

$$\mathcal{A}f(x) = \tfrac{1}{2}\sigma^2x^2f''(x) + r\,xf'(x), \tag{1.90}$$

and the backward equation $-\partial u/\partial s = \mathcal{A}u$ is, after adding the discounting term $-ru$ supplied by Feynman–Kac, exactly the Black–Scholes equation (1.114) of Section 1.4. Its adjoint,

$$\mathcal{A}^*\rho(x) = \tfrac{1}{2}\frac{\partial^2}{\partial x^2}\Big(\sigma^2x^2\rho(x)\Big) - \frac{\partial}{\partial x}\Big(rx\,\rho(x)\Big), \tag{1.91}$$

governs the forward equation (1.86), and it is the object Dupire inverts in Subsection 1.5.1 to read a local volatility function off market data. The two equations of the table are thus not an abstraction: they are the pricing equation and the calibration equation of the entire second half of this chapter.

1.3 Financial Derivatives

In this section, we introduce the main ideas behind Mathematical Finance and the role played by Mathematics in this field. We first informally introduce the concept of *derivative* and give some examples of the main quoted derivatives. The rigorous mathematical treatment of hedging and market completeness is addressed afterwards.

1.3.1 The Concept of a Derivative

In the context of modern mathematical finance, a financial derivative is formally defined as a *contingent claim*. Its value is derived from—or "contingent" upon—the state of some other underlying variables.

Let us fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$, where $\mathbb{P}$ represents the physical (or real-world) probability measure, and the filtration $(\mathcal{F}_t)_{t\geq 0}$ models the flow of information over time up to a finite time horizon T. Mathematically, a financial derivative with maturity T is an $\mathcal{F}_T$-measurable random variable, denoted by H_T. There are certain derivatives, such as perpetual bonds, which have an infinite time horizon ($T = \infty$), but these fall outside the scope of our finite-horizon models, as they lack standard liquidity and do not constitute a core part of the derivatives market.

This measurability condition implies that the value of the derivative at maturity is fully determined by the information available at time T. The function determining this value is often referred to as the *payoff function*, denoted by Φ, so that $H_T = \Phi(S_T)$ for a claim written on the terminal spot. These two symbols — H_T for the random variable, Φ for the function — are the only ones we use for a payoff. The key idea behind the subsequent theory is being able to assign *fair prices* to those contingent claims, deduced from quoted (and liquid) market data. The prices of such claims are not independent of one another. A small set of standardised, highly liquid contracts carries enough information to determine the price of a great many bespoke ones, and the principle enforcing this consistency is the absence of **arbitrage**: informally, the impossibility of a strictly positive risk-free profit from zero net initial investment.

A single example fixes the idea. Let a non-dividend-paying stock trade at S_0 and let a forward contract on it have delivery price $K > S_0e^{rT}$, with r the continuous risk-free rate. Borrowing S_0, buying the stock and shorting the forward produces, at maturity, a delivery of the stock against K and a repayment of S_0e^{rT}, for a certain profit of $K - S_0e^{rT} > 0$. Consistency therefore forces $K = S_0e^{rT}$.

Real markets impose frictions — transaction costs, bid-ask spreads, asymmetric borrowing and lending rates, taxes — which widen this identity into a band rather than a point. But for the fungible assets and electronic markets we shall be concerned with, the band is tight, and it is precisely the narrowness of that band which makes the mathematics worth doing: not only to price consistently, but to quantify the residual risk and to construct the dynamic strategies that **hedge** it.

The Underlying Asset

The process driving the derivative's value is the *underlying asset*, an $(\mathcal{F}_t)$-adapted process $S = (S_t)_{0\leq t\leq T}$; a derivative may depend on several. In practice S_t is the price of an equity or index, a bond or interest rate (often modelled through a short-rate process r_t), a commodity such as oil, gold or electricity, or an exchange rate. Derivatives are then categorised by the functional form of their terminal payoff H_T.

1. Linear Derivatives (Forwards and Futures) These instruments possess a payoff that is a linear function of the underlying asset price at maturity. A prime example is the **Forward Contract**, which is an agreement to buy or sell the asset at time T for a pre-specified delivery price K. The payoff at time T for the holder of a long position is:

$$H_T = S_T - K \tag{1.92}$$

Note that this payoff ranges over $\mathbb{R}$, implying a potential liability for the holder if the asset price falls below K.

2. Non-Linear Derivatives (Options) Options provide the *right*, but not the *obligation*, to transact. This asymmetry introduces convexity (non-linearity) into the payoff structure, truncating downside risk.

- **European Call Option:** The right to buy S_T at strike price K. The payoff is defined as:
$$H_T = (S_T - K)^+ = \max(S_T - K, 0) \tag{1.93}$$
- **European Put Option:** The right to sell S_T at strike price K. The payoff is defined as:
$$H_T = (K - S_T)^+ = \max(K - S_T, 0) \tag{1.94}$$
- **American Options:** American call or put options are similar to their European counterparts, with the key difference being that the option can be *exercised* at any time up to maturity T.

From a theoretical standpoint, derivatives serve two primary functions in the market:

1. **Hedging:** They allow agents to transfer specific risks (e.g., volatility or tail risk) to those willing to bear them.
2. **Market Completeness:** In an idealized complete market, derivatives allow for the replication of any payoff profile, effectively spanning the state space of future uncertainties.

These concepts will be rigorously defined in the next subsection. However, the main question remains: what price should these contracts hold? Before entering the mathematical machinery, we can establish bounds based on physical intuition, much like conservation laws:

- If $T = 0$, meaning we are pricing a claim that expires immediately (and only depends on the underlying at maturity, not on its trajectory), the price must equal the payoff. For a European Call Option with strike K and underlying S_0, its price must be $(S_0 - K)^+$.
- If an asset gives its buyer strictly more optionality compared to another derivative, its price must be greater than or equal to the latter. For example, the price of an American Call Option must be bounded below by the price of a European Call Option with same strike and maturity.

1.3.2 Derivative Pricing

We now turn to the rigorous problem of pricing contingent claims. While elementary treatments often assume asset prices follow geometric Brownian motions and interest rates are constant, these assumptions are insufficient for a general mathematical theory.
We work in a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \le t \le T}, \mathbb{P})$ satisfying the usual conditions (right-continuity and completeness). The time horizon T is finite.
Let the market consist of $n + 1$ assets denoted by the vector process $S = (S^0, S^1, \ldots, S^n)$. We assume that each component S^i is an **Itô process**. While the most abstract fundamental theorems of asset pricing rely on the broader class of general semimartingales, Itô processes provide the precise mathematical framework required for the diffusion models prevalent in this thesis, seamlessly connecting with the stochastic calculus developed in Section 1.2.
We distinguish the 0-th asset, S^0, as the **numeraire**.

Definition 1.3.1 (Numeraire)**.** A numeraire is a price process S^0 that is strictly positive ($S_t^0 > 0$ almost surely for all t) and non-dividend paying. It serves as the unit of account for the market.

While S^0 is often chosen to be a bank account $B_t = \exp(\int_0^t r_s ds)$, the theory holds for *any* valid Itô process numeraire (e.g., a foreign currency or a stock). The bank account is merely a convenient particular case because it explicitly models the time value of money.

We define the **discounted price process** X as the prices of the primary assets expressed in units of the numeraire:

$$X_t = \left(1, \frac{S_t^1}{S_t^0}, \ldots, \frac{S_t^n}{S_t^0}\right) \tag{1.95}$$

Because the denominator S_t^0 is strictly positive almost surely, the transformation $f(x, y) = x/y$ is smooth. By a direct application of the multidimensional Itô's formula, the discounted price processes X_t^i are themselves Itô processes.

A trading strategy is a predictable process $H = (H^0, H^1, \ldots, H^n)$, where H_t^i represents the number of units of asset i held at time t. The value of the portfolio at time t is the scalar process:

$$V_t(H) = \sum_{i=0}^{n} H_t^i S_t^i \tag{1.96}$$

For the strategy to be economically meaningful, it must be **self-financing**. In the language of stochastic integration, this implies that changes in wealth are generated solely by the fluctuations of the asset prices, with no external injection or extraction of capital:

$$dV_t(H) = \sum_{i=0}^{n} H_t^i \, dS_t^i \tag{1.97}$$

When working with discounted prices X, the self-financing condition simplifies. Applying Itô's product rule, it can be shown that a strategy is self-financing if and only if the discounted wealth process V_t/S_t^0 satisfies:

$$\frac{V_t(H)}{S_t^0} = \frac{V_0(H)}{S_0^0} + \int_0^t \sum_{i=1}^{n} H_u^i \, dX_u^i = \frac{V_0}{S_0^0} + (H \cdot X)_t \tag{1.98}$$

where $(H \cdot X)$ denotes the vector stochastic integral.

In continuous time, the standard definition of arbitrage is insufficient because "doubling strategies" (integrands with extremely large negative values) can technically generate profit from nothing in local time intervals. To preclude this, we restrict ourselves to certain strategies.

Definition 1.3.2 (Admissible Strategy)**.** A self-financing strategy H is called a-admissible if its discounted value process is uniformly bounded from below by a constant $-a$ almost surely (i.e., $(H \cdot X)_t \geq -a$ for all t). A strategy is admissible if it is a-admissible for some $a \geq 0$.

The classical definition of "No Arbitrage" fails to be closed in the topology of L^∞. Delbaen and Schachermayer [18] proved that the appropriate condition for the existence of an equivalent martingale measure is *No Free Lunch with Vanishing Risk*.

Let $\mathcal{K}$ be the set of random variables representing the terminal wealth of all admissible strategies starting with zero capital — the calligraphic letter is deliberate, K being reserved throughout for the strike:

$$\mathcal{K} = \{(H \cdot X)_T \mid H \text{ is admissible and } V_0(H) = 0\} \tag{1.99}$$

Let $\mathcal{C}$ be the cone of claims dominated by elements of $\mathcal{K}$, that is, $\mathcal{C} = \{g \in L^\infty(\mathcal{F}_T) \mid \exists f \in \mathcal{K},\ g \leq f\}$. The market satisfies **NFLVR** if:

$$\overline{\mathcal{C}} \cap L_+^\infty = \{0\} \tag{1.100}$$

where $\overline{\mathcal{C}}$ is the closure of $\mathcal{C}$ in the L^∞ norm. Intuitively, this condition prevents not just exact arbitrage, but also sequences of strategies that approximate an arbitrage opportunity with risk converging to zero.

We are now equipped to state the core result linking market topology to measure theory.

Definition 1.3.3 (Equivalent Local Martingale Measure)**.** A probability measure $\mathbb{Q}$ is an Equivalent Local Martingale Measure (ELMM) for the numeraire S^0 if:

1. $\mathbb{Q} \sim \mathbb{P}$ (they share the same null sets).
2. The discounted price process $X_t = S_t/S_t^0$ is a **local martingale** under $\mathbb{Q}$.

Theorem 1.3.4 (First Fundamental Theorem of Asset Pricing)**.** *Assume the process X is locally bounded. The market satisfies the NFLVR condition if and only if there exists at least one Equivalent Local Martingale Measure $\mathbb{Q}$.* ■

The other reasonable question to ask ourselves is whether every contingent claim can be replicated by a self-financing strategy. Before stating the second theorem, we must rigorously define market completeness.

Definition 1.3.5 (Market Completeness)**.** A market is **complete** if every contingent claim $H_T \in L^\infty(\mathcal{F}_T)$ is *attainable*. That is, for every bounded random variable H_T, there exists an admissible self-financing strategy H and an initial capital $x \in \mathbb{R}$ such that:

$$\frac{H_T}{S_T^0} = x + (H \cdot X)_T \quad \mathbb{P}\text{-almost surely.} \tag{1.101}$$

Note that the statement is made under the physical measure $\mathbb{P}$. This is deliberate: at this stage $\mathbb{Q}$ has not been shown to be unique — indeed its uniqueness is exactly what the next theorem characterises — so quantifying over "$\mathbb{Q}$-almost surely" would be ambiguous. Since every equivalent local martingale measure shares its null sets with $\mathbb{P}$, the choice is harmless, but it is the honest one.

Theorem 1.3.6 (Second Fundamental Theorem of Asset Pricing)**.** *Assume that the set of equivalent local martingale measures $\mathcal{M}^e$ is non-empty (i.e., the market is arbitrage-free). The market is complete if and only if $\mathcal{M}^e$ is a singleton set (i.e., $\mathbb{Q}$ is unique).* ■

If the market is arbitrage-free and complete (i.e., there exists a unique $\mathbb{Q}$), the fair price at time t of a generic derivative paying H_T at maturity is given by the expectation of the discounted payoff:

$$\frac{V_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}}\left[\frac{H_T}{S_T^0}\middle|\mathcal{F}_t\right] \implies V_t = S_t^0\,\mathbb{E}^{\mathbb{Q}}\left[\frac{H_T}{S_T^0}\middle|\mathcal{F}_t\right] \tag{1.102}$$

This formulation reveals that "risk-neutral probabilities" are not absolute; they are linked to the choice of numeraire.

Let N be a different numeraire (strictly positive semimartingale). Let $\mathbb{Q}^S$ be the measure associated with numeraire S^0, and $\mathbb{Q}^N$ be the measure associated with numeraire N. Prices must be consistent regardless of the unit of account. Therefore, for any asset Z:

$$\frac{Z_t}{S_t^0} = \mathbb{E}^{\mathbb{Q}^S}\left[\frac{Z_T}{S_T^0} \mid \mathcal{F}_t\right] \quad \text{and} \quad \frac{Z_t}{N_t} = \mathbb{E}^{\mathbb{Q}^N}\left[\frac{Z_T}{N_T} \mid \mathcal{F}_t\right] \tag{1.103}$$

The change of measure from $\mathbb{Q}^S$ to $\mathbb{Q}^N$ is defined by the Radon-Nikodym derivative, which turns out to be exactly the normalized ratio of the numeraires:

$$\left.\frac{d\mathbb{Q}^N}{d\mathbb{Q}^S}\right|_{\mathcal{F}_T} = \frac{N_T/S_T^0}{N_0/S_0^0} \tag{1.104}$$

This elegantly shows that changing the numeraire is mathematically equivalent to performing a change of probability measure. This technique is particularly powerful in interest rate theory (e.g., switching to the T-forward measure to efficiently price bond options and swaptions).

Finally, we can now state the Feynman-Kac formula, which provides a powerful link between the PDE approach and the probabilistic representation of derivative prices.

Theorem 1.3.7 (Feynman–Kac Formula)**.** *Let X be an n-dimensional Itô diffusion process on the interval $[t, T]$ driven, under a measure $\mathbb{Q}$, by the stochastic differential equation*

$$dX_s = \mu(s, X_s)\, ds + \sigma(s, X_s)\, dW_s, \quad s \in [t, T], \tag{1.105}$$

with initial condition $X_t = x$, where W is a standard n-dimensional $\mathbb{Q}$-Brownian motion, and the diffusion matrix is given by $\Sigma(s, x) = \sigma(s, x)\sigma(s, x)^\top$. Let $h : \mathbb{R}^n \to \mathbb{R}$ be a Borel-measurable terminal payoff function, and $r : [0, T] \times \mathbb{R}^n \to \mathbb{R}$ be a continuous function representing the continuously compounded interest rate. (The time variable is written s, and not τ as would be natural, because τ is reserved for stopping times and for the time to maturity $T - t$.)
Assume $u(t, x)$ is a sufficiently regular function (typically $u \in C^{1,2}([0, T) \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$) that satisfies the following linear parabolic partial differential equation:

$$\frac{\partial u}{\partial t}(t, x) + \sum_{i=1}^{n} \mu_i(t, x) \frac{\partial u}{\partial x_i}(t, x) + \frac{1}{2} \sum_{i,j=1}^{n} \Sigma_{ij}(t, x) \frac{\partial^2 u}{\partial x_i \partial x_j}(t, x) - r(t, x)u(t, x) = 0, \tag{1.106}$$

for all $(t, x) \in [0, T) \times \mathbb{R}^n$, subject to the terminal condition:

$$u(T, x) = h(x). \tag{1.107}$$

If $u(t, x)$ satisfies a suitable polynomial growth condition preventing the existence of explosive solutions, then $u(t, x)$ admits the stochastic representation:

$$u(t, x) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s, X_s) ds} h(X_T) \middle| X_t = x \right]. \tag{1.108}$$

■

The subscript on the expectation is not decoration. Equation (1.108) is an expectation under *the measure making X solve the stated equation* — that is, the measure under which the drift is μ — and nothing in the theorem says which measure that is. When the theorem is applied to option pricing in Section 1.4, the drift supplied is rS rather than μS, so the measure in question is the risk-neutral $\mathbb{Q}$ and emphatically not the physical $\mathbb{P}$; writing a bare $\mathbb{E}$ there would silently conflate the two measures that the whole of Subsection 1.4.1 is at pains to keep apart. We therefore name it here.

1.4 The Black-Scholes Framework

Having established the general theory of arbitrage-free pricing and martingale measures, we now specialize our market model to the most famous and widely used framework in financial history: the Black-Scholes-Merton model.

1.4.1 The Model

We assume a frictionless market: no transaction costs, no bid-ask spread, unrestricted short selling, and a constant risk-free rate r at which one may borrow and lend freely. Under the *physical* measure $\mathbb{P}$ — the measure describing what actually happens — the underlying is modelled by a geometric Brownian motion,

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t, \qquad S_0 > 0, \tag{1.109}$$

where $\mu \in \mathbb{R}$ is the expected rate of return, $\sigma > 0$ the volatility, and W a $\mathbb{P}$-Brownian motion. By Example 1.2.14 the solution is $S_t = S_0 \exp\!\big((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\big)$, which is positive for all t — as a limited-liability equity should be.

It is essential to begin under $\mathbb{P}$ and not under a risk-neutral measure. Doing so makes the central result of the theory visible rather than assumed, and that result is the following: **the price of the derivative does not depend on μ.** Two investors who disagree completely about the expected return of the stock, but agree on its volatility, must agree on the price of every option written on it. This is not intuitive, it is not a modelling convention, and it is worth seeing emerge from the computation rather than being handed the risk-neutral dynamics with μ already replaced by r.

The hedging argument

Consider a derivative whose value is a smooth function $V(t, S)$ of time and the current spot. Form the portfolio

$$\Pi_t = -V(t, S_t) + \Delta_t S_t,$$

consisting of a short position in the derivative and Δ_t units of the underlying. Assuming the position is self-financing in the sense of (1.97), its value evolves as

$$d\Pi_t = -dV_t + \Delta_t \, dS_t. \tag{1.110}$$

Itô's rule (Theorem 1.2.12) applied to $V(t, S_t)$, using $(dS_t)^2 = \sigma^2 S_t^2 \, dt$ from Table 1.1, gives

$$dV_t = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} \, dW_t. \tag{1.111}$$

Substituting (1.109) and (1.111) into (1.110) and collecting terms,

$$d\Pi_t = \underbrace{\left(-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \mu S_t \Big[\Delta_t - \frac{\partial V}{\partial S} \Big] \right)}_{\text{drift}} dt + \underbrace{\sigma S_t \left(\Delta_t - \frac{\partial V}{\partial S} \right)}_{\text{diffusion}} dW_t. \tag{1.112}$$

The portfolio is exposed to randomness only through the second term. Choosing

$$\Delta_t = \frac{\partial V}{\partial S}(t, S_t) \tag{1.113}$$

annihilates it. This is the *delta* of the derivative, and (1.113) is the hedge.

Now observe what (1.112) has become. The same choice (1.113) that kills the diffusion term also kills the bracket multiplying μ in the drift. **The expected return of the stock has cancelled**, and it has done so for a structural reason rather than by accident: μ enters dS_t and dV_t through the identical channel, so a position that is neutral to the random part of S is automatically neutral to its drift as well. What remains,

$$d\Pi_t = \left(-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt,$$

is riskless. A riskless self-financing portfolio must earn the risk-free rate — otherwise, by borrowing or lending against it, one manufactures an arbitrage in the sense of Subsection 1.3.2 — so $d\Pi_t = r\Pi_t \, dt$ with $\Pi_t = -V + S_t \partial V / \partial S$. Equating the two expressions and rearranging yields the Black–Scholes equation.

Theorem 1.4.1 (Black–Scholes–Merton)**.** *Under the dynamics* (1.109)*, the value $V(t, S)$ of a European derivative with payoff $V(T, S) = \Phi(S)$ satisfies the linear parabolic partial differential equation*

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0, \qquad (t, S) \in [0, T) \times (0, \infty), \tag{1.114}$$

together with the terminal condition $V(T, S) = \Phi(S)$. ■

Equation (1.114) contains r but not μ, which is the claim made above, now proved. It is also exactly the Feynman–Kac equation of Theorem 1.3.7 for a diffusion with drift rS and diffusion coefficient σS, whence the probabilistic representation

$$V(t,S) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}\Big[\Phi(S_T)\,\Big|\,S_t = S\Big], \qquad dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}. \tag{1.115}$$

The passage from (1.109) to the dynamics in (1.115) is precisely Girsanov's theorem with $\alpha = (r-\mu)/\sigma$, the *market price of risk*: the measure change subtracts a drift and leaves σ untouched, exactly as discussed after Theorem 1.2.35. That the hedging argument and the martingale argument give the same answer is not a coincidence but the content of Feynman–Kac, and the two viewpoints will alternate throughout what follows — the PDE when we want to compute, the expectation when we want to reason.

Finally, the market is complete: by Theorem 1.2.37 and Remark 1.2.38, the filtration is generated by the single Brownian motion driving the single traded asset, so every square-integrable claim is attainable and $\mathbb{Q}$ is unique. The delta (1.113) is the integrand φ whose abstract existence the representation theorem guarantees; here we have it explicitly.

The closed-form solution

For a European call, $\Phi(S) = (S-K)^+$, evaluating (1.115) against the log-normal density gives

$$C_{BS}(t,S) = S\,N(d_+) - Ke^{-r(T-t)}N(d_-), \qquad d_\pm = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \tag{1.116}$$

with $\tau = T-t$ and N the standard normal distribution function. The corresponding put price follows from put–call parity, Proposition 1.4.5 below.

One might instead assume **normal** (Bachelier) dynamics $dS_t = \mu\,dt + \sigma\,dW_t$, as in the original 1900 thesis [1]. This permits negative prices, which is why it was criticised for equities, but it is standard for interest rates and for spread options where negative values are economically meaningful. For equities the log-normal convention prevails, on account of the limited liability of shareholders.

The Greeks

The delta (1.113) was singled out above as the hedge, and identified in Remark 1.2.38 with the integrand φ whose existence the Martingale Representation Theorem guarantees. It is not the only sensitivity that matters, and since the whole of Subsection 1.4.2 is an argument about one of the others, we record them here. Write $\tau = T-t$ and let $N'(x) = e^{-x^2/2}/\sqrt{2\pi}$ be the standard normal density.

Proposition 1.4.2 (Black–Scholes sensitivities)**.** *For the European call* (1.116) *on a non-dividend-paying underlying,*

$$\Delta := \frac{\partial C}{\partial S} = N(d_+), \qquad \Gamma := \frac{\partial^2 C}{\partial S^2} = \frac{N'(d_+)}{S\,\sigma\sqrt{\tau}}, \tag{1.117}$$

$$\mathcal{V} := \frac{\partial C}{\partial \sigma} = S\,N'(d_+)\sqrt{\tau}, \qquad \vartheta := \frac{\partial C}{\partial t} = -\frac{S\,N'(d_+)\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_-). \tag{1.118}$$

■

The names are delta, gamma, vega and theta respectively. Two notational cautions: vega is not a Greek letter and has no standard symbol, so we write $\mathcal{V}$; and theta is written ϑ rather than Θ, the capital being reserved throughout for the spot-to-forward conversion factor Θ of (1.125) and for nothing else.

Two identities among these are worth more than the formulae themselves. The first is that Γ and $\mathcal{V}$ are proportional,

$$\mathcal{V} \;=\; S^2\,\sigma\,\tau\,\Gamma, \tag{1.119}$$

as one verifies immediately from (1.117) and (1.118). Convexity in the spot and sensitivity to volatility are therefore the *same* exposure, measured in two units — which is the analytic content of the informal statement that an option buyer is long volatility precisely because the payoff is convex, and the reason Proposition 1.1.4, item 8, has been such a persistent theme.

The second is that substituting (1.117) into the Black–Scholes equation (1.114) rearranges it into

$$\vartheta + \tfrac{1}{2}\sigma^2 S^2\,\Gamma \;=\; r\Big(C - S\,\Delta\Big), \tag{1.120}$$

which says that for a delta-hedged position (Δ neutralised, so that the right-hand side is just the financing of the residual cash) the time decay ϑ and the convexity Γ are in exact opposition. This is the whole economics of running an option book in one line: the gamma that makes the position profit from movement is paid for, day by day, by theta, and (1.120) fixes the exchange rate between them. Equation (1.114) is usually read as a pricing equation; (1.120) is the same equation read as a statement about a hedged portfolio, and it is the form a trader would recognise.

Vega is the one to keep in view. It is strictly positive, so a call is worth more when volatility is higher; and it is *not* a sensitivity to a quantity the model admits to be uncertain — in (1.109) the parameter σ is a constant. Differentiating a price with respect to a constant is a symptom that the model is being used outside its own assumptions, and the entire subject of Section 1.5 is the attempt to build models in which the quantity vega measures is a genuine state variable rather than a parameter one perturbs by hand. Subsection 1.4.2 is where that symptom becomes impossible to ignore.

Dividends, and why the forward is the right variable

The derivation above quietly assumed that holding the stock produces no cash flows. Real equities pay dividends, and the way they do so is awkward enough to force a change of variable that we shall keep for the rest of the dissertation.

Suppose first that dividends are paid *continuously* at a constant proportional rate q. The holder of the stock receives $qS_t\,dt$ over $[t, t+dt]$, so the self-financing condition (1.110) acquires the extra term $\Delta_t qS_t\,dt$; repeating the argument verbatim replaces the coefficient r of $\partial V/\partial S$ in (1.114) by $r-q$, and r by $r-q$ in $d_\pm$. Nothing structural changes — but the closed form does not follow from those two substitutions alone, and it is worth writing out, since applying them mechanically to (1.116) gives the wrong price. The discounted-spot term acquires its own factor:

$$C(S,\tau) = Se^{-q\tau}N(d_+) - Ke^{-r\tau}N(d_-), \qquad d_\pm = \frac{\log(S/K) + (r - q \pm \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}. \tag{1.121}$$

The reason is visible in the Black-76 form (1.127): what enters the formula is the forward $F = Se^{(r-q)\tau}$, and substituting it into $e^{-r\tau}[FN(d_+) - KN(d_-)]$ produces $e^{-q\tau}$ on the spot term automatically. This is a further argument for treating (1.127) rather than (1.116) as the primitive object: the dividend correction is invisible there because the forward has already absorbed it. This is the case usually presented, and it is the case that does not occur.

Actual equity dividends are *discrete cash amounts*, announced in advance and paid on known dates. If an amount D_i is paid at time t_i, then no-arbitrage forces the spot to drop by that amount across the payment:

$$S_{t_i} = S_{t_i^-} - D_i. \tag{1.122}$$

This is genuinely destructive of the framework above, and it is worth being precise about why. Equation (1.122) says S has a *predictable jump*: the process is not continuous, so Itô's rule as stated does not apply across t_i. Worse, the jump is *additive* while geometric Brownian motion is multiplicative, so the two are structurally incompatible — one cannot write $S_t = S_0\mathcal{E}(\cdot)$ for a positive $\mathcal{E}$ and reproduce (1.122). Worse still, a fixed cash dividend can in principle exceed the spot price, so a model treating D_i as deterministic does not even guarantee positivity. And since σ multiplies S_t in (1.109), a discontinuous drop in S produces a discontinuous drop in the absolute volatility, which is not what is observed.
The resolution is to stop modelling the spot and model the *forward* instead.

Definition 1.4.3 (Forward price)**.** The **forward price** $F(t,T)$ is the delivery price agreed at t for a transaction in the underlying at T which has zero value at inception. Equivalently, writing $P(t,T)$ for the price at t of a zero-coupon bond maturing at T, a forward contract struck at K has time-t value

$$V_t(K) = P(t,T)\,\mathbb{E}^{\mathbb{Q}^T}\Big[S_T - K \mid \mathcal{F}_t\Big], \tag{1.123}$$

where $\mathbb{Q}^T$ is the T-forward measure associated with the numeraire $P(\cdot,T)$. Since $P(t,T) > 0$, the requirement that the contract be costless at inception, $V_t(F(t,T)) = 0$, determines the delivery price uniquely:

$$F(t,T) = \mathbb{E}^{\mathbb{Q}^T}\Big[S_T \mid \mathcal{F}_t\Big]. \tag{1.124}$$

Identity (1.124) says that the forward price is a $\mathbb{Q}^T$-martingale — it is a conditional expectation — and this is the whole point. It holds by construction, with no assumption whatever on interest rates, on dividends, or on their nature. In particular we deliberately do *not* write $F(t,T) = S_t e^{(r-q)(T-t)}$, which is valid only for constant r, constant q, and purely proportional dividends. Instead we record the relation abstractly:

$$S_t = \Theta_t\,F(t,T), \tag{1.125}$$

where Θ_t is a strictly positive, adapted, model-independent conversion factor assembled from the discount curve, the repo or borrow cost, and the announced cash and proportional dividend schedule. Its precise construction is a market-data question, not a modelling one, and it is addressed in Chapter 3. What matters here is that (1.125) remains valid for time-dependent $r(t)$ and $q(t)$, for discrete cash dividends, for proportional dividends, and for any mixture of them.
Modelling the forward as a driftless log-normal martingale,

$$dF_t = \sigma F_t\,dW_t^T, \tag{1.126}$$

therefore has three advantages simultaneously. The dividend discontinuities are absorbed into Θ_t and never touch the dynamics, which stay diffusive. The drift vanishes, because a forward is already a martingale under its own measure — so there is no drift to specify and none to get wrong. And the resulting formula is calibrated directly against quantities the market quotes.

The Forward Measure and Black-76

Changing numeraire from the bank account to the zero-coupon bond $P(\cdot,T)$, in the manner of (1.104), converts (1.115) into an expectation of the undiscounted payoff under $\mathbb{Q}^T$, multiplied by a deterministic discount factor. With the dynamics (1.126) this gives Black's 1976 formula.

Proposition 1.4.4 (Black-76 [6]). *Under (1.126), the time-t value of a European call with strike K and maturity T is*

$$C(t) = P(t,T)\Big[F_t\, N(d_+) - K\, N(d_-)\Big], \qquad d_\pm = \frac{\ln(F_t/K) \pm \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad \tau = T - t. \tag{1.127}$$

■

Formula (1.127) is formally identical to (1.116) with S replaced by F_t and the interest rate removed from $d_\pm$ — but note that the discount factor $P(t,T)$ remains, multiplying the whole bracket. It has not disappeared; it has merely been collected outside. We shall use (1.127) as the primitive throughout, treating (1.116) as the special case obtained when $P(t,T) = e^{-r\tau}$ and $F_t = S_t e^{r\tau}$.

Put–call parity

Before turning to implied volatility we record the one relation in this subject that holds independently of any model, and which we shall invoke repeatedly.

Proposition 1.4.5 (Put–call parity). *Let $C(t)$ and $P(t)$ be the time-t prices of European call and put options on the same underlying, with the same strike K and maturity T. Then, in any arbitrage-free market,*

$$C(t) - P(t) = P(t,T)\Big(F(t,T) - K\Big). \tag{1.128}$$

Proof: The payoffs satisfy the pointwise identity $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, valid for every value of S_T. Taking $\mathbb{E}^{\mathbb{Q}^T}[\,\cdot \mid \mathcal{F}_t]$ and multiplying by $P(t,T)$, the right-hand side becomes $P(t,T)(F(t,T) - K)$ by (1.124). ■

The proof used no dynamics — not (1.126), not even continuity of paths — only the linearity of expectation and an algebraic identity between payoffs. Consequently (1.128) must hold in every model in this dissertation, and it is the standard sanity check on any implementation: a pricer that violates put–call parity is wrong, and no appeal to model choice can rescue it.

Two consequences are worth noting now. First, since $C - P$ is model-free, calls and puts of the same strike carry *identical* information about volatility; a call and a put struck at K must therefore have the same implied volatility, and the smile is a single function of strike rather than two. Second, (1.128) is how the forward is extracted from quoted option prices in practice: the call-minus-put spread is linear in K with slope $-P(t,T)$ and root $K = F(t,T)$, so a regression across strikes recovers both the discount factor and the forward without any assumption about dividends. This is exactly the construction of Θ_t in (1.125), obtained from market data rather than postulated.

The Inversion Problem and Implied Volatility

In professional markets, participants do not "price" options using Black-Scholes to find the dollar value. The market price C^{mkt} is determined by supply and demand.

Instead, Black-Scholes is used as a translation mechanism. Since the pricing function $C_{BS}(\sigma)$ is strictly monotonic in σ, we can invert it to find the unique parameter σ_{imp} that matches the market price:

$$C_{BS}(S, K, T, r, q, \sigma_{\text{imp}}) = C^{mkt} \tag{1.129}$$

This implied volatility is the true "price" quoted in the market. It represents the market's consensus on the average future volatility of the underlying asset over the life of the option.

1.4.2 Smile and Skewness

If the assumptions of Black-Scholes were satisfied physically (i.e., if returns were truly log-normal with constant variance), the implied volatility σ_{imp} would be constant across all strike prices K and maturities T.
Empirical data reveals this is not the case. When we plot $\sigma_{\text{imp}}(K)$ against K, we observe distinct shapes:

- **The Skew (Smirk):** Typical in equity markets. σ_{imp} decreases as K increases. This reflects "Crash Phobia"—investors bid up the price of low-strike Puts (insurance against market drops), implying higher volatility for downside strikes.

- **The Smile:** Typical in Forex markets. σ_{imp} is lowest at the money ($K \approx S$) and increases for both deep ITM and OTM options. This reflects "Fat Tails" (leptokurtosis) in the return distribution—extreme moves are more probable than the Gaussian model predicts.

The two shapes are not merely descriptive. Both say that the market's risk-neutral distribution is not log-normal, and the following classical result makes the connection exact.

Proposition 1.4.6 (Breeden–Litzenberger [9])**.** *Let $C(K,T)$ denote the price of a European call of strike K and maturity T, and suppose the risk-neutral law of S_T admits a density ψ_T. Then*

$$\psi_T(K) = \frac{1}{P(0,T)} \frac{\partial^2 C}{\partial K^2}(K,T). \tag{1.130}$$

Proof: Write $C(K,T) = P(0,T) \int_K^\infty (x-K)\psi_T(x)\,dx$. Differentiating once under the integral sign, the boundary term vanishes because the integrand is zero at $x = K$, leaving $\partial_K C = -P(0,T) \int_K^\infty \psi_T(x)\,dx$. Differentiating a second time gives (1.130). ∎
The result is remarkable for what it does not assume: no model, no dynamics, only that prices are discounted expectations of the payoff. A continuum of option quotes at a fixed maturity therefore determines the entire marginal distribution of S_T — the market does not merely express an opinion about volatility, it expresses a complete probability law, and the smile is the visible trace of that law failing to be log-normal.
Two consequences of (1.130) will be used constantly. First, since $\psi_T \geq 0$, arbitrage-free call prices must be *convex* in strike; a violation is a butterfly arbitrage, executable by buying two wing options and selling two at the middle strike. Second, the formula involves a *second* derivative of market data, which is numerically delicate in a way that Subsection 1.5.1 will have to confront directly.

[FIGURE HERE] Empirical implied volatility smile for a liquid index at two maturities, showing the flattening of the skew with increasing T. To be produced from the market data of Chapter 3.

1.5 Other Important Models

To account for the volatility smile and skew, we must extend the Black-Scholes framework. There are two primary approaches: making volatility a deterministic function of the state (Local Volatility) or making it a stochastic process (Stochastic Volatility).

1.5.1 Local Volatility Models

The Local Volatility (LV) framework, introduced by Dupire [19], posits that volatility is a deterministic function of the current asset price and time: $\sigma_{\text{loc}} = \sigma_{\text{loc}}(S_t, t)$. The dynamics under the risk-neutral measure are:

$$dS_t = (r - q)\, S_t\, dt + \sigma_{\text{loc}}(S_t, t)\, S_t\, dW_t \tag{1.131}$$

This model preserves the completeness of the market (there is only one source of randomness W_t) while allowing the model to fit *any* observed arbitrage-free implied volatility surface exactly.

The Dupire Formula and Gatheral's Transformation

Dupire's insight was to link the unknown function $\sigma_{\text{loc}}(K, T)$ directly to observed market option prices. Applying the Fokker–Planck equation (1.86) to the transition density of the diffusion and combining it with Breeden–Litzenberger yields the following.

Theorem 1.5.1 (Dupire)**.** *Suppose call prices $C(K, T)$ are known for a continuum of strikes and maturities and are consistent with a diffusion of the form $dS_t = (r - q)S_t\, dt + \sigma_{\text{loc}}(S_t, t)S_t\, dW_t$. Then the local volatility function is uniquely determined by*

$$\sigma^2_{\text{loc}}(K, T) = \frac{\dfrac{\partial C}{\partial T} + (r - q)\, K \dfrac{\partial C}{\partial K} + q\, C}{\frac{1}{2} K^2 \dfrac{\partial^2 C}{\partial K^2}}. \tag{1.132}$$

■

The content of (1.132) is striking and deserves to be stated plainly: *there is exactly one* local volatility diffusion consistent with a given arbitrage-free surface. The calibration problem has a unique solution, not a family of them, and the model can therefore reproduce every European quote simultaneously and exactly. Note also that the denominator is proportional to the risk-neutral density by Proposition 1.4.6, so (1.132) is well posed precisely when that density is strictly positive.

Static arbitrage and the implied-variance parameterisation

Formula (1.132) is, however, close to unusable as written. It requires one derivative in maturity and two in strike of raw market data, and market data are quoted on a sparse grid, carry bid-ask spreads, and are contaminated by microstructure noise. Numerical differentiation amplifies all three, and the practical symptom is a negative denominator — that is, a negative local variance, which is meaningless.

The remedy is not better differentiation but a change of variables, together with an interpolation that is guaranteed arbitrage-free by construction. Define the **log-moneyness** and the **total implied variance**

$$k := \log \frac{K}{F(t, T)}, \qquad w(k, T) := \sigma^2_{\text{imp}}(k, T)\, T. \tag{1.133}$$

Total variance is the natural coordinate because it, rather than implied volatility itself, is what accumulates linearly in time under Black–Scholes. We now state the conditions under which a surface w is free of *static* arbitrage — that is, admits no arbitrage exploitable by a buy-and-hold portfolio of European options, with no trading in between.

Proposition 1.5.2 (Absence of static arbitrage)**.** *A total implied variance surface $w(k, T)$ is free of static arbitrage if the following hold.*

*i) (**Calendar spread**) For each fixed k, the map $T \mapsto w(k,T)$ is non-decreasing:*

$$\frac{\partial w}{\partial T}(k,T) \geq 0. \tag{1.134}$$

*ii) (**Butterfly**) For each fixed T, the function*

$$g(k) := \left(1 - \frac{k\,\partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial^2_{kk} w}{2} \tag{1.135}$$

satisfies $g(k) \geq 0$ for all k.

*iii) (**Wings**) $w(k,T)$ grows at most linearly in $|k|$ as $|k| \to \infty$; more precisely, the asymptotic slopes are bounded by 2, which is Lee's moment formula [41].*

■

Each condition has a direct financial meaning. Condition i) says a longer-dated option cannot be cheaper than a shorter-dated one of the same moneyness — otherwise one buys the long and sells the short. Condition ii) is convexity of the call price in strike, expressed in the new coordinates, and its violation is a butterfly spread with negative cost and non-negative payoff. Condition iii) prevents the extrapolated wings from implying infinite moments of S_T.
In practice one first fits a parametric form guaranteeing i)–iii) — Gatheral's SVI parameterisation [28] being the standard choice, with SABR common in rates markets — and only then differentiates. Since the fitted surface is smooth and arbitrage-free by construction, the differentiation is stable and the local variance is automatically non-negative. This yields Gatheral's form of the Dupire formula, which is the version actually implemented.

Theorem 1.5.3 (Dupire's formula in implied-variance coordinates)**.** *With k and w as in (1.133),*

$$\sigma^2_{\text{loc}}(k,T) = \frac{\dfrac{\partial w}{\partial T}}{1 - \dfrac{k}{w}\dfrac{\partial w}{\partial k} + \dfrac{1}{4}\left(-\dfrac{1}{4} - \dfrac{1}{w} + \dfrac{k^2}{w^2}\right)\left(\dfrac{\partial w}{\partial k}\right)^2 + \dfrac{1}{2}\dfrac{\partial^2 w}{\partial k^2}}. \tag{1.136}$$

■

Formula (1.136) is worth a closer look, because the two no-arbitrage conditions of Proposition 1.5.2 are visible in it. The numerator is exactly the calendar spread quantity (1.134), and the denominator is precisely the butterfly quantity $g(k)$ of (1.135). Hence

$$\frac{\partial w}{\partial T} \geq 0 \;\text{ and }\; g(k) \geq 0 \implies \sigma^2_{\text{loc}} \geq 0,$$

so that **an implied volatility surface free of static arbitrage yields a non-negative local variance**. The implication runs one way only, and it is worth being precise about why: a quotient is non-negative whenever numerator and denominator share a sign, so $\sigma^2_{\text{loc}} \geq 0$ is also compatible with both being negative, a configuration which certainly does admit arbitrage. What is true is the chain

$$\text{no static arbitrage} \implies \frac{\partial w}{\partial T} \geq 0 \,\wedge\, g(k) \geq 0 \implies \sigma^2_{\text{loc}} \geq 0,$$

in which each of the two conditions is *separately* necessary for absence of arbitrage — $\partial_T w \geq 0$ for calendar spreads, $g(k) \geq 0$ for butterflies — and only the last arrow fails to reverse.
That the denominator of (1.136) is $g(k)$ exactly, and not merely something of the same sign, is the substantive observation here, and it is what makes the two-step procedure above more than a matter of taste: arbitrage-freeness is not an aesthetic requirement imposed on the interpolation, it is the condition under which (1.136) is guaranteed to return a usable answer. It is also the reason this parameterisation, rather than the raw form (1.132), is the one calibrated in practice.

The limitation

Local volatility fits the surface perfectly, and by Theorem 1.5.1 it is the unique diffusion that does. It also preserves market completeness, since a single Brownian motion drives a single traded asset and Theorem 1.2.37 applies unchanged. These are substantial virtues.
The defect is equally fundamental, and it concerns *dynamics* rather than fit. In a local volatility model the instantaneous volatility is a deterministic function of the spot, so the future smile is entirely determined by where the spot goes. The consequence, observed universally in practice, is that the model predicts the smile flattening as the spot moves, whereas real smiles translate with the spot and retain their shape. A model may therefore reproduce every European price today and still be wrong about every European price tomorrow. This matters not at all for pricing European options — their value depends only on the terminal marginal, which is fitted exactly — and a great deal for anything path-dependent: barriers, cliquets, forward-starting options and autocallables all depend on the joint law across dates, which local volatility gets wrong. That observation is what motivates the next subsection.

1.5.2 Stochastic Volatility Models

Throughout this subsection, and in Subsection 1.5.3 which follows it, interest rates and dividend yields are taken to be *deterministic*. The assumption is worth stating once rather than repeating: it is what makes the T-forward measure coincide with the risk-neutral one, $\mathbb{Q}^T = \mathbb{Q}$, so that expectations below may be written under $\mathbb{Q}$ while still exploiting the $\mathbb{Q}^T$-martingale property of the forward; and it is what allows the discounting term in the American pricing operator of (1.146) to be a deterministic function of time. With stochastic rates both statements require the change of numeraire of Subsection 1.4.1 to be carried through explicitly.
To capture realistic dynamics of the volatility surface itself, we introduce a second source of randomness. In Stochastic Volatility (SV) models, the variance $v_t = \sigma_t^2$ follows its own Stochastic Differential Equation. Under the risk-neutral measure $\mathbb{Q}$, the system is described by:

$$\begin{cases} dS_t = (r-q)\,S_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S \\ dv_t = \mu_v(v_t,t)\,dt + \sigma_v(v_t,t)\,dW_t^v \end{cases} \tag{1.137}$$

where μ_v and σ_v are written thus, rather than as the α and β common in the literature, because both those letters are already spoken for — α as the Girsanov integrand of Subsection 1.2.5 and as the Hölder exponent throughout Chapter 2, and α, β jointly as the exponents of Kolmogorov's criterion (1.38). A critical parameter is the correlation ρ between the two Brownian motions: $d\langle W^S, W^v\rangle_t = \rho\,dt$.

- If $\rho < 0$, a drop in price ($S\downarrow$) is associated with an increase in volatility ($v\uparrow$), naturally generating the **skew** observed in equity markets.
- The "vol of vol" parameter (in the diffusion coefficient σ_v) controls the convexity or **curvature** of the smile.

The Heston Model

The most prominent SV model is that of Heston [35], in which the variance follows the Cox–Ingersoll–Ross square-root process already analysed in Example 1.2.21:

$$\begin{cases} dS_t = (r-q)\,S_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S, \\ dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \end{cases} \qquad d\left\langle W^S, W^v\right\rangle_t = \rho\,dt, \tag{1.138}$$

where κ is the speed of mean reversion, θ the long-run variance, ξ the volatility of volatility, and ρ the correlation, understood in the sense of Definition 1.2.24. We recall from Subsection 1.2.3 that (1.138) is well posed by Yamada–Watanabe rather than by the Lipschitz theory, and that strict positivity of v requires the Feller condition $2\kappa\theta \geq \xi^2$ of Proposition 1.2.22 — a condition frequently violated by calibrated parameters, with the numerical consequences noted there.
Unlike local volatility, stochastic volatility renders the market **incomplete**, and we are now in a position to say exactly why rather than merely asserting it. The filtration is generated by two Brownian motions while only one asset is traded, so the hypothesis of Theorem 1.2.37 fails: a claim whose value depends on W^v cannot be written as a stochastic integral against the traded asset alone. By Theorem 1.3.6 the equivalent martingale measure is therefore not unique, and pricing requires either specifying a market price of volatility risk — already embedded in the $\mathbb{Q}$-dynamics (1.138) — or, as is universal in practice, calibrating $(\kappa, \theta, \xi, \rho, v_0)$ directly to liquid quotes. Volatility risk cannot be hedged away with the underlying; it can only be hedged with other options.
The reason Heston's model, rather than some other stochastic volatility model, became the industry standard is computational, and worth stating explicitly because Part II will build directly on it. The characteristic function of the log-price is available in closed form.

Theorem 1.5.4 (Heston's characteristic function)**.** *Under* (1.138)*, write* $\tau = T - t$ *and* $F_s = S_s e^{(r-q)(T-s)}$ *for the forward to* T*. Then*

$$\varphi(u,\tau) := \mathbb{E}^{\mathbb{Q}}\Big[\exp\Big(iu\log\tfrac{F_T}{F_t}\Big)\,\Big|\,\mathcal{F}_t\Big] = \exp\Big(C(u,\tau)\,\theta + D(u,\tau)\,v_t\Big), \tag{1.139}$$

the characteristic function of the centred log-forward $\log(F_T/F_t)$*, where* C *and* D *solve the system of Riccati ordinary differential equations*

$$\frac{\partial D}{\partial \tau} = \frac{\xi^2}{2}D^2 + (\rho\xi iu - \kappa)D - \frac{u^2 + iu}{2}, \qquad \frac{\partial C}{\partial \tau} = \kappa D, \tag{1.140}$$

with $C(u,0) = D(u,0) = 0$*. The system* (1.140) *admits an explicit solution in terms of elementary functions.* ■

The centring in (1.139) is deliberate and should not be skipped over. Taking the characteristic function of $\log(F_T/F_t)$ rather than of the absolute log-spot X_T strips out the factor $e^{iu\log F_t}$, and does so for two reasons. The first is arithmetical: the inversion formula below carries F_t in its prefactor and again inside the log-moneyness $k = \log(K/F_t)$, so a characteristic function still carrying F_t would introduce it a third time and the normalisation would be applied once and undone once. The second is structural: the centred φ depends only on $(v_t, \kappa, \theta, \xi, \rho, \tau)$ and not on S_t, r or q, which is exactly the separation between model parameters and market data that makes calibration well posed. This is the same reason the forward was singled out as the right variable in Subsection 1.4.1.
The centring also supplies a cheap correctness check, worth recording because the error it catches is silent. Since F is a $\mathbb{Q}^T$-martingale, $\mathbb{E}^{\mathbb{Q}}[F_T/F_t \mid \mathcal{F}_t] = 1$, so

$$\varphi(-i,\tau) = 1 \qquad \text{for every } \tau \geq 0, \tag{1.141}$$

an identity satisfied by the centred characteristic function and violated by the uncentred one. Any implementation of (1.139) should be tested against (1.141) before it is tested against prices.
Given (1.139), European option prices follow by Fourier inversion: the call price is recovered from the characteristic function through an integral of Lewis–Carr–Madan type [11],

$$C(K,T) = P(t,T)\left[F_t - \frac{\sqrt{F_t K}}{\pi}\int_0^\infty \mathrm{Re}\left(e^{iuk}\,\varphi\Big(u - \tfrac{i}{2}, \tau\Big)\right)\frac{du}{u^2 + \frac14}\right], \tag{1.142}$$

a one-dimensional integral evaluated by quadrature in microseconds, with $k = \log(K/F_t)$ the log-moneyness of Subsection 1.5.1. Calibration — which requires pricing hundreds of options at each step of an optimiser — is thereby feasible, whereas a model requiring Monte Carlo for each European quote would not be.

One implementation caveat deserves a line, since it is a genuine trap rather than a nuisance. The explicit solution of (1.140) involves a complex logarithm, and the naive branch choice makes φ discontinuous in τ once the integrand has wound sufficiently far, corrupting prices at long maturities while leaving short ones correct. The remedy is the standard one in the literature, namely to write the solution in the form whose principal branch is continuous; the phenomenon is documented in [3].

That the tractability of the model reduces to a system of Riccati equations is a point worth holding onto. The affine structure of (1.138) — drift and variance both affine functions of the state — is exactly what produces (1.140), and it is a structural property rather than a happy accident. When we come to rough volatility in Part II, the model will retain this affine character but lose the Markov property, and (1.140) will be replaced by a *fractional* Riccati equation. The pricing machinery (1.142) survives essentially unchanged; only the equation for D becomes non-local in time. Recording the classical case here is what will make that generalisation a short step rather than a fresh start.

What stochastic volatility does and does not fix

Stochastic volatility repairs the principal defect of local volatility: because the volatility has its own noise, the smile moves with the spot instead of flattening, and forward-starting products are priced far more sensibly. The correlation ρ generates skew and the vol-of-vol ξ generates curvature, both dynamically rather than by construction.

It does not, however, fit the surface exactly — five parameters cannot match hundreds of quotes — and it fails in one specific and instructive place. Empirically, the implied volatility skew of index options behaves like $T^{-H'}$ for small T with H' well below $\frac{1}{2}$, becoming very steep at short maturities. Any model driven by standard Brownian motion, Heston included, produces a short-maturity skew that flattens as $T \to 0$, because over short horizons a diffusive variance has had no time to move. No choice of $(\kappa, \theta, \xi, \rho, v_0)$ repairs this: the failure is not one of calibration but of the driving noise. This is the empirical observation from which Part II begins.

1.5.3 Path-Dependence and American Options

We conclude this section by briefly discussing the challenges posed by path-dependent derivatives, focusing on the most liquid and prominent example: American Options.

Because American options hold the right to be exercised at any time up to maturity, merely knowing a single implied volatility is insufficient to price them; a single volatility parameter is compatible with an infinite set of potential asset dynamics. This distinction is critically important because options on *single names* (i.e., individual corporate stocks) are almost exclusively American-style. Investors in single equities are reluctant to relinquish the flexibility of early exercise, particularly to capture discrete dividends or navigate corporate actions. In contrast, broad market indices are generally considered more stable and are traded via extremely liquid European options.

Before developing any machinery it is worth recording how much of the early-exercise premium is genuinely there, because for one of the two basic contracts the answer is: none. The result is due to Merton [47] and explains why dividends have kept reappearing in the discussion above. Its statement refers forward to the optimal stopping problem (1.144), written out two subsections below; the reader willing to grant that an American option is worth the supremum, over exercise times, of the expected discounted payoff will lose nothing by reading it now.

Proposition 1.5.5 (No early exercise for the American call). *Assume $r \geq 0$ and that the underlying pays no dividends before T. Then an American call is never optimally exercised before maturity, and its value coincides with that of the European call with the same strike and maturity.*

Proof: Write $C(t)$ for the European call value at time t. Under $\mathbb{Q}$ the discounted spot is a martingale, so for any $t \leq T$, using the conditional Jensen inequality (Proposition 1.1.4, item 8) applied to the convex function $x \mapsto (x-K)^+$,

$$C(t) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}\Big[(S_T - K)^+ \mid \mathcal{F}_t\Big] \;\geq\; e^{-r(T-t)}\Big(\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] - K\Big)^+ = \Big(S_t - Ke^{-r(T-t)}\Big)^+,$$

the last step because $\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = S_t e^{r(T-t)}$ in the absence of dividends. Since $r \geq 0$ gives $Ke^{-r(T-t)} \leq K$, the right-hand side is at least $(S_t - K)^+$, which is the immediate-exercise payoff. Holding therefore dominates exercising at every $t < T$.

It remains to convert this pointwise domination into a statement about stopping times, which does not follow from it directly. The chain above, read at a general t, says precisely that the discounted payoff process $Z_t := e^{-rt}(S_t - K)^+$ satisfies $\mathbb{E}^{\mathbb{Q}}[Z_T \mid \mathcal{F}_t] \geq Z_t$; that is, Z is a $\mathbb{Q}$-*sub*martingale — as it must be, being a convex function of a martingale, by Proposition 1.1.4, item 8, together with $r \geq 0$. Doob's optional stopping theorem (Theorem 1.1.28, case i — every stopping time being bounded by T under Definition 1.1.10) applied to a submartingale gives, for every $\tau \in \mathcal{T}_{[t,T]}$,

$$\mathbb{E}^{\mathbb{Q}}\Big[Z_T \,\Big|\, \mathcal{F}_\tau\Big] \;\geq\; Z_\tau.$$

Since $\tau \geq t$ we have $\mathcal{F}_t \subseteq \mathcal{F}_\tau$, so conditioning both sides on $\mathcal{F}_t$ and applying the tower property (Proposition A.1.1, item 5) gives

$$\mathbb{E}^{\mathbb{Q}}\Big[e^{-r\tau}(S_\tau - K)^+ \,\Big|\, \mathcal{F}_t\Big] \;\leq\; \mathbb{E}^{\mathbb{Q}}\Big[e^{-rT}(S_T - K)^+ \,\Big|\, \mathcal{F}_t\Big],$$

so the supremum in (1.144) is attained at $\tau = T$. ■

Two things are worth extracting. The inequality is strict whenever $r > 0$ and the option has time value, so the American call is not merely no better than the European one but strictly worse to exercise early: one would be surrendering the interest earned on the strike. And the proof uses $\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = S_t e^{r(T-t)}$, which is exactly what a dividend breaks. Once $q > 0$, the forward is $S_t e^{(r-q)(T-t)}$ and the chain of inequalities fails as soon as $q > r$; early exercise then becomes genuinely optimal in a neighbourhood of a dividend date. For the American *put* the argument fails at the first step regardless of dividends, since deep in the money the discounted payoff is bounded above while immediate exercise is not, and early exercise is optimal for low enough spot. This is why the material below is needed at all, and why it is single-name options — which pay discrete dividends — that require it.

We aim to show how a Local Volatility model can be adapted to price American options. However, we must first note a fundamental limitation: Local Volatility models do not perform perfectly for all types of path-dependent derivatives. In particular, because LV models assume volatility is a deterministic function perfectly correlated with the spot price, they often misprice derivatives with discontinuous payoffs (which often lead to ill-posed pricing problems) or derivatives whose strikes are set at a future date, such as forward-starting options and cliquets. Nevertheless, to price American options accurately we must model the entire volatility surface prior to maturity, not merely the terminal marginal. The model we use, expressed under the T-forward measure of Subsection 1.4.1, is

$$dF_t = \sigma_{\text{loc}}(F_t, t)\, F_t\, dW_t^T. \tag{1.143}$$

Working in the forward has the same three advantages set out in Subsection 1.4.1: the drift vanishes because a forward is a martingale under its own measure, the dynamics stay purely diffusive, and — crucially here — the discontinuities introduced by discrete cash dividends are absorbed into the spot-to-forward factor Θ_t of (1.125) and never enter (1.143) at all.

Optimal stopping and the Snell envelope

Before writing any partial differential equation, let us state what an American option *is* in probabilistic terms, since the PDE formulation is derived from it and is opaque without it. The holder may exercise at any stopping time of her choosing, and will choose the one maximising the value of the resulting cash flow. Writing $\mathcal{T}_{[t,T]}$ for the set of stopping times valued in $[t,T]$ and Φ for the exercise payoff, the value at time t is

$$V_t = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{[t,T]}} \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^\tau r(s)\,ds}\,\Phi(\tau,S_\tau)\,\middle|\,\mathcal{F}_t\right]. \tag{1.144}$$

The essential supremum, defined in Appendix A.7, is required because the family is uncountable; this is an optimal stopping problem, and its solution is classical.

Theorem 1.5.6 (Snell envelope)**.** *Let Y be a right-continuous adapted process of class (D) — Definition A.6.2 — representing the discounted payoff. Then the value process U defined by* (1.144) *is the* ***Snell envelope*** *of Y: that is, U is the smallest càdlàg supermartingale dominating Y. It satisfies:*

i) $U_t \geq Y_t$ for all t (domination);

ii) U is a $\mathbb{Q}$-supermartingale;

iii) U is a $\mathbb{Q}$-martingale on the continuation region $\mathfrak{C} = \{(t,\omega) : U_t > Y_t\}$*, the fraktur letter distinguishing it from the cone $\mathcal{C}$ of dominated claims in Subsection 1.3.2;*

and the stopping time $\tau^* = \inf\{s \geq t : U_s = Y_s\}$ is optimal in (1.144)*. A full treatment is given in [51].* ■

The three properties have an immediate financial reading, and it is worth pausing on it because the LCP below is nothing more than their restatement. Property i) says the option is never worth less than exercising it — otherwise one would buy the option, exercise immediately, and pocket the difference. Property ii) says that holding the option is, on average and after discounting, a losing proposition: waiting has a cost, which is what makes early exercise ever attractive. Property iii) says that this cost is zero exactly where continuation is optimal, so that in the region where one does not exercise, the option is a fair game. The optimal rule is then transparent: continue while $U > Y$, and stop the instant they meet.

The Linear Complementarity Problem (LCP)

Translating Theorem 1.5.6 into the language of partial differential equations gives a free boundary problem, most rigorously expressed as a linear complementarity problem. Write $\mathcal{L}$ for the pricing operator associated with (1.143),

$$\mathcal{L}V := \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2_{\text{loc}}(F,t)\,F^2\frac{\partial^2 V}{\partial F^2} - r(t)\,V.$$

A point of care is required with the obstacle, and it is the reason this subsection is not a routine transcription. Exercise delivers the *spot*, not the forward: the holder of an American put receives $K - S_\tau$, and by (1.125) we have $S_t = \Theta_t\,F_t$. Expressed in the forward variable, the obstacle is therefore

$$\Phi(t,F) = \left(K - \Theta_t\,F\right)^+ \quad \text{for a put,} \qquad \Phi(t,F) = \left(\Theta_t\,F - K\right)^+ \quad \text{for a call,} \tag{1.145}$$

which is **explicitly time-dependent through** Θ_t. Writing $(K-F)^+$ instead, as is sometimes done, silently assumes $\Theta_t \equiv 1$ and is simply wrong: it misplaces the exercise boundary, and it does so by the widest margin in a name paying large discrete dividends — precisely the case where early exercise matters most. Keeping Θ_t explicit costs nothing and preserves validity for time-dependent $r(t)$ and $q(t)$, for cash and proportional dividends, and for any mixture of them.

Theorem 1.5.7 (American pricing as an LCP)**.** *The value $V(t,F)$ of the American option satisfies, almost everywhere on $[0,T)\times(0,\infty)$, the three simultaneous conditions*

$$\begin{cases} \mathcal{L}V \le 0 & \text{(supermartingale property, Theorem 1.5.6(ii))} \\ V(t,F) \ge \Phi(t,F) & \text{(domination, Theorem 1.5.6(i))} \\ \Big(\mathcal{L}V\Big)\cdot\Big(V(t,F)-\Phi(t,F)\Big) = 0 & \text{(complementarity, Theorem 1.5.6(iii))} \end{cases} \tag{1.146}$$

together with the terminal condition $V(T,F)=\Phi(T,F)$ and Φ as in (1.145). ■

The correspondence with Theorem 1.5.6 is exact, line for line, and this is the point of having stated the probabilistic version first. The complementarity condition encodes the dichotomy: at each point either $V>\Phi$, in which case $\mathcal{L}V=0$ and the ordinary pricing equation holds, or $V=\Phi$ and the option is exercised. The boundary separating the two regions is not known in advance — it is part of the solution — which is what makes the problem free-boundary rather than merely parabolic, and what rules out any closed-form solution.

Calibration to Single Names

A practical hurdle arises at exactly this point. Dupire's formula (Theorem 1.5.1) requires derivatives of *European* prices, whereas single-name quotes are American, so the surface cannot be calibrated directly from the market. The standard remedy — a short iterative mapping known as **de-Americanisation** — is a calibration procedure rather than a piece of the theory, and we defer it to Section 3.3 of Chapter 3, where the numerical machinery it needs is available.

Notes and references

Sections 1.1–1.2. The theoretical framework of these sections draws primarily from two authoritative sources. The general Probability Theory concepts follow the classic exposition of P. Billingsley [4], while the construction of the Stochastic Integral is based on the comprehensive treatment by H. Kuo [40]. Additionally, references for the underlying Functional Analysis theory, implicit in the PDE treatment of diffusions, can be found in the seminal work of H. Brezis [10].
Certain advanced topics, such as the general formulation of Stochastic Integration with respect to semimartingales or deeper pathwise properties of Brownian motion, have been omitted as they lie beyond the scope of this work. However, the selected contents provide a rigorous and sufficient foundation for the financial applications discussed in subsequent sections and chapters.
Sections 1.3–1.5.
The financial framework and pricing methodologies developed in the latter half of this chapter are built upon cornerstone texts in quantitative finance and stochastic analysis. The rigorous transition from abstract probability to continuous-time asset pricing, including the formulation of equivalent martingale measures and the *No Free Lunch with Vanishing Risk* (NFLVR) condition, relies heavily on the definitive work by Delbaen and Schachermayer [18]. This

abstract measure-theoretic approach is bridged with continuous-time applications in the classic textbooks by Shreve [56], Øksendal [49], and Björk [5].
The explicit formulation of diffusion models is rooted in the foundational breakthroughs of Black and Scholes [7] and Merton [47]. To address the empirical shortcomings of this framework, Local Volatility was formalized by Dupire [19], with its practical translation into the implied volatility domain extensively covered by Gatheral [27]. For an exhaustive, practitioner-oriented treatment of volatility smile dynamics and Stochastic Volatility models (such as the Heston model [35]), Bergomi [3] serves as an indispensable reference.

Part II

Modern Approaches

Chapter 2

Rough Paths

2.1 Introduction

Remark 1.1.46 closed Part I with a diagnosis rather than a theorem. Brownian motion has finite p-variation exactly for $p > 2$; Young's integral requires $p < 2$; and the Itô integral of Section 1.2 bridges the gap not by refining an estimate but by *spending probabilistic structure* — adaptedness, an $L^2(\Omega)$ limit in place of a pathwise one, and a definition valid only up to null sets. That bargain is what Part I was built on, and it is worth being precise about the two circumstances in which it collapses.

The first is that the driving signal may be rougher than Brownian. Itô's construction is not a general theory of integration against irregular paths: it is a theory of integration against *semimartingales*, and Definition 1.1.32 was recorded in Chapter 1 precisely so that we could mark its edge. The moment the integrator leaves that class, the entire apparatus — the isometry of Theorem 1.2.5, the martingale property, the localisation of Example 1.1.12 — becomes unavailable at once, and not merely inconvenient.

The second is subtler and, in the end, more consequential. Even where Itô's theory applies, it delivers the solution of a stochastic differential equation as a measurable function on Ω, constructed through an L^2 limit that depends on the filtration and the measure. It does not tell us that the solution depends *continuously* on the driving trajectory, and in fact, as we shall see in Subsection 2.1.3, no such statement is true in the topology one would naïvely choose. This is not a technical blemish. A model whose output is a discontinuous function of its input is a model one cannot approximate, cannot discretise with confidence, and cannot calibrate stably.

Rough path theory, introduced by Lyons [43], resolves both difficulties simultaneously, and by a single device: it enlarges the notion of "path". The remainder of this chapter develops that idea. We begin, however, with the process that forces the question upon us.

2.1.1 Fractional Brownian Motion

Brownian motion is characterised, among Gaussian processes with stationary increments, by the independence of those increments. Relaxing exactly that requirement — and nothing else — yields the following one-parameter family, introduced by Kolmogorov and given its modern form by Mandelbrot and Van Ness [45].

Definition 2.1.1 (Fractional Brownian motion)**.** Let $H \in (0, 1)$. A **fractional Brownian motion** with Hurst parameter H is a centred Gaussian process $W^H = \{W^H_t\}_{t\geq 0}$ with $W^H_0 = 0$ almost surely and covariance

$$\mathbb{E}\Big[W^H_s W^H_t\Big] \;=\; R_H(s,t) \;:=\; \tfrac{1}{2}\Big(s^{2H} + t^{2H} - |t-s|^{2H}\Big), \qquad s,t \geq 0. \tag{2.1}$$

That (2.1) is a legitimate covariance — that is, symmetric and positive semi-definite — is a genuine constraint on H, and it is the reason the parameter range is $(0,1)$ and not larger; a proof is in [46, Ch. 7]. Granting it, the process exists by the standard Gaussian construction, and Kolmogorov's criterion (Theorem 1.1.38) supplies a continuous modification, with which we always work in accordance with Remark 1.1.8.

Proposition 2.1.2 (Elementary properties)**.** *Let W^H be a fractional Brownian motion with Hurst parameter $H \in (0,1)$. Then:*

a) W^H has stationary increments, *with*

$$\mathbb{E}\Big[\,|W^H_t - W^H_s|^2\,\Big] = |t-s|^{2H}, \qquad \text{so } W^H_t - W^H_s \sim \mathcal{N}\Big(0, |t-s|^{2H}\Big). \tag{2.2}$$

b) W^H is self-similar of index H*: for every $\lambda > 0$, the processes $\{W^H_{\lambda t}\}_{t\geq 0}$ and $\{\lambda^H W^H_t\}_{t\geq 0}$ have the same law.*

c) Almost every sample path is locally γ-Hölder continuous for every $\gamma < H$, and for no $\gamma \geq H$.

d) Almost surely, $V_p(W^H)_t < \infty$ for every $p > 1/H$ and $V_p(W^H)_t = \infty$ for every $p \leq 1/H$: the critical exponent of Remark 1.1.46 is $1/H$.

e) $H = \frac{1}{2}$ recovers Brownian motion. For $H > \frac{1}{2}$ the increments over disjoint intervals are positively correlated, for $H < \frac{1}{2}$ negatively, and in neither case are they independent.

Proof: a) Expanding and using (2.1),

$$\mathbb{E}\Big[(W^H_t - W^H_s)^2\Big] = R_H(t,t) - 2R_H(s,t) + R_H(s,s) = t^{2H} - \Big(s^{2H} + t^{2H} - |t-s|^{2H}\Big) + s^{2H} = |t-s|^{2H}.$$

Since the process is centred and Gaussian, the law of an increment is determined by this variance alone, and depends on s,t only through $t-s$.
b) $R_H(\lambda s, \lambda t) = \lambda^{2H} R_H(s,t)$ is immediate from (2.1). Two centred Gaussian processes with the same covariance have the same law.
c) The Gaussian moments give $\mathbb{E}[|W^H_t - W^H_s|^{2m}] = c_m |t-s|^{2mH}$, so (1.38) holds with $\alpha = 2m$ and $\beta = 2mH - 1$, and Theorem 1.1.38 yields Hölder exponents $\gamma < H - 1/2m$, which approach H as $m \to \infty$. The negative half is the exact analogue of Theorem 1.1.45 i) and is proved by the same Paley–Wiener–Zygmund argument, adapted in [46, Ch. 7].
d) Follows from c) together with the same competition between p-variation and modulus of continuity that produced Proposition 1.1.43; see [26, Ch. 15] for the details, which we do not repeat.
e) For $H = \frac{1}{2}$, (2.1) collapses to $\min\{s,t\}$, which is Proposition 1.1.37 a); by Definition 1.1.33 and the Gaussian characterisation this identifies the process. For the sign of the correlation, take $0 \leq s \leq t \leq u \leq v$ and compute directly from (2.1),

$$2\,\mathbb{E}\Big[(W^H_t - W^H_s)(W^H_v - W^H_u)\Big] = |v-s|^{2H} + |u-t|^{2H} - |u-s|^{2H} - |v-t|^{2H},$$

which is the second difference of the function $\tau \mapsto \tau^{2H}$, hence positive when that function is convex ($H > \frac{1}{2}$) and negative when it is concave ($H < \frac{1}{2}$). It vanishes identically only at $H = \frac{1}{2}$. ■

Part e) is the modelling content of the definition and deserves its name. For $H > \frac{1}{2}$ the process exhibits *persistence*: an increase tends to be followed by an increase, and the correlations decay so slowly that they are not summable — the phenomenon known as *long memory*. For $H < \frac{1}{2}$

it exhibits *anti-persistence*, or mean reversion at every scale, and the paths are correspondingly rougher than Brownian. Part d) makes "rougher" precise: the critical exponent $1/H$ exceeds 2 exactly when $H < \frac{1}{2}$.
That last observation is not a curiosity but an obstruction, and we state it as such.

Theorem 2.1.3 (Rogers)**.** *Fractional Brownian motion is a semimartingale in the sense of Definition 1.1.32 if and only if $H = \frac{1}{2}$.* ■

The proof is in [54], and the mechanism is worth recording even though we do not reproduce the argument. For $H > \frac{1}{2}$ the quadratic variation of Definition 1.1.36 vanishes, while the total variation is infinite by part d); a continuous semimartingale with zero quadratic variation must be of finite variation, by the uniqueness of the decomposition, and this is a contradiction. For $H < \frac{1}{2}$ the quadratic variation is instead infinite, which is incompatible with the decomposition from the other direction. Only the balance point $H = \frac{1}{2}$ escapes.
The consequence is severe and complete. **For $H \neq \frac{1}{2}$ there is no Itô integral against W^H**, no Itô formula, no martingale representation, and no Girsanov theorem — not because the constructions are difficult but because every one of them takes the semimartingale property as its hypothesis. This was anticipated in Chapter 1, where Definition 1.1.32 was described as marking "the outer limit of the theory developed in this chapter". We have now crossed it.

2.1.2 Volatility is Rough

Why should one wish to cross it? The answer, for this dissertation, is empirical, and although a thesis in mathematics is not the place for an econometric study, the finding that motivates everything below can be stated in one paragraph.
Let σ_t denote spot volatility, estimated from high-frequency returns over a grid of spacing Δ, and consider the q-th absolute moment of the increments of its logarithm,

$$m(q,\Delta) \;:=\; \frac{1}{N}\sum_{i=1}^{N}\Big|\log\sigma_{i\Delta} - \log\sigma_{(i-1)\Delta}\Big|^{q}. \tag{2.3}$$

Gatheral, Jaisson and Rosenbaum [29] observe that, across equity indices and thousands of individual assets, $m(q,\Delta) \propto \Delta^{\zeta_q}$ with $\zeta_q = qH$ — a linear, *monofractal* scaling with no curvature in q — and that the increments are close to Gaussian. Comparison with (2.2) is immediate: these are exactly the scaling relations of a fractional Brownian motion. The estimated exponent is $H \approx 0.1$, and it is remarkably stable across assets and across decades.

[FIGURE HERE] Log–log plot of $m(q,\Delta)$ against Δ for several values of q, showing the parallel straight lines whose slopes are ζ_q, alongside the plot of ζ_q against q exhibiting the linear (monofractal) relation $\zeta_q = qH$. This is the visual statement of the empirical finding of [29].

This inverts an earlier consensus. The fractional stochastic volatility model of Comte and Renault [15] also used fractional Brownian motion, but with $H > \frac{1}{2}$, in order to reproduce the apparent long memory of realised volatility. The finding above says that the correct exponent is not merely below $\frac{1}{2}$ but very close to 0: volatility is not persistent, it is *rough*. The resulting model, which we shall meet again in Chapter 3, replaces the Heston dynamics of Subsection 1.5.2 by

$$\sigma_t = \exp(X_t), \qquad dX_t = \nu\, dW_t^H - a\Big(X_t - m\Big)dt, \tag{2.4}$$

with a so small that, over any horizon of practical interest, X is indistinguishable from a fractional Brownian motion.[1]

[1] The mean-reversion speed is written α in [29]; we use a, since α is reserved throughout this chapter for the Hölder exponent fixed in (2.15).

For our purposes the decisive consequence is the one that concerns the object Chapter 1 spent its final sections constructing: the implied volatility surface. A model of the form (2.4) generates an at-the-money skew whose term structure behaves like

$$\psi(\tau) \ := \ \left| \frac{\partial \sigma_{\text{imp}}}{\partial k}(\tau, k) \right|_{k=0} \ \sim \ \tau^{H-1/2} \qquad \text{as } \tau \downarrow 0, \tag{2.5}$$

a power-law explosion at short maturities. Every model driven by a semimartingale volatility factor — Heston included, and by Theorem 2.1.3 that is precisely the case $H = \frac{1}{2}$ — produces a skew that flattens as $\tau \downarrow 0$, and no choice of parameters repairs this. The observed surface does not flatten. Roughness is what the market implies, and (2.5) is the cleanest statement of why.

Remark 2.1.4. The reader should note what (2.4) does and does not require. The equation is driven by W^H but is *linear* in X, so it can be solved explicitly, by a pathwise Riemann–Stieltjes argument, without any integration theory for W^H at all. The difficulty arrives one level up, when one wishes to price — that is, to integrate σ_t, a nonlinear functional of W^H, against the Brownian motion driving the asset price. That is a genuine stochastic integral against a rough signal, and it is the problem this chapter exists to address.

2.1.3 What Fails, and Why

Before constructing anything we should be clear that the obstruction is real, and in particular that it is not removed by any sharpening of the classical estimates. There are three layers to this, of increasing depth.

Young integration is out of reach. Recall from Remark 1.1.46 that $\int g\, df$ exists pathwise as a limit of Riemann sums whenever f has finite p-variation, g finite q-variation, and $1/p+1/q > 1$. In Hölder terms, if $f \in \mathcal{C}^\alpha$ and $g \in \mathcal{C}^\beta$ the condition reads $\alpha + \beta > 1$. Now the integrands we care about are of the form $g = F(f)$, which inherits the regularity α of f itself, so the condition becomes $2\alpha > 1$, that is $\alpha > \frac{1}{2}$. By Proposition 2.1.2 c) this holds for W^H precisely when $H > \frac{1}{2}$. Young's theory therefore covers exactly the persistent case that the data reject, and none of the rough one.

Itô integration is out of reach. By Theorem 2.1.3, immediately and for all $H \neq \frac{1}{2}$.

And the obstruction is not a matter of technique. One might still hope that the map $(f, g) \mapsto \int g\, df$, manifestly well defined for smooth paths, extends by continuity to the irregular ones — that some cleverer topology on path space would do what the uniform topology fails to do. The following example, due to Lyons, shows that the uniform topology cannot.

Example 2.1.5 (The oscillation counterexample)**.** For $n \in \mathbb{N}$ define the smooth planar paths

$$x_t^{(n)} := \frac{1}{n} \sin\!\left(n^2 t\right), \qquad y_t^{(n)} := \frac{1}{n} \cos\!\left(n^2 t\right), \qquad t \in [0, T]. \tag{2.6}$$

Both converge to the zero path uniformly, since $|x^{(n)}|, |y^{(n)}| \leq 1/n$. Yet

$$\int_0^t y_s^{(n)}\, dx_s^{(n)} = \int_0^t \frac{\cos(n^2 s)}{n} \cdot n \cos(n^2 s)\, ds = \int_0^t \cos^2(n^2 s)\, ds = \frac{t}{2} + \frac{\sin(2n^2 t)}{4n^2} \ \longrightarrow \ \frac{t}{2},$$

which is not the integral of the zero path against itself. The integral is therefore *not* continuous with respect to uniform convergence.

The picture behind the computation is worth holding on to, because it is the picture behind the whole theory. The paths $\left(x^{(n)}, y^{(n)}\right)$ trace circles of radius $1/n$ traversed n^2 times per unit

time. The circles shrink to a point, which is why the paths converge uniformly to zero; but the *area* swept out is π/n^2 per revolution and there are $n^2/2\pi$ revolutions per unit time, so the total area accumulated per unit time is a constant, independent of n. Uniform convergence sees the shrinking radius. It does not see the accumulating area. And the integral, as the computation shows, depends on the area.
One might now hope for some other norm. There is none.

Proposition 2.1.6 (Lyons)**.** *Let $\mathcal{W}$ denote the space of continuous paths $[0, T] \to \mathbb{R}^d$ and let $E^\infty \subseteq \mathcal{W}$ be the smooth paths. There is no norm $\|\cdot\|$ on E^∞ such that, simultaneously, the closure E of E^∞ carries full Wiener measure and the map $(f, g) \mapsto \int g\, df$ extends continuously from $E^\infty \times E^\infty$ to $E \times E$.* ■

A proof is given in [25, Ch. 1]. The statement is a genuine impossibility theorem, not an admission of ignorance, and it forces a change of question. If no topology on *paths* makes the integral continuous, then the integral is not a function of the path alone, and the object we should be topologising is not the path.
Example 2.1.5 tells us exactly what is missing. What distinguishes $\left(x^{(n)}, y^{(n)}\right)$ from the zero path is the area it encloses — equivalently, the iterated integral $\iint dx\, dy$. So we shall carry that quantity along as part of the data, postulate it where it cannot be constructed, and topologise the enlarged object. That the resulting theory is not merely a device but the *correct* formulation is the content of Theorem 2.3.19, and everything between here and there is the construction.

One word on how the chapter is to be read, since its architecture is deliberate and easy to miss. What follows is not a survey of an unrelated theory placed alongside Part I. Three of the central results of Chapter 1 — Itô's formula, the Itô–Stratonovich correction, and the Wong–Zakai approximation theorem — will be *re-derived* here on different foundations, and in each case we shall state explicitly what the new derivation costs in hypotheses and what it buys in conclusions. That accounting is the real argument of this chapter. The results themselves are not new, and the rough-path versions are in two of the three cases strictly weaker than the classical ones; what is new is that they are obtained pathwise, deterministically, and with a continuity statement attached — and that the second-order term which Chapter 1 presented as an irreducible feature of stochastic calculus turns out to be a consequence of a choice. The coda closing the chapter collects the three and draws the moral.

2.2 Rough Paths

2.2.1 Tensor Spaces

The iterated integrals we must carry are indexed by pairs of coordinates, and the algebraic home for such objects is the tensor product. We recall only what is needed; the systematic treatment, which requires free Lie algebras and the theory of the free nilpotent group, is developed in [30, Ch. 3] and [26, Ch. 7], and we shall not need it.
Throughout this chapter $V := \mathbb{R}^d$ with its standard basis $\{e_1, \ldots, e_d\}$.

Definition 2.2.1 (Tensor product)**.** The **tensor product** $V \otimes V$ is the vector space generated by the symbols $v \otimes w$, for $v, w \in V$, subject to bilinearity:

$$(av_1 + v_2) \otimes w = a(v_1 \otimes w) + v_2 \otimes w, \qquad v \otimes (aw_1 + w_2) = a(v \otimes w_1) + v \otimes w_2, \tag{2.7}$$

for all $a \in \mathbb{R}$ and $v, v_i, w, w_i \in V$.

Concretely $V \otimes V$ has dimension d^2 with basis $\{e_i \otimes e_j\}_{1 \le i,j \le d}$, so that it may be identified with the space of $d \times d$ matrices, an element being written $\xi = \sum_{i,j} \xi^{ij}\, e_i \otimes e_j$. The point of the notation is that the product is *not* commutative: $e_1 \otimes e_2 \neq e_2 \otimes e_1$. This is essential, since the two iterated integrals $\iint dx^1 dx^2$ and $\iint dx^2 dx^1$ are genuinely different numbers, and it is exactly their difference that Example 2.1.5 detected.

Definition 2.2.2 (Truncated tensor algebra)**.** The **truncated tensor algebra of order two** over V is

$$T^{(2)}(V) \ :=\ \mathbb{R} \oplus V \oplus (V \otimes V), \tag{2.8}$$

whose elements we write $\xi = (\xi_0, \xi_1, \xi_2)$, equipped with the multiplication

$$(\xi \otimes \eta)_n \ :=\ \sum_{k=0}^{n} \xi_k \otimes \eta_{n-k}, \qquad n = 0, 1, 2, \tag{2.9}$$

all terms of order higher than two being discarded. We write $T_1^{(2)}(V) := \{\xi \in T^{(2)}(V) : \xi_0 = 1\}$. More generally, the truncation at order N is $T^{(N)}(V) := \bigoplus_{n=0}^{N} V^{\otimes n}$, with the same multiplication truncated at order N, and the full tensor algebra — in which no truncation is performed and elements are therefore infinite sequences — is $T((V)) := \prod_{n \geq 0} V^{\otimes n}$. Only $T^{(2)}$ is used until Section 2.4.

Written out, (2.9) says $\xi \otimes \eta = (\xi_0\eta_0,\ \xi_0\eta_1 + \xi_1\eta_0,\ \xi_0\eta_2 + \xi_1 \otimes \eta_1 + \xi_2\eta_0)$. The algebra is associative with unit $\mathbf{1} = (1, 0, 0)$, and although a general element has no inverse, every element of $T_1^{(2)}(V)$ does, namely

$$(1, \xi_1, \xi_2)^{-1} = \Big(1,\ -\xi_1,\ \xi_1 \otimes \xi_1 - \xi_2\Big), \tag{2.10}$$

as one checks by multiplying out. Thus $T_1^{(2)}(V)$ is a group, and this is the structure in which the whole chapter takes place.
The motivation is the following computation, which the reader should regard as the definition of everything that follows.

Example 2.2.3 (The canonical lift of a smooth path)**.** Let $x : [0, T] \to V$ be continuously differentiable. For $s \leq t$ set

$$X_{s,t} := x_t - x_s \in V, \qquad \mathbb{X}_{s,t} := \int_s^t \Big(x_r - x_s\Big) \otimes dx_r = \int_s^t \int_s^r dx_u \otimes dx_r \in V \otimes V, \tag{2.11}$$

both integrals being ordinary Riemann–Stieltjes integrals. In coordinates, $\mathbb{X}^{ij}_{s,t} = \int_s^t (x^i_r - x^i_s)\, \dot{x}^j_r\, dr$. The pair $\mathbf{X}_{s,t} := \Big(1, X_{s,t}, \mathbb{X}_{s,t}\Big)$ takes values in $T_1^{(2)}(V)$.

Two features of (2.11) will survive the passage to irregular paths, and one will not. The two that survive are an algebraic identity, developed in the next subsection, and an analytic bound. The one that does not is the following, which is worth isolating now.

Proposition 2.2.4 (The symmetric part is determined)**.** *With the notation of Example 2.2.3,*

$$\operatorname{Sym} \mathbb{X}_{s,t} \ :=\ \tfrac{1}{2}\Big(\mathbb{X}_{s,t} + \mathbb{X}_{s,t}^{\top}\Big) \ =\ \tfrac{1}{2}\, X_{s,t} \otimes X_{s,t}. \tag{2.12}$$

Proof: In coordinates the claim is $\mathbb{X}^{ij}_{s,t} + \mathbb{X}^{ji}_{s,t} = X^i_{s,t} X^j_{s,t}$. Since x is differentiable, the ordinary product rule gives

$$\frac{d}{dr}\Big[\Big(x^i_r - x^i_s\Big)\Big(x^j_r - x^j_s\Big)\Big] = \Big(x^i_r - x^i_s\Big)\dot{x}^j_r + \Big(x^j_r - x^j_s\Big)\dot{x}^i_r,$$

and integrating from s to t, where the left-hand side vanishes at $r = s$, yields exactly the claim.
■

So for a smooth path the symmetric part of the iterated integral carries no information beyond the increment itself: only the *antisymmetric* part, the Lévy area

$$A_{s,t} := \tfrac{1}{2}\Big(\mathbb{X}_{s,t} - \mathbb{X}_{s,t}^{\top}\Big), \tag{2.13}$$

is new. Geometrically $A^{ij}_{s,t}$ is the signed area enclosed between the chord joining (x^i_s, x^j_s) to (x^i_t, x^j_t) and the arc of the path between them — precisely the quantity that Example 2.1.5 showed to be invisible to uniform convergence. Whether (2.12) continues to hold once x is no longer differentiable is exactly the distinction between the two classes of rough path introduced in Subsection 2.2.3, and it is the precise mathematical location of the difference between Itô and Stratonovich calculus.

2.2.2 The Multiplicative Property

The algebraic identity announced above is the following, and it is the axiom on which the entire theory rests.

Proposition 2.2.5 (Chen's relation for smooth paths)**.** *With the notation of Example 2.2.3, for all $s \leq u \leq t$,*

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} \;=\; X_{s,u} \otimes X_{u,t}, \tag{2.14}$$

equivalently $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ in the group $T_1^{(2)}(V)$.

Proof: Split the outer integral in (2.11) at u:

$$\mathbb{X}_{s,t} = \int_s^u (x_r - x_s) \otimes dx_r + \int_u^t (x_r - x_s) \otimes dx_r = \mathbb{X}_{s,u} + \int_u^t \Big[(x_r - x_u) + (x_u - x_s)\Big] \otimes dx_r.$$

The first term of the bracket integrates to $\mathbb{X}_{u,t}$. The second is constant in r, so it factors out of the integral, leaving $X_{s,u} \otimes \int_u^t dx_r = X_{s,u} \otimes X_{u,t}$. That the resulting three identities — one at each level — are precisely the statement $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ is immediate from (2.9). ■

Relation (2.14) is worth reading twice. At the first level it is the triviality $X_{s,t} = X_{s,u} + X_{u,t}$, the additivity of increments. At the second level it is a genuine constraint, and it says that $\mathbb{X}$ is *not* additive: concatenating two intervals produces a cross-term $X_{s,u} \otimes X_{u,t}$ which is the area contributed by the interaction of the two pieces. The relation is thus the algebraic residue of the additivity of integration, and it is what makes $\mathbb{X}$ a coherent object rather than an arbitrary two-parameter function.

Notation 2.2.6. Throughout, $\mathbb{X}$ is a function of *two* time variables and is never a path. There is no "$\mathbb{X}_t$". This is not a notational accident: by (2.14) the increments of $\mathbb{X}$ do not telescope, so no single-variable function generates it. The first level, by contrast, may be regarded either way, and we write $X_{s,t} = x_t - x_s$ when convenient.

We can now give the central definition. From this point onwards we fix

$$\alpha \in \Big(\tfrac{1}{3}, \tfrac{1}{2}\Big], \tag{2.15}$$

a restriction whose necessity will become visible in Theorem 2.3.5 and which we discuss in Remark 2.3.6.

A word on why the exponent is a Hölder one. Lyons' original theory [43], and the systematic treatment of [26], are formulated in p-variation — the scale of Definition 1.1.44, in which Proposition 2.1.2 d) and Remark 1.1.46 are also stated, so that a reader arriving here expects it. The two settings are essentially interchangeable, the dictionary being $p = 1/\alpha$: a path of finite p-variation is α-Hölder after reparametrisation by its own variation, and every α-Hölder path has finite $1/\alpha$-variation. We work in Hölder norms because they make the estimates of Subsection 2.3.1 additive in the exponent and therefore easier to read, and because (2.15) then corresponds to $p \in [2, 3)$ — exactly the range in which two levels suffice.

Definition 2.2.7 (Rough path)**.** An α-**Hölder rough path** over V is a pair $\mathbf{X} = (X, \mathbb{X})$ consisting of $X \in \mathcal{C}^\alpha\big([0,T], V\big)$ and a two-parameter function $\mathbb{X} : \{0 \le s \le t \le T\} \to V \otimes V$ such that:

i) (**Chen's relation**) for all $s \le u \le t$,

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}; \tag{2.16}$$

ii) (**analytic bound**) the seminorms

$$\|X\|_\alpha := \sup_{s \ne t} \frac{|X_{s,t}|}{|t-s|^\alpha}, \qquad \|\mathbb{X}\|_{2\alpha} := \sup_{s \ne t} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} \tag{2.17}$$

are both finite.

We write $\mathbf{X} \in \mathbf{C}^\alpha\big([0,T], V\big)$, abbreviated $\mathbf{C}^\alpha$, and set $|||\mathbf{X}|||_\alpha := \|X\|_\alpha + \|\mathbb{X}\|_{2\alpha}^{1/2}$.

The exponent 2α in (2.17) is not a convention but a necessity: $\mathbb{X}$ is formally a double integral of increments of size $|t-s|^\alpha$, so any object deserving the name must scale that way. Note also that the two conditions pull in opposite directions — i) is algebraic and rigid, ii) is analytic and merely quantitative — and that neither alone would suffice.

Definition 2.2.8 (The rough path metric)**.** For $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (\widetilde{X}, \widetilde{\mathbb{X}})$ in $\mathbf{C}^\alpha$ set

$$\varrho_\alpha\big(\mathbf{X}, \widetilde{\mathbf{X}}\big) \;:=\; \big\|X - \widetilde{X}\big\|_\alpha \;+\; \big\|\mathbb{X} - \widetilde{\mathbb{X}}\big\|_{2\alpha}. \tag{2.18}$$

The second level enters (2.18) to the first power, whereas it entered $|||\mathbf{X}|||_\alpha$ in Definition 2.2.7 under a square root, so that $\varrho_\alpha(\mathbf{X}, \mathbf{0}) \ne |||\mathbf{X}|||_\alpha$. This is not a misprint. The metric is *inhomogeneous* and is the one in which the estimates of Subsection 2.3.4 are linear; the norm is *homogeneous*, meaning that it scales correctly under the dilation $(X, \mathbb{X}) \mapsto (\lambda X, \lambda^2 \mathbb{X})$ which the exponent 2α of (2.17) forces. Both are standard, and each is used where its own scaling behaviour is wanted.

Remark 2.2.9 ($\mathbf{C}^\alpha$ is not a vector space)*.* The set of rough paths is closed under neither addition nor scalar multiplication, since (2.16) is quadratic in X: if $\mathbf{X}$ and $\widetilde{\mathbf{X}}$ satisfy it, their sum does not. It is a complete metric space under ϱ_α, and a closed subset of the Banach space $\mathcal{C}^\alpha \oplus \mathcal{C}_2^{2\alpha}$ — the subscript 2 denoting functions of *two* time variables, as $\mathbb{X}$ is — in which it naturally sits, but it is neither a linear space nor separable. The correct picture is a nonlinear "fibre bundle": over each α-Hölder path X sits the set of admissible second levels, which we now describe. See [25, Ch. 2].

Proposition 2.2.10 (The lift is not unique)**.** *Let* $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ *and let* $f : [0,T] \to V \otimes V$ *belong to* $\mathcal{C}^{2\alpha}$*. Then* $\widetilde{\mathbf{X}} := \big(X,\ \mathbb{X} + \delta f\big)$*, where* $\delta f_{s,t} := f_t - f_s$*, is again an element of* $\mathbf{C}^\alpha$*. Conversely, if* $(X, \mathbb{X})$ *and* $(X, \widetilde{\mathbb{X}})$ *both lie in* $\mathbf{C}^\alpha$*, then* $\widetilde{\mathbb{X}} - \mathbb{X} = \delta f$ *for some* $f \in \mathcal{C}^{2\alpha}$*.*

Proof: For the first claim, note that increments telescope: $\delta f_{s,t} - \delta f_{s,u} - \delta f_{u,t} = 0$, so adding δf to $\mathbb{X}$ leaves (2.16) unchanged, while $\|\delta f\|_{2\alpha} = \|f\|_{2\alpha} < \infty$ preserves the bound. Conversely, set $\Delta_{s,t} := \widetilde{\mathbb{X}}_{s,t} - \mathbb{X}_{s,t}$. Subtracting the two instances of (2.16), which share the same right-hand side because the first levels agree, gives $\Delta_{s,t} = \Delta_{s,u} + \Delta_{u,t}$ for all $s \le u \le t$. Taking $s = u$ shows $\Delta_{s,s} = 0$, and then $f_t := \Delta_{0,t}$ satisfies $\delta f_{s,t} = \Delta_{0,t} - \Delta_{0,s} = \Delta_{s,t}$, with $\|f\|_{2\alpha} = \|\Delta\|_{2\alpha} < \infty$ by the triangle inequality. ■

This is the single most important structural fact in the subject, and it should be dwelt on. **A rough path is not a path.** The second level is genuine additional data, undetermined by the trajectory, and the space of admissible choices over a fixed X is an affine copy of $\mathcal{C}^{2\alpha}$. The theory does not *discover* $\mathbb{X}$; it takes $\mathbb{X}$ as given and proceeds. Different choices are different rough paths, and — as Remark 2.3.9 will make precise — they yield different integrals and different solutions to differential equations. Far from being a defect, this is what allows a single formalism to contain both the Itô and the Stratonovich calculus of Chapter 1 as special cases, distinguished by nothing other than the choice of $\mathbb{X}$.

2.2.3 Types of Rough Paths

Proposition 2.2.4 held automatically for smooth paths. Promoting it to a definition isolates the class on which ordinary calculus survives.

Definition 2.2.11 (Geometric rough paths)**.** A rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ is **weakly geometric** if

$$\operatorname{Sym} \mathbb{X}_{s,t} = \tfrac{1}{2}\, X_{s,t} \otimes X_{s,t} \qquad \text{for all } s \le t, \tag{2.19}$$

and **geometric** if it is the limit, in the metric ϱ_α, of canonical lifts of smooth paths in the sense of Example 2.2.3. We write $\mathbf{C}^\alpha_{wg}$ and $\mathbf{C}^\alpha_g$ respectively.

Proposition 2.2.12. $\mathbf{C}^\alpha_g \subseteq \mathbf{C}^\alpha_{wg}$.

Proof: Each canonical lift satisfies (2.19) by Proposition 2.2.4. The condition is a closed one: if $\varrho_\alpha(\mathbf{X}^{(n)}, \mathbf{X}) \to 0$ then in particular $\mathbb{X}^{(n)}_{s,t} \to \mathbb{X}_{s,t}$ and $X^{(n)}_{s,t} \to X_{s,t}$ for each fixed pair s, t, and both sides of (2.19) pass to the limit. ■

The inclusion is strict, but only just, and for our purposes harmlessly: a weakly geometric α-Hölder rough path is a geometric β-Hölder rough path for every $\beta < \alpha$, a result of Friz and Victoir [26, Ch. 8]. Since our hypotheses always tolerate a slightly smaller exponent, we shall use the two notions interchangeably and say simply *geometric*, flagging the distinction only where it does work.

The content of (2.19) is best seen through what it makes true. If $\mathbf{X}$ is geometric then the chain rule takes its classical form, with no second-order correction: geometric rough paths are exactly those for which the calculus is ordinary calculus. We are not yet in a position to say this precisely — the statement involves an integral we have not built — but it is proved as Proposition 2.3.12 below, and the reader may wish to read that result as the true definition of "geometric". Non-geometric rough paths are those for which the calculus is not ordinary — and the canonical example is one we have already met at length.

Example 2.2.13 (Brownian motion as a rough path)**.** Let W be a d-dimensional Brownian motion. Define, for $s \le t$,

$$\mathbb{W}^{\text{Itô}}_{s,t} := \int_s^t \Big(W_r - W_s\Big) \otimes dW_r, \tag{2.20}$$

the integral being the Itô integral of Definition 1.2.4, and

$$\mathbb{W}^{\text{Strat}}_{s,t} := \mathbb{W}^{\text{Itô}}_{s,t} + \tfrac{1}{2}(t-s)\,\mathrm{Id}, \tag{2.21}$$

where $\mathrm{Id} = \sum_i e_i \otimes e_i$. Then, for every $\alpha \in (\frac{1}{3}, \frac{1}{2})$, both $\mathbf{W}^{\text{Itô}} := (W, \mathbb{W}^{\text{Itô}})$ and $\mathbf{W}^{\text{Strat}} := (W, \mathbb{W}^{\text{Strat}})$ are almost surely rough paths in the sense of Definition 2.2.7. The second is geometric; the first is not.

Proof: Chen's relation (2.16) for $\mathbb{W}^{\text{Itô}}$ is the computation of Proposition 2.2.5 verbatim, the splitting of the outer integral at u and the extraction of the constant $W_u - W_s$ being valid for the Itô integral by linearity. For (2.21) it holds because the added term $\frac{1}{2}(t-s)\mathrm{Id}$ is an increment and therefore telescopes, exactly as in Proposition 2.2.10. The analytic bound

(2.17) follows from the Itô isometry (Theorem 1.2.5), which gives $\mathbb{E}\Big[|\mathbb{W}^{\text{Itô}}_{s,t}|^2\Big] \le C|t-s|^2$, hence $\mathbb{E}\Big[|\mathbb{W}_{s,t}|^{2m}\Big] \le C_m|t-s|^{2m}$ by Gaussian hypercontractivity, followed by a two-parameter version of Kolmogorov's criterion; the details are in [25, Ch. 3].
For the geometric assertion, apply the multidimensional Itô rule (Theorem 1.2.23) to the product $W^i_{s,t}W^j_{s,t}$, using the Itô multiplication table of Chapter 1:

$$W^i_{s,t}W^j_{s,t} = \int_s^t W^i_{s,r}\,dW^j_r + \int_s^t W^j_{s,r}\,dW^i_r + \delta_{ij}(t-s),$$

the last term being $\langle W^i, W^j\rangle_{s,t}$ by Proposition 1.1.47. In tensor notation this reads $2\,\mathrm{Sym}\,\mathbb{W}^{\text{Itô}}_{s,t} = W_{s,t}\otimes W_{s,t} - (t-s)\mathrm{Id}$, so (2.19) fails for the Itô lift by exactly $-\frac{1}{2}(t-s)\mathrm{Id}$ — and holds for the Stratonovich lift, which was defined by adding precisely that discrepancy back. ■

Remark 2.2.14 (The law of the Lévy area). Since $\mathbb{W}$ has just been constructed, it is worth recording what its distribution is. For $d = 2$ the antisymmetric part (2.13) is the scalar $A_{s,t} = \frac{1}{2}\Big(\mathbb{X}^{12}_{s,t} - \mathbb{X}^{21}_{s,t}\Big)$, the signed area swept by the planar Brownian path against its chord, and Lévy's formula gives its conditional characteristic function

$$\mathbb{E}\Big[e^{iuA_{s,t}}\,\Big|\,W_{s,t} = z\Big] \;=\; \frac{u(t-s)/2}{\sinh\!\Big(u(t-s)/2\Big)}\,\exp\left(\frac{|z|^2}{2(t-s)}\Big[1 - \tfrac{u(t-s)}{2}\coth\tfrac{u(t-s)}{2}\Big]\right). \tag{2.22}$$

Unconditionally, $A_{0,t}$ is distributed as $\frac{t}{2}\,C$ with C having characteristic function $1/\cosh(u/2)$. Note that the law is *not* Gaussian, which is the first sign that the second level is a genuinely new object rather than a function of the first; the derivation, via the Cameron–Martin formula for a quadratic functional of Brownian motion, is in [26, Ch. 13]. We do not use (2.22), but it is what one would simulate from if constructing $\mathbb{W}$ directly rather than by discretisation.

The reader should pause over what has just happened. The Itô–Stratonovich correction of Chapter 1, which appeared there as a consequence of the left-endpoint convention in the construction of the integral, has reappeared here as the difference between two *elements of the same space* $\mathbf{C}^\alpha$, lying over the same first level and differing by the increment of $f_t = \frac{1}{2}t\,\mathrm{Id} \in \mathcal{C}^1 \subset \mathcal{C}^{2\alpha}$. Proposition 2.2.10 predicted that such pairs exist; Example 2.2.13 says that the two calculi of Part I are an instance. The choice between them is not forced by the trajectory. It never was.
We turn to the process that motivated the chapter.

Theorem 2.2.15 (Coutin–Qian)**.** *Let W^H be a d-dimensional fractional Brownian motion with $H \in (\frac{1}{3}, \frac{1}{2}]$ and independent components, and let $W^{H,(n)}$ denote its piecewise-linear interpolation over the dyadic partition of mesh 2^{-n}, canonically lifted as in Example 2.2.3. Then, for every $\alpha < H$, the lifts converge in probability in the metric ϱ_α to a limit $\mathbf{B}^H = \Big(W^H, \mathbb{X}^H\Big) \in \mathbf{C}^\alpha_g$, which is therefore a geometric rough path.* ■

The boldface letter is $\mathbf{B}$ and not $\mathbf{W}$ deliberately: the first level of $\mathbf{B}^H$ is the fractional Brownian motion W^H, but $\mathbf{W}^{\text{Itô}}$ and $\mathbf{W}^{\text{Strat}}$ of Example 2.2.13 are lifts of an ordinary Brownian motion, and the two families are never the same object.
The result is due to Coutin and Qian [16]; a modern proof, via the ϱ-variation of the covariance (2.1), is given in [25, Ch. 10]. The mechanism is instructive: one shows that R_H has finite two-dimensional ϱ-variation with $\varrho = 1/(2H)$, and the construction requires $\varrho < \frac{3}{2}$, which is exactly $H > \frac{1}{3}$.

Remark 2.2.16 (The empirical exponent lies outside). Theorem 2.2.15 covers $H > \frac{1}{3}$; by admitting a third iterated integral — a level we have deliberately excluded from Definition

2.2.7 — the range extends to $H > \frac{1}{4}$, and no further, since for $H \leq \frac{1}{4}$ the piecewise-linear lifts fail to converge at all. The exponent $H \approx 0.1$ reported in Subsection 2.1.2 therefore lies outside the reach of the canonical construction, at any number of levels. The precise statement worth remembering is not that no lift exists — non-canonical liftings of W^H below $\frac{1}{4}$ have been constructed by other means, the Lyons–Victoir extension theorem [44] guaranteeing by a Hahn–Banach argument that *every* α-Hölder path admits at least one lift, at the cost of the axiom of choice and with no canonical selection among them — but that no lift exists which is the limit of the piecewise-linear approximations. That is the property one actually needs: a lift that no discretisation converges to is a lift that cannot be simulated, cannot be estimated from data, and carries no guarantee that a numerical scheme is approximating it rather than something else. The obstruction is thus not an absence of objects but an absence of *canonical* ones.

This is not a gap in the exposition but a real boundary, and it would be dishonest to leave it unmarked. Section 2.6 says what lies beyond it. Two remarks temper the difficulty in the meantime. First, the model (2.4) is linear in X and needs no integration theory for W^H; the rough integral is required for the *pricing* equation, where the integrator is the Brownian motion driving the asset and H does not enter the regularity of the integrator at all. Second, and more importantly for Chapter 3, the signature of Section 2.4 is a transform of an *observed* data stream, and observed data streams are piecewise linear. No lifting theorem is needed to compute it, and no value of H obstructs it.

2.3 Fundamental Results

2.3.1 Integration

We now construct the integral. The strategy has three steps: an abstract lemma which converts local estimates into a global object; a class of admissible integrands; and the theorem that puts them together.

Step one: sewing. Every construction of an integral in this dissertation has had the same shape — propose a Riemann sum, show it converges. The following lemma, due in this form to Gubinelli, extracts the mechanism once and for all.

Lemma 2.3.1 (Sewing Lemma). *Let $\beta > 1$ and let $\Xi : \{0 \leq s \leq t \leq T\} \to \mathbb{R}^m$ be continuous with $\Xi_{t,t} = 0$ and*

$$\left|\delta\Xi_{s,u,t}\right| \;\leq\; C\,|t-s|^{\beta}, \qquad \textit{where } \delta\Xi_{s,u,t} := \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t}, \tag{2.23}$$

for all $s \leq u \leq t$. Then there exists a unique function $\mathcal{I}\Xi$, of the same two variables, which is additive *— that is, $(\mathcal{I}\Xi)_{s,t} = (\mathcal{I}\Xi)_{s,u} + (\mathcal{I}\Xi)_{u,t}$ for all $s \leq u \leq t$ — and satisfies*

$$\left|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}\right| \;\leq\; \frac{C}{1 - 2^{\,1-\beta}}\,|t-s|^{\beta}. \tag{2.24}$$

Moreover $(\mathcal{I}\Xi)_{s,t} = \lim_{|\mathcal{P}|\to 0} \sum_{[u,v]\in\mathcal{P}} \Xi_{u,v}$, the limit being over partitions $\mathcal{P}$ of $[s,t]$ with mesh tending to zero.

Proof (uniqueness): Suppose A and $\widetilde{A}$ are both additive and both satisfy (2.24), with constants C_1 and C_2. Then $D := A - \widetilde{A}$ is additive and $|D_{s,t}| \leq (C_1 + C_2)|t-s|^{\beta} =: K|t-s|^{\beta}$. Fix $s < t$ and let $\mathcal{P}_n$ be the uniform partition of $[s,t]$ into n pieces. Additivity gives

$$|D_{s,t}| = \Bigg|\sum_{[u,v]\in\mathcal{P}_n} D_{u,v}\Bigg| \leq \sum_{[u,v]\in\mathcal{P}_n} K\,|v-u|^{\beta} = n\cdot K\Big(\frac{t-s}{n}\Big)^{\beta} = K\,|t-s|^{\beta}\,n^{1-\beta}.$$

Since $\beta > 1$ the right-hand side tends to 0 as $n \to \infty$, so $D \equiv 0$. The existence half, which proceeds by dyadic refinement, is proved in Appendix B.7. ■

The hypothesis (2.23) is the exact quantitative statement of "Ξ is almost additive", and the conclusion is that an almost additive function is uniformly close to a unique genuinely additive one. The strictness of $\beta > 1$ is what the counting argument above consumes, and it is not improvable: at $\beta = 1$ the factor $n^{1-\beta}$ is 1 and the argument collapses. Everything in this section is an application of the lemma with a specific Ξ, and the whole difficulty is arranging for $\beta > 1$ to hold.

Step two: admissible integrands. We cannot integrate an arbitrary path against $\mathbf{X}$; the integrand must be compatible with the integrator in a sense that Chapter 1 would have called adaptedness, but which is here entirely deterministic.

Definition 2.3.2 (Controlled rough path)**.** Let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$. A pair (Y, Y') of paths $Y \in \mathcal{C}^\alpha([0,T], W)$ and $Y' \in \mathcal{C}^\alpha\Big([0,T], \mathcal{L}(V,W)\Big)$ is a path **controlled by X** if the remainder defined by

$$R^Y_{s,t} \;:=\; Y_{s,t} - Y'_s\, X_{s,t}, \qquad Y_{s,t} := Y_t - Y_s, \tag{2.25}$$

satisfies $\|R^Y\|_{2\alpha} < \infty$. We write $(Y, Y') \in \mathcal{D}^{2\alpha}_X$ and call Y' the **Gubinelli derivative** of Y with respect to X.

Equation (2.25) is a first-order Taylor expansion of Y along X: the path Y is required to look, locally, like a linear image of X, with an error of order 2α rather than the α that mere Hölder continuity would give. The pair $\Big(\mathcal{D}^{2\alpha}_X, \|\cdot\|\Big)$ with $\|(Y,Y')\|_{X,2\alpha} := \|Y'\|_\alpha + \|R^Y\|_{2\alpha}$ is a Banach space once the initial values are fixed [25, Ch. 4].

Remark 2.3.3 (Y' is data, not a derivative). The notation Y' is suggestive but must not be taken literally: in general Y' is *not* determined by Y and X. If X happens to be smooth, say $X_t = t$, then any Y' whatsoever satisfies (2.25) with a remainder of the required order, since $Y'_s X_{s,t} = O(|t-s|)$ is itself 2α-small when $2\alpha \le 1$. Uniqueness holds only when X is genuinely irregular — "truly rough" in the terminology of [25, Ch. 6] — which is the case almost surely for Brownian motion and for W^H. Like the second level $\mathbb{X}$ itself, Y' is part of the given data. The theory is consistently one in which one carries more information than the classical objects contain.

Example 2.3.4. i) The integrator controls itself: $(X, \mathrm{Id}) \in \mathcal{D}^{2\alpha}_X$, with $R^X \equiv 0$.
ii) If $F \in \mathcal{C}^2_b(W, \widetilde{W})$ and $(Y,Y') \in \mathcal{D}^{2\alpha}_X$, then $\Big(F(Y),\, DF(Y)Y'\Big) \in \mathcal{D}^{2\alpha}_X$. Indeed, Taylor's theorem with integral remainder gives

$$F(Y_t) - F(Y_s) = DF(Y_s)Y_{s,t} + O\Big(|Y_{s,t}|^2\Big) = DF(Y_s)Y'_s X_{s,t} + DF(Y_s)R^Y_{s,t} + O\Big(|t-s|^{2\alpha}\Big),$$

and the last two terms are both of order 2α. This closes the class under the nonlinearities that appear in differential equations, and is the reason Definition 2.3.2 is the right one.

Step three: the integral. The naïve Riemann sum $\sum Y_u X_{u,v}$ does not converge: its defect is of order $|t-s|^{2\alpha}$, and $2\alpha \le 1$, so Lemma 2.3.1 does not apply. The remedy is to add a second-order term, and the expansion (2.25) tells us precisely which one.

Theorem 2.3.5 (Gubinelli)**.** *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, let $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ and let $(Y,Y') \in \mathcal{D}^{2\alpha}_X$. Then the **compensated Riemann sums***

$$\int_s^t Y_r\, d\mathbf{X}_r \;:=\; \lim_{|\mathcal{P}|\to 0} \sum_{[u,v]\in\mathcal{P}} \Big(Y_u\, X_{u,v} \;+\; Y'_u\, \mathbb{X}_{u,v}\Big) \tag{2.26}$$

converge, and the limit satisfies

$$\left|\int_s^t Y_r\,d\mathbf{X}_r - Y_sX_{s,t} - Y'_s\mathbb{X}_{s,t}\right| \;\le\; C\Big(\|X\|_\alpha\|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha}\|Y'\|_\alpha\Big)\,|t-s|^{3\alpha}, \tag{2.27}$$

for a constant $C = C(\alpha)$. Moreover the indefinite integral is itself controlled by $\mathbf{X}$:

$$\Big(\int_0^\cdot Y_r\,d\mathbf{X}_r,\; Y\Big) \;\in\; \mathcal{D}_X^{2\alpha}. \tag{2.28}$$

The proof, an application of Lemma 2.3.1 to $\Xi_{s,t} := Y_sX_{s,t} + Y'_s\mathbb{X}_{s,t}$, is given in Appendix B.8.

Remark 2.3.6 (Where $\alpha > \frac{1}{3}$ comes from). The exponent 3α in (2.27) is the whole story. Computing $\delta\Xi_{s,u,t}$ for the choice above, and using Chen's relation (2.16) to cancel the leading terms — this is the computation in the appendix, and it is the only place where (2.16) is used — one finds

$$-\delta\Xi_{s,u,t} = R^Y_{s,u}X_{u,t} + \Big(Y'_u - Y'_s\Big)\mathbb{X}_{u,t}, \tag{2.29}$$

whose two terms are of order $2\alpha + \alpha$ and $\alpha + 2\alpha$ respectively. So $\beta = 3\alpha$, and Lemma 2.3.1 demands $\beta > 1$, that is $\alpha > \frac{1}{3}$. Below that threshold the second level no longer suffices and a third iterated integral must be carried; below $\frac{1}{4}$, a fourth; and in general $\lfloor 1/\alpha \rfloor$ levels are required. We have restricted to (2.15) precisely to avoid the algebra that the general case demands, at no cost to the ideas.

Theorem 2.3.7 (Stability of the integral)**.** *Fix $M < \infty$ and consider $\mathbf{X}, \widetilde{\mathbf{X}} \in \mathbf{C}^\alpha$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, $(\widetilde{Y}, \widetilde{Y'}) \in \mathcal{D}_{\widetilde{X}}^{2\alpha}$, all of norm at most M. Then, with $C = C(M, \alpha)$,*

$$\left\|\int_0^\cdot Y\,d\mathbf{X} - \int_0^\cdot \widetilde{Y}\,d\widetilde{\mathbf{X}}\right\|_\alpha \;\le\; C\Big(\varrho_\alpha\big(\mathbf{X}, \widetilde{\mathbf{X}}\big) + \big|Y_0 - \widetilde{Y}_0\big| + \big|Y'_0 - \widetilde{Y'}_0\big| + \|Y, Y'; \widetilde{Y}, \widetilde{Y'}\|_{X,\widetilde{X},2\alpha}\Big), \tag{2.30}$$

where the last term is the distance between two controlled paths lying over different *rough paths,*

$$\|Y, Y'; \widetilde{Y}, \widetilde{Y'}\|_{X,\widetilde{X},2\alpha} \;:=\; \Big\|Y' - \widetilde{Y'}\Big\|_\alpha \;+\; \Big\|R^Y - R^{\widetilde{Y}}\Big\|_{2\alpha}, \tag{2.31}$$

with $R^Y_{s,t} = Y_{s,t} - Y'_sX_{s,t}$ and $R^{\widetilde{Y}}_{s,t} = \widetilde{Y}_{s,t} - \widetilde{Y'}_s\widetilde{X}_{s,t}$ as in (2.25)*.* ∎

The quantity (2.31) requires a word, since (Y, Y') and $(\widetilde{Y}, \widetilde{Y'})$ live in the *different* Banach spaces $\mathcal{D}_X^{2\alpha}$ and $\mathcal{D}_{\widetilde{X}}^{2\alpha}$ and cannot simply be subtracted. What is compared instead is the pair of Gubinelli derivatives, which live in a common space, together with the pair of remainders, which also do; the two rough paths enter only through the definitions of R^Y and $R^{\widetilde{Y}}$. This is the standard device, and it is the reason the estimate carries $\varrho_\alpha(\mathbf{X}, \widetilde{\mathbf{X}})$ as a separate term rather than absorbing it.

This is proved in [25, Thm. 4.17] by re-running the proof of Theorem 2.3.5 on differences. We record it because it, and not Theorem 2.3.5 itself, is what makes the continuity statement of Subsection 2.3.4 possible.

The construction would be worthless if it disagreed with the theory of Part I where both apply. It does not.

Proposition 2.3.8 (Consistency)**.** *Let W be a Brownian motion and let $F \in \mathcal{C}_b^3$. Then, almost surely,*

$$\int_0^t F(W_r)\,d\mathbf{W}_r^{\text{Itô}} = \int_0^t F(W_r)\,dW_r, \qquad \int_0^t F(W_r)\,d\mathbf{W}_r^{\text{Strat}} = \int_0^t F(W_r)\circ dW_r, \tag{2.32}$$

the right-hand sides being the Itô integral of Definition 1.2.4 and the Stratonovich integral of Definition 1.2.9 respectively. If X is smooth, the rough integral against its canonical lift is the Riemann–Stieltjes integral. ∎

See [25, Ch. 5]. The first identity is the one to appreciate: the left-hand side is a deterministic limit of Riemann sums, computed path by path with no reference to the filtration, and the right-hand side is an $L^2(\Omega)$ limit that exists only up to null sets. That they agree means the Itô integral, for this class of integrands, admits a pathwise description after all — at the price of carrying $\mathbb{W}^{\text{Itô}}$, an object which is itself defined only through Itô's construction. Rough path theory does not eliminate probability; it isolates it into a single step, performed once, after which everything is deterministic.

Remark 2.3.9 (The integral depends on the lift). We can now make precise the claim of Subsection 2.2.3, that geometric rough paths are those on which ordinary calculus survives. The following quantity measures the failure of (2.19), and turns out to be exactly the correction term in the chain rule.

Definition 2.3.10 (Bracket of a rough path)**.** For $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ define

$$[\mathbf{X}]_t \;:=\; X_{0,t} \otimes X_{0,t} \;-\; 2\,\mathrm{Sym}\,\mathbb{X}_{0,t} \;\in\; V \otimes V. \tag{2.33}$$

Lemma 2.3.11. *The bracket is a genuine path, in the sense that its increments are given by the corresponding two-parameter expression:*

$$[\mathbf{X}]_t - [\mathbf{X}]_s \;=\; X_{s,t} \otimes X_{s,t} - 2\,\mathrm{Sym}\,\mathbb{X}_{s,t}, \tag{2.34}$$

and $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$.

Proof: Expanding $X_{0,t} = X_{0,s} + X_{s,t}$ in the first term of (2.33),

$$X_{0,t} \otimes X_{0,t} - X_{0,s} \otimes X_{0,s} = X_{0,s} \otimes X_{s,t} + X_{s,t} \otimes X_{0,s} + X_{s,t} \otimes X_{s,t},$$

while Chen's relation (2.16) gives $\mathbb{X}_{0,t} - \mathbb{X}_{0,s} = \mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t}$, so that

$$2\,\mathrm{Sym}\,\mathbb{X}_{0,t} - 2\,\mathrm{Sym}\,\mathbb{X}_{0,s} = 2\,\mathrm{Sym}\,\mathbb{X}_{s,t} + X_{0,s} \otimes X_{s,t} + X_{s,t} \otimes X_{0,s}.$$

Subtracting, the two cross terms cancel and (2.34) follows. The regularity is then immediate from Definition 2.2.7, both terms on the right of (2.34) being of order $|t-s|^{2\alpha}$. ∎

By construction $\mathbf{X}$ is weakly geometric precisely when $[\mathbf{X}] \equiv 0$; Definition 2.2.11 could equally well have been phrased that way.

Proposition 2.3.12 (Rough chain rule)**.** *Let* $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ *with* $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, *and let* $F \in \mathcal{C}_b^3(V, W)$. *Then* $\big(F(X), DF(X)\big) \in \mathcal{D}_X^{2\alpha}$, *the rough integral* $\int_0^t DF(X_r)\, d\mathbf{X}_r$ *is defined by Theorem 2.3.5, and*

$$F(X_t) - F(X_0) \;=\; \int_0^t DF(X_r)\, d\mathbf{X}_r \;+\; \frac{1}{2} \int_0^t D^2 F(X_r)\, d[\mathbf{X}]_r, \tag{2.35}$$

the second integral being a Young integral, well defined since $D^2F(X) \in \mathcal{C}^\alpha$, $[\mathbf{X}] \in \mathcal{C}^{2\alpha}$ *and* $3\alpha > 1$.
In particular, if $\mathbf{X}$ *is weakly geometric then* $[\mathbf{X}] \equiv 0$ *and* (2.35) *reduces to the classical chain rule*

$$F(X_t) - F(X_0) \;=\; \int_0^t DF(X_r)\, d\mathbf{X}_r. \tag{2.36}$$

∎

A proof is in [25, Ch. 7]; the mechanism is a second-order Taylor expansion of F along the partition, in which the symmetric part of $\mathbb{X}$ is exactly what the quadratic term of the expansion produces, so that the two agree and cancel precisely when (2.19) holds.
The corollary is the one we have been aiming at since Subsection 2.2.3.

Corollary 2.3.13 (Itô's formula)**.** *Let W be a d-dimensional Brownian motion and $F \in \mathcal{C}_b^3$. Applying* (2.35) *to the two lifts of Example 2.2.13:*

i) for $\mathbf{W}^{\mathrm{Strat}}$, which is geometric, one obtains $F(W_t) - F(W_0) = \int_0^t DF(W_r) \circ dW_r$, the classical chain rule of Stratonovich calculus;

ii) for $\mathbf{W}^{\mathrm{Itô}}$ one computes from (2.34) *and the identity* $2\,\mathrm{Sym}\,\mathbb{W}^{\mathrm{Itô}}_{s,t} = W_{s,t} \otimes W_{s,t} - (t-s)\mathrm{Id}$ *established in Example 2.2.13 that $[\mathbf{W}^{\mathrm{Itô}}]_t = t\,\mathrm{Id}$, whence*

$$F(W_t) - F(W_0) = \int_0^t DF(W_r)\,dW_r + \frac{1}{2}\int_0^t \Delta F(W_r)\,dr, \tag{2.37}$$

which is the case $\theta(t,x) = F(x)$, $X = W$ of Theorem 1.2.23 of Chapter 1.

The qualification matters, and the accounting is worth making explicit. Theorem 1.2.23 covers $\theta(t, X_t)$ for a general n-dimensional Itô process and asks only that $\theta \in \mathcal{C}^{1,2}$; what we have recovered is the time-homogeneous case, driven by Brownian motion, under the stronger hypothesis $F \in \mathcal{C}_b^3$. The rough-path route is therefore not a strengthening of Itô's formula but a *re-derivation of a special case of it on different foundations*, and the extra derivative is the price of those foundations, exactly as in Remark 2.3.18. What is bought in exchange is that (2.37) now holds pathwise, for a fixed trajectory and its lift, with both sides continuous in the rough path metric — neither of which the classical statement provides.

It is worth being explicit about what has just been shown, because it is the cleanest summary of the chapter's thesis. In Chapter 1, the second-order term of Itô's formula arose from the non-vanishing quadratic variation of Brownian motion, and appeared as an irreducible feature of stochastic calculus — the price of integrating against a martingale. Here it has arisen from the bracket (2.33) of a particular *lift*, and the Stratonovich lift of the same trajectory has no such term at all. The correction was never a property of Brownian motion. It was a property of a choice, made implicitly in Subsection 1.2.1 when the integrand was evaluated at the left endpoint of each subinterval, and made explicitly here by writing down $\mathbb{W}^{\mathrm{Itô}}$ rather than $\mathbb{W}^{\mathrm{Strat}}$. Let $\mathbf{X} = (X, \mathbb{X})$ and $\widetilde{\mathbf{X}} = (X, \mathbb{X} + \delta f)$ be two lifts of the same path, as in Proposition 2.2.10. Directly from (2.26),

$$\int_s^t Y\,d\widetilde{\mathbf{X}} = \int_s^t Y\,d\mathbf{X} + \int_s^t Y'_r\,df_r, \tag{2.38}$$

the last integral existing in the Young sense since $Y' \in \mathcal{C}^\alpha$, $f \in \mathcal{C}^{2\alpha}$ and $3\alpha > 1$. Taking $f_t = \frac{1}{2}t\,\mathrm{Id}$ and $Y' = DF(W)$ recovers, by Example 2.2.13, the classical relation $\int F(W) \circ dW = \int F(W)\,dW + \frac{1}{2}\int F'(W)\,dr$ of Chapter 1. The Itô–Stratonovich correction is thus not a feature of stochastic calculus at all: it is the general formula (2.38) evaluated at a particular f.

2.3.2 Differential Equations

With an integral in hand, differential equations are defined in the only way available: as integral equations. What requires care is that the unknown is now a *pair*.

Definition 2.3.14 (Rough differential equation)**.** Let $\mathbf{X} \in \mathbf{C}^\alpha$, let $\xi \in W$ and let $f : W \to \mathcal{L}(V, W)$. A pair $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ is a **solution** of the rough differential equation

$$dY_t = f(Y_t)\,d\mathbf{X}_t, \qquad Y_0 = \xi, \tag{2.39}$$

if $Y' = f(Y)$ and, for every $t \in [0,T]$,

$$Y_t = \xi + \int_0^t f(Y_r)\,d\mathbf{X}_r, \tag{2.40}$$

the integral being that of Theorem 2.3.5, which is well defined because $\big(f(Y), Df(Y)Y'\big) \in \mathcal{D}_X^{2\alpha}$ by Example 2.3.4 ii).

Remark 2.3.15 (Why $Y' = f(Y)$ is imposed). The constraint is not cosmetic; without it uniqueness fails outright. Consider $X \in \mathcal{C}^1$ lifted canonically, so that $|X_{s,t}| \lesssim |t-s|$. Then for *any* choice of Y' the term $Y'_s X_{s,t}$ is $O(|t-s|)$ and can be absorbed into a 2α-remainder, so every Y' makes (Y, Y') controlled and (2.40) carries no information about it. Requiring $Y' = f(Y)$ pins the derivative to the equation and restores well-posedness. The reader may compare Definition 1.2.16, where the analogous role is played by the requirement that the solution be adapted.

There is an equivalent formulation, due to Davie, which dispenses with controlled paths entirely and which is the one a numerical analyst would write down.

Proposition 2.3.16 (Davie's formulation)**.** *A path $Y \in \mathcal{C}^\alpha$ solves* (2.39) *in the sense of Definition 2.3.14 if and only if*

$$Y_{s,t} \; = \; f(Y_s)\, X_{s,t} \; + \; Df(Y_s)f(Y_s)\, \mathbb{X}_{s,t} \; + \; o\Big(|t-s|\Big) \tag{2.41}$$

uniformly in $s \leq t$. ∎

The proof is in [25, Prop. 8.8]. Expression (2.41) is worth recognising: truncated after the first term it is the Euler scheme, and as it stands it is the *Milstein scheme*, with $\mathbb{X}_{s,t}$ playing the role of the iterated integral that the Milstein correction requires. Rough path theory therefore does not merely license the numerical schemes used in practice; it identifies the object those schemes were always approximating, and Theorem 2.3.19 will supply their convergence. We shall use this in Chapter 3.

2.3.3 Existence

Theorem 2.3.17 (Local existence and uniqueness)**.** *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, let $\mathbf{X} \in \mathbf{C}^\alpha([0,T], V)$ and let $f \in \mathcal{C}^3_b\Big(W, \mathcal{L}(V, W)\Big)$. Then for every $\xi \in W$ there exist $T_0 \in (0, T]$ and a unique $(Y, Y') \in \mathcal{D}^{2\alpha}_X$ on $[0, T_0]$ solving* (2.39)*. The time T_0 depends only on α, $\|f\|_{\mathcal{C}^3_b}$ and $\|\!|\mathbf{X}|\!\|_\alpha$, and* not *on ξ; consequently the solution extends to a unique global solution on $[0, T]$.*

The proof is given in Appendix B.9. Because the mechanism matters more than the estimates, we describe its shape here.
One defines the map

$$\mathcal{M}(Y, Y') \; := \; \Big(\xi + \int_0^{\cdot} f(Y_r)\, d\mathbf{X}_r, \; f(Y)\Big) \tag{2.42}$$

on the Banach space $\mathcal{D}^{2\alpha}_X$, whose fixed points are by Definition 2.3.14 exactly the solutions. One then shows that $\mathcal{M}$ maps a suitable closed ball into itself and is a contraction on it, provided the time horizon is small; the Banach fixed point theorem does the rest. This is precisely the Picard iteration by which Theorem 1.2.18 was proved in Chapter 1, and the analogy is exact — but the argument is now entirely deterministic, and takes place in the space of controlled paths rather than in $L^2(\Omega)$.
The contraction factor comes from the horizon. Every estimate in Theorem 2.3.5 carries a factor T_0^α when restricted to $[0, T_0]$, and choosing T_0 small makes that factor as small as desired. The subtle point — and the reason the theorem is stated as it is — is that the admissible T_0 must be *independent of the initial condition*, since otherwise the successive restarts used to reach T might have horizons shrinking to zero and the solution would fail to be global. That independence is secured by the fact, visible in (2.27), that the estimates involve $\|Y'\|_\alpha$ and $\|R^Y\|_{2\alpha}$ but never $|Y_0|$, together with the boundedness of f. With T_0 uniform, one solves on $[0, T_0]$, restarts at Y_{T_0}, and repeats finitely often.

Remark 2.3.18 (On the regularity of f). Three bounded derivatives are required, one more than the two that Example 2.3.4 ii) needs to make $f(Y)$ controlled. The extra derivative is spent on the contraction estimate, where one must compare $f(Y)$ with $f(\widetilde{Y})$ *as controlled paths*, which involves Df and its Lipschitz constant. This is the analogue of the Lipschitz hypothesis in Definition 1.2.17; that a rougher driver should demand a smoother coefficient is the recurring trade of this subject. Without boundedness one still obtains a maximal solution, which may explode in finite time exactly as in the ODE case.

2.3.4 Uniqueness and Continuity

We arrive at the theorem for which the whole apparatus was built. It answers the question raised in Subsection 2.1.3, and it answers it affirmatively.

Theorem 2.3.19 (Universal Limit Theorem; continuity of the Itô–Lyons map)**.** *Let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $f \in \mathcal{C}_b^3$. For $\mathbf{X} \in \mathbf{C}^\alpha$ and $\xi \in W$ let $\Phi(\xi, \mathbf{X}) := Y$ denote the unique solution of* (2.39) *given by Theorem 2.3.17. Then Φ is* locally Lipschitz continuous*: for every $M < \infty$ there is $C = C(M, \alpha, f)$ such that, whenever $\|\!|\mathbf{X}|\!\|_\alpha, \|\!|\widetilde{\mathbf{X}}|\!\|_\alpha \leq M$,*

$$\left\|\Phi(\xi, \mathbf{X}) - \Phi(\tilde{\xi}, \widetilde{\mathbf{X}})\right\|_\alpha \;\leq\; C\Big(\left|\xi - \tilde{\xi}\right| \;+\; \varrho_\alpha\big(\mathbf{X}, \widetilde{\mathbf{X}}\big)\Big). \tag{2.43}$$

The proof is given in Appendix B.10; it combines the stability of rough integration (Theorem 2.3.7) with the stability of composition, applied to the fixed point characterisation (2.42).

The significance of (2.43) is best measured against what Part I could say. There, the solution of an SDE was produced as an equivalence class of random variables, obtained through an $L^2(\Omega)$ limit; the map from the driving Brownian path to the solution path existed only as a measurable map defined up to null sets, and Example 2.1.5 showed that it could not be continuous in the uniform topology. Theorem 2.3.19 says that once the driving signal is *enhanced* — once we carry $\mathbb{X}$ alongside X and measure distance with ϱ_α — the solution map becomes not merely continuous but locally Lipschitz, and entirely deterministic. All the probability in the theory of stochastic differential equations has been compressed into the single measurable step of constructing $\mathbb{W}$, performed once in Example 2.2.13. Everything downstream of it is analysis.
Three consequences deserve mention.

Corollary 2.3.20 (Wong–Zakai)**.** *Let W be a Brownian motion, $W^{(n)}$ its piecewise-linear interpolation over a partition of mesh tending to zero, and let $Y^{(n)}$ solve the ordinary differential equation $dY^{(n)} = f(Y^{(n)})\, dW^{(n)}$. Then $Y^{(n)}$ converges, almost surely and in α-Hölder norm, to the solution of the* Stratonovich *equation $dY = f(Y) \circ dW$.*

Proof: The canonical lifts of $W^{(n)}$ converge in ϱ_α to $\mathbf{W}^{\text{Strat}}$ — this is the content of [25, Ch. 3], and the Stratonovich rather than the Itô lift appears because each $W^{(n)}$ is smooth, hence its lift is geometric, hence so is the limit by Proposition 2.2.12. Theorem 2.3.19 then transfers the convergence to the solutions, and Proposition 2.3.8 identifies the limit. ■

This is a genuinely non-trivial classical theorem, and here it is a two-line corollary. It also explains the phenomenon that made it famous: approximating Brownian motion by smooth paths and solving the resulting ODEs does *not* converge to the Itô solution, because the lifts converge to the geometric one. In the language of this chapter, the limit remembers the area, and the area of a smooth approximation is the Stratonovich one.
Second, Theorem 2.3.19 makes numerical analysis possible. The Milstein scheme (2.41) converges because the map is continuous, and the rate is controlled by (2.43); no such argument is available within Itô theory, where discretisation error must be estimated afresh in $L^2(\Omega)$ for each scheme.

Third — and this is the reason the theory is called *universal* — nothing in Theorem 2.3.19 refers to Brownian motion, to a filtration, or to a probability measure. Any process that can be lifted to $\mathbf{C}^\alpha$ inherits the entire theory at once: fractional Brownian motion for $H > \frac{1}{3}$ by Theorem 2.2.15, Gaussian processes with sufficiently regular covariance, Markov processes, and the piecewise-linear paths formed from observed data. The existence, uniqueness and continuity results of this section were obtained *once*, and they apply to all of them.

2.4 The Signature Transform

The rough paths of Definition 2.2.7 carry two levels because two are what an $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ *integrator* requires — the regularity of the integrand is governed instead by the controlled-path condition (2.25). Nothing prevents us from carrying more, and if we carry all of them we obtain an object of independent interest, and the natural direction in which the present work extends.

Theorem 2.4.1 (Lyons' extension theorem)**.** *Let* $\mathbf{X} = (X, \mathbb{X}) \in \mathbf{C}^\alpha$ *with* $\alpha \in (\frac{1}{3}, \frac{1}{2}]$*. Then there is a unique extension of* $\mathbf{X}$ *to a family* $\left(1, X^1_{s,t}, X^2_{s,t}, X^3_{s,t}, \dots\right)$*, with* $X^1 = X$*,* $X^2 = \mathbb{X}$ *and* $X^n_{s,t} \in V^{\otimes n}$*, which satisfies Chen's relation at every level,*

$$X^n_{s,t} = \sum_{k=0}^{n} X^k_{s,u} \otimes X^{n-k}_{u,t}, \qquad s \le u \le t, \tag{2.44}$$

and the bounds $|X^n_{s,t}| \le C_n \, |t-s|^{n\alpha}$ *for every* n*.* ■

The theorem is Lyons' [43]; see also [30, Ch. 3]. Its content is that the higher levels are *not* additional data: once two are given, the rest are determined, and determined by a construction which is again an instance of the sewing lemma. This is the precise sense in which Definition 2.2.7 loses nothing by truncating.

Definition 2.4.2 (Signature)**.** The **signature** of a rough path $\mathbf{X}$ over $[0, T]$ is the full collection of its iterated integrals over the whole interval,

$$\mathrm{S}(\mathbf{X}) \;:=\; \left(1,\; X^1_{0,T},\; X^2_{0,T},\; X^3_{0,T},\; \dots\right), \tag{2.45}$$

and its **truncation at level** N, written $\mathrm{S}^N(\mathbf{X})$, is the finite collection $\left(1, X^1_{0,T}, \dots, X^N_{0,T}\right) \in T^{(N)}(V)$. For a path x of bounded variation these are the ordinary iterated integrals

$$X^n_{0,T} = \int_{0<t_1<\cdots<t_n<T} dx_{t_1} \otimes \cdots \otimes dx_{t_n}. \tag{2.46}$$

Example 2.4.3. For $d = 1$ the tensor products are ordinary products and (2.46) evaluates to $X^n_{0,T} = (x_T - x_0)^n / n!$, independently of the trajectory. The signature of a scalar path is therefore $\exp(x_T - x_0)$ and carries no information beyond the total increment. All the content of the transform lies in the non-commutativity of $\otimes$, which is to say in dimension $d \ge 2$.

Three properties make the signature useful, and they are worth stating separately because each answers a different practical objection.

Proposition 2.4.4 (Factorial decay)**.** *For a path* x *of bounded variation with total variation* $L := V_1(x)_T$*,*

$$\left|X^n_{0,T}\right| \;\le\; \frac{L^n}{n!}. \tag{2.47}$$

Proof: Bounding (2.46) termwise, the integrand has modulus at most $|dx_{t_1}| \cdots |dx_{t_n}|$, and the domain of integration is the simplex $0 < t_1 < \cdots < t_n < T$, which is the $1/n!$ fraction of the cube on which the unrestricted product integrates to L^n. ■

The estimate answers the objection that (2.45) is an infinite object: the terms decay factorially, so a truncation at modest N — in practice $N \in \{3, 4, 5\}$ — retains almost all of the information. This is what makes S^N a usable feature set.

Proposition 2.4.5 (Shuffle product and linearisation)**.** *Let* $\mathbf{X}$ *be geometric. Write* $X^{n;I}_{0,T}$ *for the coordinate of* $X^n_{0,T}$ *along the basis vector* $e_{i_1} \otimes \cdots \otimes e_{i_n}$*, indexed by the word* $I = (i_1, \ldots, i_n)$*. Then for any two words* $I = (i_1, \ldots, i_m)$ *and* $J = (j_1, \ldots, j_n)$ *in the letters* $\{1, \ldots, d\}$*,*

$$X^{m;I}_{0,T} \cdot X^{n;J}_{0,T} \;=\; \sum_{K \,\in\, I ш J} X^{m+n;K}_{0,T}, \tag{2.48}$$

the sum being over all $\binom{m+n}{m}$ *interleavings of* I *and* J *preserving the internal order of each.* ■

Relation (2.48) — the *shuffle product formula*, proved in [30, Lem. 1.15] by splitting the product of two simplices into the disjoint union of the simplices indexed by the shuffles — has a consequence out of all proportion to its appearance. It says that the product of two coordinates of the signature is a *linear combination* of other coordinates. Hence every polynomial in the entries of $\mathrm{S}(\mathbf{X})$ is again a linear functional of $\mathrm{S}(\mathbf{X})$, and any continuous functional of the path can be approximated, by Stone–Weierstrass, by a *linear* functional of its signature. A nonlinear problem on path space becomes a linear problem on signature space. This is the theoretical basis for the use of signatures as features in statistical learning; see [13] and [42].
It remains to be shown that nothing is lost.

Definition 2.4.6 (Tree-like paths)**.** A *real tree* is a metric space in which any two points are joined by a unique non-self-intersecting path, which is moreover a geodesic. A continuous path $x : [0, T] \to V$ is **tree-like** if there exist a real tree $\mathcal{T}$ and continuous maps $\phi : [0, T] \to \mathcal{T}$ and $\psi : \mathcal{T} \to V$ with $\phi(0) = \phi(T)$ and $x = \psi \circ \phi$. A path is **tree-reduced** if it contains no tree-like piece.

The second notion is the one that makes Theorem 2.4.7 an honest bijection: the signature is injective not on all paths but on the tree-reduced ones, which are exactly the representatives of the equivalence classes the theorem describes.
Informally, a tree-like path is one that retraces its own steps and thereby cancels itself: it goes out along some route and returns along the same route, enclosing no area. Such excursions contribute nothing to any iterated integral, and the following theorem says that they are the *only* thing the signature forgets.

Theorem 2.4.7 (Uniqueness of the signature)**.** *Let* $\mathbf{X}$ *be a weakly geometric rough path. Then* $\mathrm{S}(\mathbf{X}) = \mathbf{1}$ *if and only if* $t \mapsto X_{0,t}$ *is tree-like. Consequently — applying this to* $\mathbf{X} \star \widetilde{\mathbf{X}}^{-1}$*, the concatenation of* $\mathbf{X}$ *with the time-reversal of* $\widetilde{\mathbf{X}}$*, whose signature is* $\mathrm{S}(\mathbf{X}) \otimes \mathrm{S}(\widetilde{\mathbf{X}})^{-1}$ *by the group structure of* $T((V))$ *— two weakly geometric rough paths have the same signature if and only if they coincide up to tree-like pieces.* ■

The bounded-variation case is due to Hambly and Lyons [34]; the extension to rough paths is due to Boedihardjo, Geng, Lyons and Yang [8]. A sketch of the argument, which proceeds by showing that the space of signatures is itself a real tree, is in [30, §4.3].

We can now close a loop opened in Chapter 1. There, discussing the log-normal distribution, we observed that a probability law need not be determined by its moments — Heyde's construction, recorded in Appendix A.3, exhibits distinct densities sharing every moment with the log-normal

— and remarked that "knowing every moment of an object is strictly weaker than knowing the object", promising that for *paths* the analogous statement could be made to go the right way. Theorem 2.4.7 is that statement. The iterated integrals are the moments of a path; unlike the moments of a random variable, they determine it, and the only ambiguity is the geometrically trivial one of tree-like excursions, which can be removed outright by the standard device of *time augmentation* — replacing x_t by (t, x_t), whose first coordinate is strictly increasing and which therefore admits no tree-like piece at all. The same device removes the invariance of (2.46) under reparametrisation, which is otherwise a genuine loss of information whenever the speed at which a path is traversed carries meaning — as it does, emphatically, for a price series.

The analogy can in fact be pushed one step further, and it is worth doing so, because the promise made in Chapter 1 was about *laws* and Theorem 2.4.7 is about a fixed path. The statement that genuinely corresponds to "the moments determine the distribution" is obtained by averaging.

Definition 2.4.8 (Expected signature)**.** Let $\mathbf{X}$ be a random geometric rough path, that is, a random variable with values in $\mathbf{C}_g^\alpha([0,T],V)$. Its **expected signature** is the element of the full tensor algebra $T((V)) = \prod_{n\geq 0} V^{\otimes n}$ of Definition 2.2.2, obtained by averaging the signature level by level,

$$\mathbb{E}\Big[\mathrm{S}(\mathbf{X})\Big] \;=\; \Big(1,\; \mathbb{E}\Big[X^1_{0,T}\Big],\; \mathbb{E}\Big[X^2_{0,T}\Big],\; \ldots\Big), \tag{2.49}$$

whenever each expectation exists.

Theorem 2.4.9 (Chevyrev–Lyons)**.** *Let* $\mathbf{X}$ *be a random geometric rough path whose expected signature* (2.49) *has infinite radius of convergence, in the sense that* $\sum_{n\geq 0} \lambda^n \left|\mathbb{E}[X^n_{0,T}]\right| < \infty$ *for every* $\lambda > 0$*. Then* $\mathbb{E}[\mathrm{S}(\mathbf{X})]$ *determines the law of* $\mathbf{X}$*, up to the tree-like equivalence of Theorem 2.4.7.* ■

The result is due to Chevyrev and Lyons [14]. It is the precise analogue, for measures on path space, of the elementary fact that a compactly supported law on $\mathbb{R}$ is determined by its moment sequence — and the growth hypothesis plays exactly the role that Carleman's condition played in Chapter 1. The parallel is worth stating in full, since it closes the loop opened there on the nose rather than approximately:

At the level of individual objects the correspondence is between the power X^k of a random variable and the iterated integral $X^n_{0,T}$ of a path; there it is Theorem 2.4.7 that has no counterpart on the left, since the powers of a single real number determine it trivially. The table below is therefore stated one level up, at the level of *laws*, which is where the two theories are genuinely parallel.

Random variables (Ch. 1)	**Paths (this chapter)**
moments $\mathbb{E}[X^k]$	expected iterated integrals $\mathbb{E}\Big[X^n_{0,T}\Big]$
moment generating function	expected signature $\mathbb{E}[\mathrm{S}(\mathbf{X})]$
Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$	infinite radius of convergence of $\mathbb{E}[\mathrm{S}(\mathbf{X})]$
fails for the log-normal (App. A.3)	holds for a wide class, including Brownian motion on a bounded interval

The right-hand column is deliberately headed by $\mathbb{E}[\mathrm{S}(\mathbf{X})]$ and not by $\mathrm{S}(\mathbf{X})$: the signature is a property of a single path, whereas the moment generating function is a property of a law, and it is precisely the averaging of Definition 2.4.8 that puts the two on the same footing. That is the distinction Theorem 2.4.9 exists to make.

One word on the direction of the two criteria, since the parallel is exact only when it is stated correctly. Carleman's condition is *sufficient* for determinacy, not necessary; the log-normal moments grow as $e^{k^2\sigma^2/2}$ and therefore violate it, but that violation establishes only that the criterion is silent. What establishes indeterminacy is Heyde's explicit construction (Appendix A.3), and it is that construction, not the growth rate, which does the work in Chapter 1. On the right-hand side the corresponding criterion — infinite radius of convergence — is likewise sufficient, and here it is *satisfied*: the signature decays factorially by Proposition 2.4.4, so the series converges for every λ whenever the path length has enough integrability, and Theorem 2.4.9 applies. The two situations are thus parallel in form and opposite in outcome, and that is the sense in which, for paths, the analogous statement goes the right way.

Remark 2.4.10 (Why this matters for calibration)*.* The three results above are what make the signature usable as a feature map for path-dependent problems. Theorem 2.4.7 guarantees that no information is discarded by the transform; Proposition 2.4.4 guarantees that truncation is cheap; Proposition 2.4.5 guarantees that any continuous functional of the path is, to arbitrary accuracy, a *linear* functional of the truncated signature, so that a learning problem on path space becomes a regression. And, as noted in Remark 2.2.16, none of the three requires a lifting theorem: an observed data stream, joined by line segments, is a bounded-variation path, and (2.46) applies to it directly, whatever the regularity of the process it samples. We do not pursue signature-based calibration in this dissertation — Chapter 3 takes a different route — but we record the transform here because it is the point at which rough path theory becomes a practical tool rather than a structural one, and because it is the natural direction in which the present work extends.

2.5 Rough Volatility Models

Subsection 2.1.2 recorded the empirical finding and wrote down the RFSV model (2.4), which is the natural time-series description of a rough log-volatility. For pricing one wants something else: a model specified directly under the pricing measure, rich enough to fit an implied volatility surface, and tractable enough to be calibrated. The model that meets these requirements is the one that pays the debt incurred in Subsection 1.5.2, where the affine structure of the Heston model was described as "a structural property rather than a happy accident" and its rough generalisation promised.

Definition 2.5.1 (Rough Heston)**.** Let $H \in (0, \frac{1}{2})$ and set $\alpha_H := H + \frac{1}{2} \in (\frac{1}{2}, 1)$. The **rough Heston model** is

$$\begin{cases} dS_t = (r-q)\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^S, \\ v_t = v_0 + \dfrac{1}{\Gamma(\alpha_H)} \displaystyle\int_0^t (t-s)^{\alpha_H - 1} \Big[\kappa(\theta - v_s)\, ds + \xi \sqrt{v_s}\, dW_s^v\Big], \end{cases} \tag{2.50}$$

with $d\langle W^S, W^v\rangle_t = \rho\, dt$ in the sense of Definition 1.2.24.

Setting $\alpha_H = 1$, that is $H = \frac{1}{2}$, collapses the convolution kernel to a constant and recovers the classical Heston system (1.138) exactly. For $\alpha_H < 1$ the kernel $(t-s)^{\alpha_H - 1}$ is singular at $s = t$, and it is this singularity — a fractional integral of order α_H — that roughens the variance path to Hölder regularity H^-, in accordance with Proposition 2.1.2 c).
Two structural consequences follow immediately, and they pull in opposite directions.
The first is a loss. The convolution in (2.50) makes v_t depend on the entire history of W^v and not merely on v_s, so **the variance process is no longer Markov**, and neither is the pair (S, v). Every tool of Subsection 1.2.6 therefore becomes unavailable at once: there is no infinitesimal generator, no Kolmogorov equation, no Feynman–Kac representation. This is precisely the failure anticipated at the end of that subsection, where the semigroup identity

$P_{t+s} = P_t \circ P_s$ was described as a restatement of the Markov property and as "the first of the three approaches to fail" under memory. It has now failed.
The second is a survival, and it is what makes the model usable. The *affine* structure — the fact that the drift and the squared diffusion coefficient are affine in v — is untouched by the convolution. Affineness was what produced the Riccati system (1.140) in the classical case, and it produces its analogue here.

Theorem 2.5.2 (El Euch–Rosenbaum)**.** *Let F denote the forward price and write $\tau = T - t$. Under* (2.50) *the characteristic function of the centred log-forward is*

$$\varphi(u,\tau) \;=\; \mathbb{E}^{\mathbb{Q}}\Big[\exp\Big(iu\log\tfrac{F_T}{F_t}\Big)\,\Big|\,\mathcal{F}_t\Big] \;=\; \exp\Big(\kappa\theta\int_0^\tau \psi(u,s)\,ds \;+\; v_t\,I^{1-\alpha_H}\psi(u,\tau)\Big), \tag{2.51}$$

where $I^{1-\alpha_H}$ denotes the Riemann–Liouville fractional integral of order $1-\alpha_H$ and $\psi(u,\cdot)$ is the unique continuous solution of the ***fractional Riccati equation***

$$D^{\alpha_H}\psi(u,\tau) \;=\; \frac{\xi^2}{2}\,\psi(u,\tau)^2 \;+\; \Big(\rho\xi\,iu-\kappa\Big)\psi(u,\tau) \;-\; \frac{u^2+iu}{2}, \qquad I^{1-\alpha_H}\psi(u,0)=0, \tag{2.52}$$

D^{α_H} being the Riemann–Liouville fractional derivative of order α_H. ■

The result is due to El Euch and Rosenbaum [21]. Comparison with Chapter 1 is the point of stating it. Equation (2.52) is (1.140) *term for term*: the same quadratic in ψ, the same linear coefficient $\rho\xi iu - \kappa$, the same inhomogeneity $-\frac{1}{2}(u^2+iu)$. The single difference is that the ordinary derivative ∂_τ has been replaced by a fractional derivative of order $\alpha_H = H + \frac{1}{2}$, and at $H = \frac{1}{2}$ the two coincide. The promise made in Subsection 1.5.2 — that the generalisation would be "a short step rather than a fresh start" — is discharged by this observation, and the Fourier inversion (1.142) carries over verbatim, since it depends on the characteristic function and on nothing else.

Remark 2.5.3 (What is gained and what it costs)*.* Three comments, in decreasing order of comfort.
The model reproduces the skew. By construction the variance has regularity H, so the short-maturity at-the-money skew behaves as $\tau^{H-1/2}$ by (2.5). This is the empirical shape, and no classical stochastic volatility model produces it.
The equation is non-local, hence expensive. Because D^{α_H} is an integral operator, (2.52) cannot be advanced by a local time-stepping scheme: each step requires the entire computed history of ψ, so a discretisation over n steps costs $O(n^2)$ rather than $O(n)$. The standard solver is the Adams predictor–corrector scheme. Since calibration evaluates (2.51) at hundreds of strikes for every trial parameter vector, this cost is the practical bottleneck, and it is the main reason rough Heston is harder to deploy than its classical counterpart.
It is not a theorem of this chapter. Theorem 2.5.2 is proved by techniques from affine Volterra processes and fractional calculus, not by rough path theory, and nothing in Sections 2.2 to 2.4 is used in its proof. This is not an inconsistency but a division of labour, and Section 2.6 explains why it is unavoidable: at $H \approx 0.1$ the lifting theorems are not available, so a route that does not require them is not merely convenient but necessary. What rough path theory supplies here is the vocabulary — the regularity exponent, the p-variation scale, the precise sense in which the Markov property has been traded away — rather than the machinery.

2.6 The Boundary of the Theory

It would be a poor advertisement for a chapter on roughness to conclude without saying where its own methods give out, particularly since Remark 2.2.16 has already located the point.

Rough path theory lifts fractional Brownian motion for $H > \frac{1}{4}$. The volatility data suggest $H \approx 0.1$. The gap is not an artefact of the level-two restriction (2.15) that we adopted for clarity: for $H \leq \frac{1}{4}$ the piecewise-linear approximations fail to converge at *any* level, and the canonical lift does not exist.

The difficulty is visible in the pricing problem itself. Writing $\widehat{W}_t = \int K^H(t-s)\,dW_s$ for the Riemann–Liouville form of the driving noise in (2.4), with kernel $K^H(u) = u^{H-1/2}\mathbb{1}_{u>0}$, one must make sense of integrals of the type

$$\int_0^T F\Big(\widehat{W}_t\Big)\,dW_t. \tag{2.53}$$

The integrator W is a perfectly good Brownian rough path. The integrand, however, has Hölder regularity H^-, which for $H < \frac{1}{2}$ is *worse* than that of W, so $F(\widehat{W})$ is not a path controlled by W in the sense of Definition 2.3.2, and Theorem 2.3.5 does not apply. The Taylor expansion (2.25) is simply the wrong ansatz.

What replaces it is Hairer's theory of *regularity structures*, of which rough path theory is the one-dimensional special case. There, the fixed local expansion (2.25) in powers of $X_{s,t}$ is replaced by an expansion in an abstract, problem-dependent basis of symbols, and the products which fail to converge are repaired by *renormalisation* — subtracting divergent constants in a manner constrained by the algebra rather than chosen by hand. Applied to (2.53) the construction requires four symbols, corresponding to the powers $(\widehat{W}_{s,t})^m$ for $m \leq 3$, and yields a renormalised Wong–Zakai theorem of the form

$$\int_0^t F\Big(\widehat{W}^\varepsilon\Big)\,dW^\varepsilon \;-\; c^\varepsilon\int_0^t F'\Big(\widehat{W}^\varepsilon\Big)\,ds \;\longrightarrow\; \int_0^t F\Big(\widehat{W}\Big)\,dW, \tag{2.54}$$

valid for $H > \frac{1}{8}$ and with a constant c^ε which *diverges* as the mollification parameter ε tends to zero.

The reader who has been keeping score will object that $\frac{1}{8} = 0.125$ exceeds the $H \approx 0.1$ reported in Subsection 2.1.2, so that this construction misses the empirical exponent as well. That is correct, and the resolution is favourable: the threshold $\frac{1}{8}$ is an artefact of stopping at four symbols, not a limit of the method. The number of symbols required is governed by the smallest M with $(M+1)H + \frac{1}{2} > 1$, so admitting more symbols lowers the threshold, and the construction extends to every $H \in (0, \frac{1}{2})$ at the cost of a regularity structure whose size grows as $H \downarrow 0$. Roughness is therefore expensive rather than prohibitive — which is the honest summary, and a different statement from the genuine obstruction of Remark 2.2.16, where no number of levels suffices because the lift itself fails to exist. At $H = \frac{1}{2}$ the constant is $\frac{1}{2}$ and (2.54) degenerates into the Itô–Stratonovich correction of Chapter 1, which is a pleasing consistency check and a fair summary of the whole subject: the corrections that stochastic calculus treats as fixed constants are, at lower regularity, divergent quantities held in place by an algebraic structure.

We shall not develop any of this. The construction is set out in [25, Ch. 13–14], and the rough volatility literature built on it is surveyed in [2] and [48]. It is recorded here for the same reason Definition 1.1.32 was recorded in Chapter 1: to mark honestly the edge of what has been proved. The applications of Chapter 3 are constructed so as not to depend on a lifting theorem that is unavailable at the exponent the data demand — the local volatility surface of Subsection 1.5.1 being a classical diffusion, and the rough Heston model of Definition 2.5.1 being tractable through the affine route of Theorem 2.5.2 rather than through rough integration.

Coda: the three re-derivations

It is worth gathering, before closing, the promise made at the end of Section 2.1. Three results of Chapter 1 have been obtained again in this chapter on wholly different foundations, and the accounting in each case was stated where the result appeared:

i) **Itô's formula** (Corollary 2.3.13). Recovered from the rough chain rule (2.35) applied to the Itô lift, whose bracket (2.33) is $t\,\mathrm{Id}$. *Cost:* $F \in \mathcal{C}_b^3$ rather than $\theta \in \mathcal{C}^{1,2}$, and time-homogeneity. *Gain:* the identity holds pathwise, for a fixed trajectory and its lift, with both sides continuous in ϱ_α.

ii) **The Itô–Stratonovich correction** (2.38). Recovered not as a theorem about Brownian motion but as the general formula for the difference between two lifts of the *same* path, evaluated at $f_t = \frac{1}{2}t\,\mathrm{Id}$. *Cost:* none of substance. *Gain:* the correction is exhibited as a consequence of choosing $\mathbb{W}^{\text{Itô}}$ over $\mathbb{W}^{\text{Strat}}$, rather than as a feature of the integral.

iii) **Wong–Zakai** (Corollary 2.3.20). Recovered in two lines from the continuity of the Itô–Lyons map, Theorem 2.3.19. *Cost:* the lifting theorem for W, performed once in Example 2.2.13. *Gain:* the reason the limit is Stratonovich and not Itô becomes visible — smooth approximations carry geometric lifts, and the limit remembers the area.

Taken together these say something the classical theory cannot. In Chapter 1 the second-order term of Itô's rule, the $\frac{1}{2}\int F'$ of the Itô–Stratonovich dictionary, and the failure of smooth approximation to converge to the Itô solution appear as three separate facts, each proved by its own argument and each attributable to the non-vanishing quadratic variation of Brownian motion. Here they are one fact, seen three times: the trajectory does not determine its own iterated integrals, a choice must be made, and every one of the three phenomena is that choice becoming visible. Equation (2.54) above is the same statement pushed to lower regularity, where the constant held in place by the algebra is no longer $\frac{1}{2}$ but divergent.

Chapter 3

Applications

3.1 Calibration of a Volatility Surface

3.1.1 The Model

3.2 Joint Calibration (?)

3.2.1 The Challenge (?)

3.2.2 Calibration of a (???)

3.3 Calibration to Single Names: De-Americanisation

The local volatility surface of Subsection 1.5.1 was constructed from European quotes. Single-name options are American, and bridging that gap is the first practical step in any calibration on individual equities. This section carries out the procedure deferred in Chapter 1.
A major practical hurdle arises during calibration: Dupire's formula (Theorem 1.5.1) requires the partial derivatives of *European* option prices, but single-name market quotes are American. We cannot apply Dupire directly to American prices.
Therefore, calibrating the local volatility surface to American options requires a short, iterative mapping procedure known as **De-Americanisation**:

1. **Quote Extraction:** Obtain the market prices of the liquid American options and calculate their Black-Scholes implied volatilities using an American pricer (such as a Binomial Tree or a PDE solver).

2. **Pseudo-European Mapping:** Assume these implied volatilities approximate the volatilities of European options with the exact same strikes and maturities. Use these volatilities to generate a grid of "pseudo-European" option prices.

3. **Local Volatility Construction:** Apply a smoothing parameterization (e.g., SVI) to the pseudo-European surface to ensure it is free of static arbitrage, and then apply Gatheral's form (1.136) of the Dupire formula to construct the local volatility surface $\sigma_{\text{loc}}(F, t)$.

4. **Repricing and Verification:** Substitute this $\sigma_{\text{loc}}(F, t)$ back into the LCP (Equation (1.146)) and solve it backwards using a finite difference method (such as Crank-Nicolson combined with a Projected Successive Over-Relaxation (PSOR) algorithm) to verify that the model accurately recovers the original American market quotes.

This iterative procedure effectively bridges the theoretical gap between the European foundations of Local Volatility and the practical reality of American single-name markets, ensuring that our models reflect true market dynamics without introducing arbitrage.
However, the de-Americanisation process can be inverted, and this must be of no surprise at this stage, by the so called PINNs (*Physics-Informed Neural Networks*). We will dive further into this topic in the final chapter of the present dissertation.

3.4 Implementation

3.5 Results

Chapter 4

Conclusions

Theorem 4.0.1. *This is the main result.*

As we saw in Theorem 4.0.1, ...

$$E = mc^2 \tag{4.1}$$

We substitute into equation (4.1). also 4.1

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Appendix A

Auxiliary Results

This appendix collects results which are used in the main text but whose statements would have interrupted the argument there. Nothing here is original, and proofs are given only where they run to a line or two; references are given in every case. The organising principle is simple: a result belongs in the body if the argument cannot be followed without it, and belongs here if the argument merely *invokes* it. Results which we do prove, but whose proofs are too long to sit comfortably in the body, are deferred instead to Appendix B.

A.1 Complements on conditional expectation

Throughout, $\mathcal{G} \subseteq \mathcal{F}$ is a sub-σ-algebra of a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and all identities are understood to hold almost surely. Proofs may be found in [4] or [20]. Items 8 and 9, which are used constantly in the main text, are stated there as Proposition 1.1.4; the numbering below is consistent with it.

Proposition A.1.1 (Properties of conditional expectation)**.** *Let $X, Y \in L^1(\Omega)$. Then:*

1. $\mathbb{E}[\,\mathbb{E}[X \mid \mathcal{G}]\,] = \mathbb{E}[X]$*; that is,* $\mathbb{E}[X \mid \mathcal{G}]$ *and* X *have the same expectation.*

2. *If* X *is* $\mathcal{G}$*-measurable, then* $\mathbb{E}[X \mid \mathcal{G}] = X$.

3. *If* X *and* $\mathcal{G}$ *are independent — that is,* $\{X \in U\}$ *and* A *are independent events for every Borel* $U \subseteq \mathbb{R}$ *and every* $A \in \mathcal{G}$ *— then* $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$.

4. *(**Taking out what is known**) If* Y *is* $\mathcal{G}$*-measurable and* $\mathbb{E}[|XY|] < \infty$*, then* $\mathbb{E}[XY \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$.

5. *(**Tower property**) If* $\mathcal{H}$ *is a sub-σ-field of* $\mathcal{G}$*, then* $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[\,\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}\,]$.

6. *If* $X \le Y$*, then* $\mathbb{E}[X \mid \mathcal{G}] \le \mathbb{E}[Y \mid \mathcal{G}]$*; consequently* $|\mathbb{E}[X \mid \mathcal{G}]| \le \mathbb{E}[\,|X| \mid \mathcal{G}]$.

7. *For all* $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\,\mathbb{E}[X \mid \mathcal{G}] + b\,\mathbb{E}[Y \mid \mathcal{G}]$*. In particular* $\mathbb{E}[\,\cdot \mid \mathcal{G}]$ *is a bounded linear operator from* $L^1(\Omega, \mathcal{F})$ *into* $L^1(\Omega, \mathcal{G})$*, a fact used to pass to limits in Theorem 1.2.6.*

The convergence theorems also carry over to the conditional setting, in exactly the form one would expect.

Proposition A.1.2 (Conditional Fatou lemma)**.** *Let* $X_n \ge 0$ *with* $X_n \in L^1(\Omega)$ *for every* $n \ge 1$*, and suppose* $\liminf_{n\to\infty} X_n \in L^1(\Omega)$*. Then*

$$\mathbb{E}\Big[\liminf_{n\to\infty} X_n \,\Big|\, \mathcal{G}\Big] \le \liminf_{n\to\infty} \mathbb{E}[X_n \mid \mathcal{G}] \qquad \textit{almost surely.}$$

Proposition A.1.3 (Conditional monotone convergence)**.** *Let* $0 \leq X_1 \leq X_2 \leq \cdots$ *and suppose* $X = \lim_{n\to\infty} X_n \in L^1(\Omega)$*. Then* $\mathbb{E}[X \mid \mathcal{G}] = \lim_{n\to\infty} \mathbb{E}[X_n \mid \mathcal{G}]$ *almost surely.*

Proposition A.1.4 (Conditional dominated convergence)**.** *Suppose* $X_n \to X$ *almost surely and* $|X_n| \leq Y$ *for every* n*, with* $Y \in L^1(\Omega)$*. Then* $\mathbb{E}[X_n \mid \mathcal{G}] \to \mathbb{E}[X \mid \mathcal{G}]$ *almost surely and in* $L^1(\Omega)$*.*

A.2 Characteristic functions

Recall that the characteristic function of a real random variable X is $\varphi_X(u) = \mathbb{E}[e^{iuX}]$, $u \in \mathbb{R}$. The following elementary properties are used without comment in the main text.

Proposition A.2.1. *For any real random variable* X*:*

i) $\varphi_X(0) = 1$ *and* $|\varphi_X(u)| \leq 1$ *for all* $u \in \mathbb{R}$*;*

ii) φ_X *is uniformly continuous on* $\mathbb{R}$*;*

iii) $\varphi_X(-u) = \overline{\varphi_X(u)}$ *(Hermitian symmetry);*

iv) $\varphi_{aX+b}(u) = e^{iub}\varphi_X(au)$ *for all* $a, b \in \mathbb{R}$*;*

v) if X *and* Y *are independent, then* $\varphi_{X+Y}(u) = \varphi_X(u)\varphi_Y(u)$*;*

vi) if $\mathbb{E}[|X|^n] < \infty$*, then* φ_X *is* n *times differentiable and*
$$\varphi_X^{(k)}(0) = i^k\, \mathbb{E}[X^k], \qquad 1 \leq k \leq n.$$
Conversely, the existence of $2n$ *continuous derivatives of* φ_X *in a neighbourhood of the origin implies* $\mathbb{E}[|X|^{2n}] < \infty$*.*

Property ii) deserves a word, since it is the reason the continuity hypothesis in the converse half of Lévy's continuity theorem (Theorem 1.1.15) is a genuine restriction rather than an automatic one: it constrains the *limit* function, which is not assumed in advance to be a characteristic function.

A.3 Moment indeterminacy of the log-normal law

The following construction, due to Heyde [36], is referred to in Subsection 1.1.1 and shows that a distribution possessing moments of all orders need not be determined by them.

Proposition A.3.1 (Heyde)**.** *Let* f_X *denote the density of the standard log-normal law, that is, the density of* e^Z *with* $Z \sim \mathcal{N}(0,1)$*. For* $a \in [-1, 1]$ *define*
$$f_a(x) = f_X(x)\Big(1 + a\sin(2\pi \log x)\Big), \qquad x > 0. \tag{A.1}$$
Then every f_a *is a probability density on* $(0, \infty)$*, the laws are mutually distinct for distinct* a*, and all of them share the moment sequence of the log-normal distribution.*

Non-negativity holds because $|a\sin(\cdot)| \leq 1$, and the moment identity reduces to showing that
$$\int_0^\infty x^k f_X(x)\sin(2\pi\log x)\, dx = 0, \qquad k \in \mathbb{N}.$$
Substituting $\log x = t + k$ turns the integrand into an odd function of t multiplied by the Gaussian weight, and the integral vanishes. Taking $k = 0$ shows in particular that each f_a integrates to 1.

The underlying obstruction is that the log-normal moments grow like $e^{k^2/2}$, which violates Carleman's condition $\sum_k m_{2k}^{-1/2k} = \infty$; when Carleman's condition holds, the moment problem is determinate. See [20] for the general theory.

A.4 Grönwall's inequality

The following is the standard comparison tool for integral inequalities. It is purely deterministic, and in the stochastic setting it is applied pathwise or to expectations.

Lemma A.4.1 (Grönwall's inequality)**.** *Let $g, \varphi \in L^1[0,T]$ and $\lambda > 0$ satisfy*

$$\varphi(t) \le g(t) + \lambda \int_0^t \varphi(s)\, ds, \qquad 0 \le t \le T.$$

Then

$$\varphi(t) \le g(t) + \lambda \int_0^t g(s)\, e^{\lambda(t-s)}\, ds, \qquad 0 \le t \le T.$$

In particular, if $g \equiv \kappa$ is constant, then $\varphi(t) \le \kappa e^{\lambda t}$ for all $0 \le t \le T$; and if $\kappa = 0$, then $\varphi \equiv 0$.

The final sentence is the one actually used. Uniqueness arguments for differential equations almost invariably reduce to showing that a suitable distance between two candidate solutions satisfies a Grönwall inequality with $\kappa = 0$, whence it vanishes identically.

A.5 Radon–Nikodym and the Jordan decomposition

For completeness we record the decomposition underlying Corollary 1.1.3.

Theorem A.5.1 (Jordan decomposition)**.** *Every finite signed measure ν on a measurable space $(\Omega, \mathcal{F})$ admits a unique decomposition $\nu = \nu^+ - \nu^-$ into mutually singular finite positive measures. The measure $|\nu| := \nu^+ + \nu^-$ is called the total variation of ν, and $\nu \ll \lambda$ if and only if $|\nu| \ll \lambda$.*

See [55] for a proof.

A.6 Uniform integrability and the class (D)

Two integrability notions are invoked in the main text — in the hypotheses of Theorem 1.1.28 and of Theorem 1.5.6 respectively — and are recorded here.

Definition A.6.1 (Uniform integrability)**.** A family $\{X_i\}_{i \in I}$ of integrable random variables is **uniformly integrable** if

$$\lim_{c \to \infty} \sup_{i \in I} \mathbb{E}\Big[\, |X_i|\, \mathbb{1}_{\{|X_i| > c\}} \Big] \;=\; 0. \tag{A.2}$$

The condition says that no member of the family carries a non-negligible amount of its mass arbitrarily far out in the tail, *uniformly in i*. It is strictly stronger than $\sup_i \mathbb{E}|X_i| < \infty$ and strictly weaker than domination by a single integrable variable: if $|X_i| \le Y$ for all i with $Y \in L^1$, then (A.2) follows from dominated convergence, which is why hypothesis iii) of Theorem 1.1.28 implies hypothesis ii). Its role is that it is exactly the extra assumption converting convergence in probability into convergence in L^1, and exactly what is needed for a martingale to be closed by a terminal variable. See [20, Ch. 4].

Definition A.6.2 (Class (D))**.** A process X is of **class (D)** if the family $\{X_\tau : \tau \text{ a stopping time}\}$, indexed by all stopping times of the filtration, is uniformly integrable.

The class is named for Doob. It is the hypothesis under which the Doob–Meyer decomposition (Theorem 1.2.28) produces an *integrable* increasing process rather than merely a local one, and under which the Snell envelope of Theorem 1.5.6 is well defined. Any process dominated by an integrable variable is of class (D); in particular, so is any bounded process, which covers every application in this dissertation.

A.7 The essential supremum of a family of random variables

Definition A.7.1 (Essential supremum)**.** Let $\{X_i\}_{i\in I}$ be a family of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, indexed by an arbitrary and possibly uncountable set I. An **essential supremum** of the family, written $\operatorname{ess\,sup}_{i\in I} X_i$, is a random variable X^* such that

i) $X_i \leq X^*$ almost surely, for every $i \in I$; and

ii) if Y satisfies i) in place of X^*, then $X^* \leq Y$ almost surely.

Such an X^* exists and is unique up to null sets.

The construction, for which see [39, Appendix A], produces X^* as the pointwise supremum of a *countable* subfamily, and that is the whole point of the notion. The pointwise supremum $\sup_{i\in I} X_i(\omega)$ over an uncountable I need not be measurable, and even when it is it may exceed every X_i on a set of positive probability in a way that carries no information — an uncountable union of null sets need not be null, which is the same phenomenon that separates modification from indistinguishability in Remark 1.1.8. The essential supremum repairs this by taking the supremum in the almost-sure order rather than pointwise, and it is the correct object in (1.144), where the family is indexed by the uncountably many stopping times available to the holder of an American option.

Appendix B

Deferred Proofs

This appendix collects those proofs which are given in full but whose length would have interrupted the argument in the main text. The organising principle is the complement of the one governing Appendix A: a result whose proof is short enough to be read in passing is proved where it is stated, a result we merely invoke is recorded without proof in Appendix A, and a result we genuinely use but whose proof runs to more than a few lines is stated in the body and proved here.

Each section below carries a pointer back to the statement it establishes. The arguments of Sections B.1–B.6 follow the treatment of [40]; those of Sections B.7–B.10, which concern Chapter 2, follow [25] and [30].

B.1 Doob's submartingale inequality

We prove Proposition 1.1.29 of Chapter 1.

Proof: Let Q be an enumeration of the rationals in $[0,T]$. Right-continuity gives $\sup_{0\leq t\leq T} Z_t = \sup_{r\in Q} Z_r$ almost surely, so it suffices to bound the maximum over a finite set $\{r_1 < \cdots < r_k\}$ and let $k\to\infty$. On such a set, decompose $\{\max_\nu Z_{r_\nu}\geq\varepsilon\}$ into the disjoint events A_ν on which r_ν is the *first* index at which the level is reached. Each $A_\nu\in\mathcal{F}_{r_\nu}$, and on A_ν we have $Z_{r_\nu}\geq\varepsilon$, so the submartingale property gives

$$\varepsilon\,\mathbb{P}(A_\nu)\;\leq\;\mathbb{E}\Big[Z_{r_\nu}\mathbb{1}_{A_\nu}\Big]\;\leq\;\mathbb{E}\Big[\mathbb{E}[Z_T\mid\mathcal{F}_{r_\nu}]\,\mathbb{1}_{A_\nu}\Big]\;=\;\mathbb{E}\Big[Z_T\,\mathbb{1}_{A_\nu}\Big]\;\leq\;\mathbb{E}\Big[\max\{Z_T,0\}\,\mathbb{1}_{A_\nu}\Big].$$

Summing over ν and using disjointness bounds $\varepsilon\,\mathbb{P}(\max_\nu Z_{r_\nu}\geq\varepsilon)$ by $\mathbb{E}[\max\{Z_T,0\}]$, uniformly in k. Letting $k\to\infty$ gives the claim. The martingale case follows by applying this to the submartingale $|X_t|$, which is one by the conditional Jensen inequality. ■

B.2 Density of step processes in $L^2_{ad}([0,T]\times\Omega)$

We prove Lemma 1.2.3 of Chapter 1.

Proof: The argument proceeds by successive approximation in three steps, each enlarging the class of integrands. Write $\|f\|^2 := \int_0^T\mathbb{E}[f(t)^2]\,dt$ for the norm of $L^2_{ad}([0,T]\times\Omega)$, and recall that it suffices to approximate an arbitrary f within ε by a step process, since the step processes then form a dense subspace.

Step 1: f bounded with continuous sample paths. Let $|f|\leq M$ and let $\mathcal{P}_n=\{0=t_0^n<\cdots<t_{k_n}^n=T\}$ be partitions with $\|\mathcal{P}_n\|\to 0$. Put

$$f_n(t) := \sum_{i=1}^{k_n} f(t_{i-1}^n)\,\mathbb{1}_{[t_{i-1}^n,t_i^n)}(t).$$

Each f_n is a step process, and it is *adapted* because the coefficient $f(t^n_{i-1})$ is $\mathcal{F}_{t^n_{i-1}}$-measurable and multiplies the indicator of an interval beginning at t^n_{i-1} — this is the left-endpoint choice, and it is the only reason adaptedness survives. For almost every ω the path $t \mapsto f(t,\omega)$ is continuous on the compact $[0,T]$, hence uniformly continuous, so $f_n(t,\omega) \to f(t,\omega)$ for every t. Since $|f_n - f|^2 \leq 4M^2$, which is integrable on $[0,T]\times\Omega$ (a finite measure space), dominated convergence gives $\|f_n - f\| \to 0$.

Step 2: f bounded, sample paths not assumed continuous. Let $|f| \leq M$. Extend f by 0 outside $[0,T]$ and mollify in time: for $n \in \mathbb{N}$ set

$$g_n(t) := n\int_{t-1/n}^{t} f(s)\,ds = \int_0^1 f\Big(t - \tfrac{u}{n}\Big)du.$$

Each g_n has continuous sample paths, since it is an indefinite integral, and $|g_n| \leq M$. Adaptedness is preserved because $g_n(t)$ averages f over $[t-1/n,\,t]$ only, using no information after t; this is where averaging *backwards* rather than symmetrically matters. By the Lebesgue differentiation theorem, $g_n(t,\omega) \to f(t,\omega)$ for almost every t, for almost every ω, and dominated convergence again gives $\|g_n - f\| \to 0$. Step 1 now approximates each g_n by step processes, and a diagonal argument produces step processes converging to f.

Step 3: general $f \in L^2_{ad}([0,T]\times\Omega)$. Truncate at level n by setting $f^{(n)} := (f\wedge n)\vee(-n)$. Each $f^{(n)}$ is bounded and remains adapted, being a composition of f with a continuous deterministic function. Moreover $|f^{(n)} - f|^2 \leq f^2$ pointwise and $f^{(n)} \to f$ pointwise, so dominated convergence — with dominating function f^2, integrable precisely because $f \in L^2_{ad}$ — yields $\|f^{(n)} - f\| \to 0$. Combining with Step 2 completes the proof.

Adaptedness is the only delicate point throughout, and it is worth noting that each of the three steps preserves it for a different reason: the left-endpoint evaluation in Step 1, the backward averaging window in Step 2, and the deterministic truncation in Step 3. ∎

B.3 Continuity of the sample paths of an Itô integral

We prove Theorem 1.2.7 of Chapter 1.

Proof: Let $\{f_n\}$ be step processes converging to f in $L^2_{ad}([0,T]\times\Omega)$; the associated processes $X^n_t = \int_0^t f_n\,dW$ have continuous paths by construction, being finite sums of continuous terms. By Theorem 1.2.6 the difference $X^n - X^m$ is a martingale, so Doob's L^2 maximal inequality (Proposition 1.1.30) together with Itô's isometry gives

$$\mathbb{E}\Big[\sup_{0\leq t\leq T}\Big|X^n_t - X^m_t\Big|^2\Big] \;\leq\; 4\,\mathbb{E}\Big[|X^n_T - X^m_T|^2\Big] \;=\; 4\int_0^T \mathbb{E}\Big[(f_n - f_m)^2\Big]\,dt \longrightarrow 0.$$

Passing to a subsequence for which the right-hand side is summable, the Borel–Cantelli lemma yields $\sup_{t\leq T}|X^{n_k}_t - X_t| \to 0$ almost surely. A uniform limit of continuous functions is continuous, so almost all paths of X are continuous on $[0,T]$. ∎

B.4 The Itô integral as a limit of Riemann sums

We prove Theorem 1.2.8 of Chapter 1.

Proof: Let $f_{\mathcal{P}}(t) = \sum_i f(t_{i-1})\mathbb{1}_{[t_{i-1},t_i)}(t)$ be the left-endpoint step approximation, so that the Riemann sum is exactly $I(f_{\mathcal{P}})$. By Itô's isometry it suffices to show that $\int_0^T \mathbb{E}[(f - f_{\mathcal{P}})^2]\,dt \to 0$. Expanding the square, this integral is a sum of terms of the form $\mathbb{E}[f(t)^2] - 2\mathbb{E}[f(t)f(t_{i-1})] + \mathbb{E}[f(t_{i-1})^2]$, each built from the function $(s,t) \mapsto \mathbb{E}[f(t)f(s)]$ evaluated at points within $\|\mathcal{P}\|$ of the diagonal. That function is continuous by hypothesis, hence uniformly continuous on the compact $[0,T]^2$, so the whole expression tends to 0 uniformly as $\|\mathcal{P}\| \to 0$. ∎

B.5 Lévy's characterisation theorem

We prove Theorem 1.2.29 of Chapter 1.
Proof: The one-dimensional case is proved by applying Itô's rule to $t \mapsto e^{iuX_t}$ for fixed $u \in \mathbb{R}$. Since X is a continuous local martingale with $\langle X \rangle_t = t$, the second-order term contributes $-\frac{1}{2}u^2 e^{iuX_t}\,dt$, and after localisation and taking conditional expectations one obtains, for $s < t$,

$$\mathbb{E}\Big[e^{iu(X_t - X_s)} \mid \mathcal{F}_s\Big] = 1 - \tfrac{1}{2}u^2 \int_s^t \mathbb{E}\Big[e^{iu(X_r - X_s)} \mid \mathcal{F}_s\Big]\,dr.$$

This is *not* yet a deterministic equation: for each r the quantity $\mathbb{E}[e^{iu(X_r - X_s)} \mid \mathcal{F}_s]$ is an $\mathcal{F}_s$-measurable random variable, so the unknown is a random function and no uniqueness theorem for integral equations applies to it as it stands. The standard device removes the randomness by testing. Fix $A \in \mathcal{F}_s$, multiply through by $\mathbb{1}_A$, take expectations, and interchange expectation and the dr integral, which is legitimate by Fubini since the integrand has modulus one. Writing

$$h_A(t) := \mathbb{E}\Big[\mathbb{1}_A\, e^{iu(X_t - X_s)}\Big],$$

we obtain the genuinely deterministic linear integral equation

$$h_A(t) = \mathbb{P}(A) - \tfrac{1}{2}u^2 \int_s^t h_A(r)\,dr, \qquad t \geq s,$$

whose unique solution is $h_A(t) = \mathbb{P}(A)\, e^{-u^2(t-s)/2}$. Since this holds for every $A \in \mathcal{F}_s$ and the right-hand side is the integral over A of the constant $e^{-u^2(t-s)/2}$, we conclude

$$\mathbb{E}\Big[e^{iu(X_t - X_s)} \mid \mathcal{F}_s\Big] = e^{-u^2(t-s)/2} \qquad \text{almost surely.}$$

The conditional characteristic function of $X_t - X_s$ given $\mathcal{F}_s$ is therefore deterministic and equal to that of $\mathcal{N}(0, t-s)$, which is simultaneously independence of the increment from $\mathcal{F}_s$ and the correct Gaussian law; Proposition 1.1.34 then supplies independence of increments. The multidimensional case is identical, applied to $e^{iu^\top X_t}$ with the cross-variations $\langle X^i, X^j \rangle_t = \delta_{ij} t$ making the exponent $-\frac{1}{2}\|u\|^2(t-s)$. ■

B.6 Girsanov's theorem

We prove Theorem 1.2.35 of Chapter 1.
Proof: By Theorem 1.2.29 it suffices to show that, under $\mathbb{Q}$, the process Z is a continuous local martingale null at the origin with $\langle Z \rangle_t = t$. Continuity and $Z_0 = 0$ are clear.
For the martingale property we use Lemma 1.2.34, which reduces the claim to showing that $Z(t)\mathcal{E}_\alpha(t)$ is a $\mathbb{P}$-martingale. Itô's product formula gives

$$d\Big(Z_t \mathcal{E}_\alpha(t)\Big) = \mathcal{E}_\alpha(t)\,dZ_t + Z_t\,d\mathcal{E}_\alpha(t) + dZ_t\,d\mathcal{E}_\alpha(t).$$

Now $dZ_t = dW_t - \alpha(t)\,dt$ and $d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)\,dW_t$, so by the Itô table $dZ_t\,d\mathcal{E}_\alpha(t) = \alpha(t)\mathcal{E}_\alpha(t)\,dt$. Substituting, the two dt terms cancel exactly:

$$d\Big(Z_t\mathcal{E}_\alpha(t)\Big) = \mathcal{E}_\alpha(t)\Big(dW_t - \alpha(t)dt\Big) + Z_t\alpha(t)\mathcal{E}_\alpha(t)\,dW_t + \alpha(t)\mathcal{E}_\alpha(t)\,dt = \Big(1 + Z_t\alpha(t)\Big)\mathcal{E}_\alpha(t)\,dW_t.$$

The right-hand side has no drift, so $Z\mathcal{E}_\alpha$ is a $\mathbb{P}$-local martingale, and under the stated integrability it is a martingale; hence Z is a $\mathbb{Q}$-martingale.
Finally, $\langle Z \rangle_t = \langle W \rangle_t = t$, because subtracting the finite-variation term $\int_0^t \alpha(s)ds$ does not change the quadratic variation, and because quadratic variation is a pathwise limit and is therefore unaffected by the passage to the equivalent measure $\mathbb{Q}$. The same argument applied to $Z_t^2 - t$ confirms the compensator. Lévy's theorem now identifies Z as a $\mathbb{Q}$-Brownian motion.
■

B.7 The sewing lemma

We prove the existence half of Lemma 2.3.1 of Chapter 2; uniqueness was established where the lemma was stated.

Proof: Fix $s < t$ and let $\mathcal{P}^k$ denote the dyadic partition of $[s,t]$ into 2^k intervals of equal length $2^{-k}(t-s)$. Write

$$I^k_{s,t} := \sum_{[u,v]\in\mathcal{P}^k} \Xi_{u,v}$$

for the associated approximation, so that $I^0_{s,t} = \Xi_{s,t}$. The whole proof consists in showing that $(I^k_{s,t})_k$ is Cauchy, and the mechanism is that passing from $\mathcal{P}^k$ to $\mathcal{P}^{k+1}$ splits each interval exactly once, so that the difference of consecutive approximations is a sum of defects of the form (2.23).

Step 1: consecutive approximations. Each interval $[u,v]$ of $\mathcal{P}^k$ is divided by $\mathcal{P}^{k+1}$ at its midpoint $w = (u+v)/2$, and the corresponding contribution to $I^{k+1}_{s,t}$ is $\Xi_{u,w} + \Xi_{w,v}$ in place of $\Xi_{u,v}$. Hence

$$I^{k+1}_{s,t} - I^k_{s,t} = -\sum_{[u,v]\in\mathcal{P}^k} \delta\Xi_{u,w,v},$$

and by the hypothesis (2.23), since each of the 2^k intervals has length $2^{-k}(t-s)$,

$$\left|I^{k+1}_{s,t} - I^k_{s,t}\right| \;\le\; 2^k \cdot C\Big(2^{-k}(t-s)\Big)^{\beta} \;=\; C\,|t-s|^{\beta}\,2^{-k(\beta-1)}. \tag{B.1}$$

Step 2: convergence. Because $\beta > 1$ the exponent $-(\beta-1)$ is negative, so the bounds (B.1) form a convergent geometric series. Summing them,

$$\left|I^m_{s,t} - I^k_{s,t}\right| \;\le\; C\,|t-s|^{\beta}\sum_{j\ge k} 2^{-j(\beta-1)} \;=\; C\,|t-s|^{\beta}\,\frac{2^{-k(\beta-1)}}{1-2^{-(\beta-1)}}, \qquad m > k,$$

so $(I^k_{s,t})_k$ is Cauchy and converges; call the limit $(\mathcal{I}\Xi)_{s,t}$. Taking $k = 0$ in the display and recalling $I^0_{s,t} = \Xi_{s,t}$ gives

$$\left|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}\right| \;\le\; \frac{C\,|t-s|^{\beta}}{1-2^{-(\beta-1)}} \;=\; \frac{C}{1-2^{\,1-\beta}}\,|t-s|^{\beta},$$

which is exactly (2.24). Note that it is here, and only here, that $\beta > 1$ is consumed: at $\beta = 1$ the series in Step 2 diverges, and correspondingly the constant $(1-2^{1-\beta})^{-1}$ blows up.

Step 3: a uniform bound over arbitrary partitions. The dyadic construction produces a limit along dyadic partitions; to know that every partition sequence produces the *same* limit — which is what additivity requires — we need an estimate valid for all partitions at once. Let $\mathcal{P} = \{s = t_0 < t_1 < \cdots < t_n = t\}$ be arbitrary, with $n \ge 2$. Since

$$\sum_{j=1}^{n-1}\Big(t_{j+1} - t_{j-1}\Big) = \Big(t_n + t_{n-1}\Big) - \Big(t_0 + t_1\Big) \le 2(t-s),$$

some interior point t_j satisfies $t_{j+1} - t_{j-1} \le 2(t-s)/(n-1)$. Deleting it changes the sum by exactly one defect,

$$\left|\sum_{\mathcal{P}}\Xi - \sum_{\mathcal{P}\setminus\{t_j\}}\Xi\right| = \left|\delta\Xi_{t_{j-1},t_j,t_{j+1}}\right| \le C\Big(\frac{2(t-s)}{n-1}\Big)^{\beta}.$$

Repeating until only $\{s,t\}$ remains and summing the resulting bounds,

$$\left|\sum_{\mathcal{P}}\Xi - \Xi_{s,t}\right| \;\le\; C\,\Big(2|t-s|\Big)^{\beta}\sum_{m\ge 1} m^{-\beta} \;=\; 2^{\beta}\zeta(\beta)\,C\,|t-s|^{\beta}, \tag{B.2}$$

the series converging precisely because $\beta > 1$. The bound (B.2) holds for every partition of every subinterval, with no reference to the dyadic scheme.

Step 4: partition-independence, additivity, and the Riemann-sum limit. Let $\mathcal{P}$ be any partition of $[s,t]$ and $\mathcal{Q}$ any refinement of it. Applying (B.2) on each interval $[u,v] \in \mathcal{P}$ to the partition induced there by $\mathcal{Q}$,

$$\Big|\sum_{\mathcal{Q}} \Xi - \sum_{\mathcal{P}} \Xi\Big| \le 2^\beta \zeta(\beta)\, C \sum_{[u,v]\in\mathcal{P}} |v-u|^\beta \le 2^\beta \zeta(\beta)\, C\, |\mathcal{P}|^{\beta-1} |t-s|.$$

Given two partitions, comparing each with their common refinement shows that the sums $\sum_{\mathcal{P}} \Xi$ form a Cauchy family as $|\mathcal{P}| \to 0$. The limit therefore exists, does not depend on the sequence of partitions chosen, and in particular coincides with the dyadic limit $(\mathcal{I}\Xi)_{s,t}$ constructed in Step 2 — so the estimate (2.24), with its better constant, is retained. This also proves the final assertion of the lemma, that $(\mathcal{I}\Xi)_{s,t} = \lim_{|\mathcal{P}|\to 0} \sum_{\mathcal{P}} \Xi$.
Additivity is now immediate: for $s \le u \le t$, concatenating partitions of $[s,u]$ and of $[u,t]$ yields partitions of $[s,t]$ whose mesh tends to zero, and whose sums converge to $(\mathcal{I}\Xi)_{s,u} + (\mathcal{I}\Xi)_{u,t}$ by construction and to $(\mathcal{I}\Xi)_{s,t}$ by the preceding paragraph. ∎

B.8 Gubinelli's integration theorem

We prove Theorem 2.3.5 of Chapter 2.
Proof: The entire proof consists in applying Lemma 2.3.1 to the two-parameter function

$$\Xi_{s,t} \;:=\; Y_s\, X_{s,t} \;+\; Y'_s\, \mathbb{X}_{s,t}, \tag{B.3}$$

which is exactly the summand appearing in the compensated Riemann sums (2.26). We must verify the hypothesis (2.23) with $\beta = 3\alpha > 1$.

Step 1: computing the defect. Fix $s \le u \le t$. By definition,

$$\delta\Xi_{s,u,t} = \Xi_{s,t} - \Xi_{s,u} - \Xi_{u,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} - Y_s X_{s,u} - Y'_s \mathbb{X}_{s,u} - Y_u X_{u,t} - Y'_u \mathbb{X}_{u,t}.$$

Now use the two structural facts at our disposal. Additivity of the first level, $X_{s,t} = X_{s,u} + X_{u,t}$, collapses the terms in Y_s to $Y_s X_{u,t}$; and Chen's relation (2.16), in the form $\mathbb{X}_{s,t} - \mathbb{X}_{s,u} = \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$, collapses the terms in Y'_s to $Y'_s \mathbb{X}_{u,t} + Y'_s X_{s,u} X_{u,t}$. Therefore

$$\delta\Xi_{s,u,t} = \Big(Y_s - Y_u\Big) X_{u,t} + Y'_s X_{s,u} X_{u,t} - \Big(Y'_u - Y'_s\Big)\mathbb{X}_{u,t} = -\Big(Y_{s,u} - Y'_s X_{s,u}\Big) X_{u,t} - \Big(Y'_u - Y'_s\Big)\mathbb{X}_{u,t},$$

and recognising the first bracket as the remainder $R^Y_{s,u}$ of Definition 2.3.2 gives precisely (2.29):

$$-\delta\Xi_{s,u,t} = R^Y_{s,u}\, X_{u,t} + \Big(Y'_u - Y'_s\Big)\mathbb{X}_{u,t}.$$

It is worth pausing on this line, since it is the whole reason the theory is built as it is. The second-order term $Y'_s \mathbb{X}_{s,t}$ was inserted into (B.3) for no other purpose than to produce, via Chen's relation, the cross-term $Y'_s X_{s,u} X_{u,t}$ which cancels the leading part of $(Y_s - Y_u) X_{u,t}$. Without it the defect would be of order $\alpha + \alpha = 2\alpha \le 1$ and the sewing lemma would not apply — which is exactly the failure of the naïve Riemann sums.

Step 2: the estimate. Bounding the two terms separately, using Definitions 2.2.7 and 2.3.2 and $u - s, t - u \le t - s$,

$$\Big|\delta\Xi_{s,u,t}\Big| \le \|R^Y\|_{2\alpha} |u-s|^{2\alpha}\, \|X\|_\alpha |t-u|^\alpha + \|Y'\|_\alpha |u-s|^\alpha\, \|\mathbb{X}\|_{2\alpha} |t-u|^{2\alpha} \le C_\Xi\, |t-s|^{3\alpha},$$

with $C_\Xi := \|X\|_\alpha \|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha}\|Y'\|_\alpha$. Since $\alpha > \frac{1}{3}$ we have $3\alpha > 1$, so Lemma 2.3.1 applies and yields a unique additive $\mathcal{I}\Xi$ with $|(\mathcal{I}\Xi)_{s,t} - \Xi_{s,t}| \le C\, C_\Xi |t-s|^{3\alpha}$. By the last assertion of that lemma, $(\mathcal{I}\Xi)_{s,t}$ is the limit of $\sum_{\mathcal{P}} \Xi_{u,v}$, which is (2.26); and the displayed bound is (2.27).

Step 3: the integral is controlled. Write $Z_t := (\mathcal{I}\Xi)_{0,t}$, which is well defined by additivity. From the estimate just obtained,

$$Z_{s,t} - Y_s X_{s,t} = Y'_s \mathbb{X}_{s,t} + O\Big(|t-s|^{3\alpha}\Big),$$

so $R^Z_{s,t} := Z_{s,t} - Y_s X_{s,t}$ obeys $|R^Z_{s,t}| \le \|Y'\|_\infty \|\mathbb{X}\|_{2\alpha}|t-s|^{2\alpha} + C\, C_\Xi |t-s|^{3\alpha}$, which is $O(|t-s|^{2\alpha})$ since $3\alpha > 2\alpha$. Hence $\|R^Z\|_{2\alpha} < \infty$ and $(Z,Y) \in \mathcal{D}^{2\alpha}_X$, which is (2.28). That Z is itself α-Hölder follows from $|Z_{s,t}| \le |Y_s||X_{s,t}| + |R^Z_{s,t}|$. ■

B.9 Existence and uniqueness for rough differential equations

We prove Theorem 2.3.17 of Chapter 2.

Proof: Throughout, C denotes a constant depending only on α and $\|f\|_{C^3_b}$, allowed to change from line to line, and we abbreviate $M := |||\mathbf{X}|||_\alpha$. Recall from Definition 2.3.14 that solutions on $[0,\tau]$ are exactly the fixed points of

$$\mathcal{M}(Y,Y') \;=\; \Big(\xi + \int_0^\cdot f(Y_r)\, d\mathbf{X}_r,\ f(Y)\Big),$$

which is well defined on $\mathcal{D}^{2\alpha}_X([0,\tau])$ by Example 2.3.4 ii) together with Theorem 2.3.5.

Step 1: the ball. Since the space $\mathcal{D}^{2\alpha}_X$ is affine rather than linear once the initial data are fixed, we centre the ball at the canonical first-order approximation

$$\mathcal{W}_t := \xi + f(\xi) X_{0,t}, \qquad \mathcal{W}'_t := f(\xi), \qquad t \in [0,\tau],$$

which lies in $\mathcal{D}^{2\alpha}_X$ with $R^{\mathcal{W}} \equiv 0$, and set

$$B_\tau(R) := \Big\{ (Y,Y') \in \mathcal{D}^{2\alpha}_X([0,\tau]) \ :\ Y_0 = \xi,\ Y'_0 = f(\xi),\ \|(Y,Y') - (\mathcal{W},\mathcal{W}')\|_{X,2\alpha} \le R \Big\},$$

a closed subset of a Banach space, hence a complete metric space.

Step 2: invariance. Let $(Y,Y') \in B_\tau(R)$. Applying (2.27) to the integrand $\Big(f(Y), Df(Y)Y'\Big)$, whose controlled norm is bounded by $C(1+R)$ by Example 2.3.4 ii), one obtains

$$\Big\|\mathcal{M}(Y,Y') - (\mathcal{W},\mathcal{W}')\Big\|_{X,2\alpha} \;\le\; C\,\Big|f(\xi)\Big|\, M \;+\; C\,(1+R)\, M\, \tau^\alpha.$$

The first summand does not involve τ and is bounded by $C\|f\|_\infty M$; choose $R := 2C\|f\|_\infty M$, so that this summand is at most $R/2$. Then choose τ small enough that $C(1+R)M\tau^\alpha \le R/2$. With these choices $\mathcal{M}\Big(B_\tau(R)\Big) \subseteq B_\tau(R)$.

Two features of this step are what make the theorem global, and they should be noted explicitly. First, the estimate (2.27) involves $\|Y'\|_\alpha$ and $\|R^Y\|_{2\alpha}$ but never $|Y_0|$; and f is bounded, so $|f(\xi)| \le \|f\|_\infty$ irrespective of ξ. Hence neither R nor τ depends on the initial condition. Second, the factor τ^α is produced by restricting the time horizon, and it is the only source of smallness available — which is why the argument is local in time and not in space.

Step 3: contraction. For $(Y,Y'), (\widetilde{Y}, \widetilde{Y'}) \in B_\tau(R)$, the same estimate applied to the difference of the integrands, together with the local Lipschitz property of $Y \mapsto f(Y)$ as a map of controlled paths — this is where the third derivative of f is used, and it is proved in [25, Lem. 7.3] — gives

$$\left\|\mathcal{M}(Y,Y') - \mathcal{M}(\widetilde{Y},\widetilde{Y'})\right\|_{X,2\alpha} \;\le\; C\,(1+R)\,M\,\tau^{\alpha}\,\left\|(Y,Y') - (\widetilde{Y},\widetilde{Y'})\right\|_{X,2\alpha}.$$

Reducing τ further so that $C(1+R)M\tau^{\alpha} \le \frac{1}{2}$ makes $\mathcal{M}$ a contraction of modulus $\frac{1}{2}$ on $B_\tau(R)$. The Banach fixed point theorem supplies a unique fixed point, that is, a unique solution on $[0,\tau]$, and $\tau =: T_0$ depends only on α, $\|f\|_{\mathcal{C}_b^3}$ and M, as claimed.

Step 4: patching. Because T_0 is independent of the initial condition, the solution on $[0,T_0]$ may be restarted at Y_{T_0} to produce a solution on $[T_0,2T_0]$, and so on; after $\lceil T/T_0\rceil$ steps the interval $[0,T]$ is exhausted. That the concatenation is again a controlled path, and that it solves (2.40) across the junctions, follows from the additivity of the rough integral.
For global uniqueness, suppose $(\widetilde{Y},\widetilde{Y'})$ is another solution on $[0,T]$ and let $\sigma := \sup\{t : \widetilde{Y} = Y \text{ on } [0,t]\}$. The two agree at σ by continuity, so applying Step 3 on $[\sigma,\sigma+\tau]$ — legitimate since τ does not depend on the initial value Y_σ — forces them to agree there also, contradicting the maximality of σ unless $\sigma = T$. ∎

B.10 Continuity of the Itô–Lyons map

We prove Theorem 2.3.19 of Chapter 2.
Proof: Let $\mathbf{X},\widetilde{\mathbf{X}} \in \mathbf{C}^{\alpha}$ with $|||\mathbf{X}|||_{\alpha}, \left|\left|\left|\widetilde{\mathbf{X}}\right|\right|\right|_{\alpha} \le M$, and let $(Y,f(Y))$ and $(\widetilde{Y},f(\widetilde{Y}))$ be the corresponding solutions started at ξ and $\widetilde{\xi}$. The difficulty of the statement is that the two solutions live in *different* Banach spaces, $\mathcal{D}^{2\alpha}_X$ and $\mathcal{D}^{2\alpha}_{\widetilde{X}}$, so no norm compares them directly; this is why Theorem 2.3.7 was stated in terms of the quantity

$$d_{X,\widetilde{X},2\alpha}\Big(Y,Y';\widetilde{Y},\widetilde{Y'}\Big) := \left\|Y' - \widetilde{Y'}\right\|_{\alpha} + \left\|R^{Y} - R^{\widetilde{Y}}\right\|_{2\alpha},$$

which is not a metric but does compare the two, and it is this quantity that we estimate.

Step 1: transferring the estimate through the fixed point. By Theorem 2.3.17 the solutions are the fixed points of the respective maps (2.42), so that $(Y,f(Y)) = \mathcal{M}(Y,f(Y))$ and likewise with tildes. Applying the stability of rough integration, Theorem 2.3.7, to the two integrals, and writing $\Xi := f(Y)$, $\widetilde{\Xi} := f(\widetilde{Y})$ for the integrands,

$$d_{X,\widetilde{X},2\alpha}\Big(Y,f(Y);\widetilde{Y},f(\widetilde{Y})\Big) \;\le\; C\Big(\varrho_\alpha\Big(\mathbf{X},\widetilde{\mathbf{X}}\Big) + \Big|\xi-\widetilde{\xi}\Big| + \tau^{\alpha}\, d_{X,\widetilde{X},2\alpha}\Big(\Xi,\Xi';\widetilde{\Xi},\widetilde{\Xi}'\Big)\Big). \tag{B.4}$$

The last term is the price of the nonlinearity, and it is handled by the local Lipschitz estimate for composition with $f \in \mathcal{C}_b^3$ [25, Thm. 7.6], which bounds it by

$$d_{X,\widetilde{X},2\alpha}\Big(\Xi,\Xi';\widetilde{\Xi},\widetilde{\Xi}'\Big) \;\le\; C\Big(\varrho_\alpha\Big(\mathbf{X},\widetilde{\mathbf{X}}\Big) + \Big|\xi-\widetilde{\xi}\Big| + d_{X,\widetilde{X},2\alpha}\Big(Y,f(Y);\widetilde{Y},f(\widetilde{Y})\Big)\Big).$$

Step 2: absorbing. Substituting the second display into (B.4) gives, with a new constant,

$$d_{X,\widetilde{X},2\alpha}\Big(Y,f(Y);\widetilde{Y},f(\widetilde{Y})\Big) \le C\Big(\varrho_\alpha\Big(\mathbf{X},\widetilde{\mathbf{X}}\Big) + \Big|\xi-\widetilde{\xi}\Big|\Big) + C\tau^{\alpha}\, d_{X,\widetilde{X},2\alpha}\Big(Y,f(Y);\widetilde{Y},f(\widetilde{Y})\Big).$$

Choosing τ small enough that $C\tau^{\alpha} \le \frac{1}{2}$ — possible with τ depending only on M, α and f, exactly as in Theorem 2.3.17 — the final term may be absorbed into the left-hand side, leaving

$$d_{X,\widetilde{X},2\alpha}\Big(Y,f(Y);\widetilde{Y},f(\widetilde{Y})\Big) \;\le\; 2C\Big(\varrho_\alpha\Big(\mathbf{X},\widetilde{\mathbf{X}}\Big) + \Big|\xi-\widetilde{\xi}\Big|\Big)$$

on $[0,\tau]$. This is the required bound in the comparison quantity.

Step 3: from the comparison quantity to the Hölder norm, and iteration. Writing $Y_{s,t} - \widetilde{Y}_{s,t} = f(Y_s)X_{s,t} - f(\widetilde{Y}_s)\widetilde{X}_{s,t} + R^Y_{s,t} - R^{\widetilde{Y}}_{s,t}$ and using the elementary inequality $|ab - \tilde{a}\tilde{b}| \le |a||b - \tilde{b}| + |a - \tilde{a}||\tilde{b}|$ on the first pair of terms, the bound of Step 2 converts into

$$\left\|Y - \widetilde{Y}\right\|_\alpha \;\le\; C\Big(\varrho_\alpha\big(\mathbf{X}, \widetilde{\mathbf{X}}\big) + \left|\xi - \tilde{\xi}\right|\Big)$$

on $[0, \tau]$, which is (2.43) on that interval. Since τ depends only on M and not on the initial conditions, the argument may be restarted at τ with initial values Y_τ and $\widetilde{Y}_\tau$, whose difference is already controlled by the right-hand side; after finitely many such steps the estimate covers $[0, T]$, with a constant that has been multiplied by a fixed factor at each step. ∎